

The Welfare of Nations: Social Preferences and the Macroeconomy*

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We present a Bewley-type theory of the Social Welfare Function (SWF). Agents have heterogeneous perceptions of how the planner should value the welfare of other agents, which results in so-called Individual Welfare Functions (IWFs), shaped by their life experiences. The SWF is constructed as an aggregation of IWFs, weighted by the political weights of each group of agents. We then propose an estimation strategy to compute the SWF and IWFs using the inverse optimal approach in a heterogeneous-agent model. We apply our methodology to France and the United States. France's SWF places greater emphasis on individuals with lower incomes, contrasting with the United States' SWF, which assigns greater weight to individuals with higher incomes. Finally, we simulate the fiscal system of the United States under the assumption of adopting the French SWF. Our findings indicate that the SWF significantly shapes the optimal steady-state fiscal system and equilibrium inequality.

Keywords : Social Welfare Function, Inequality, Fiscal systems.

JEL codes : E61, E62, E32. papers

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1 Introduction

Countries differ widely in terms of their fiscal systems and inequalities: The levels of social transfers, labor and capital taxes are very different across OECD countries. For instance, the average mandatory levies in France were 43% of its GDP from 1995 to 2007, while they were around 26% in the US for the same period. Various explanations can rationalize these international differences. First, production technologies can be different across countries, and so do the distortions generated by taxation. Second, in addition to facing different trade-offs, preferences can be heterogeneous. Individual preferences over consumption and leisure, as well as social preferences regarding redistribution, can differ across countries. Third, even if technology and preferences are the same, the political system that selects and implements actual policies can differ across countries, generating outcomes that are observationally equivalent to alternative social preferences. The goal of this paper is to disentangle these three explanations and to identify the contribution of social preferences to the design of actual tax systems.

To do so, we present a theory of the aggregation of individual preferences into social preferences, which can then be estimated in the data. This Bewley-type aggregation theory is based on four assumptions. First, we assume that individuals have their own view of how the planner should care about the welfare of all people in society. This corresponds to the so-called ethical preferences (Arrow, 1951, Harsanyi, 1955, or Sen, 1977), which can be seen as the individual view of an aggregate welfare criterion. To quote Harsanyi (1955), ethical preferences are indeed defined as an agent choosing “what he prefers only in those possibly rare moments when he forces a special impartial and impersonal attitude on himself.” These ethical preferences are represented by Individual Welfare Functions (IWFs), which are agent-specific. IWFs are thus heterogeneous, consistent with empirical investigations (Gaertner and Schokkaert, 2012, Fehr et al., 2013 or Stantcheva, 2021). Our second assumption is about the source of heterogeneity for IWFs. Following the Bewley tradition, we assume that agents’ relevant life experience is their economic history. Our construction could easily manage other dimensions of heterogeneity, but this representation is already sufficiently rich for us to discuss a wide variety of political implications of our estimation. Third, we assume that the ethical preferences are of the weighted utilitarianism type. Agents value the utility of other agents by attributing some weights to the individual utility of the latter. This representation is known to be flexible enough to embed moral and political concerns (including Libertarian, Egalitarian or Utilitarian ones) into the planner’s view (Saez and Stantcheva, 2016 or Fleurbaey and Maniquet, 2018).

Our fourth assumption is that the Social Welfare Function (SWF) that the planner uses to design the fiscal policies is the (possibly biased) aggregation of the heterogeneous IWFs. Indeed, agents can have heterogeneous political power in their ability to affect the planners’ decisions. Consistently with the literature on political inequality (e.g., Cage, 2024), this could for instance be explained by lobbying powers, participation rates in elections, or the representation of voters and their representatives. This construction yields a SWF that is stable, although agents’ ethical preferences can change with their own history. The SWF can thus be estimated in the data. A final remark is that we do not require the SWF to fulfill the Pareto principle. Since Sen (1970), it is indeed known that allowing the planner’s objective to account for moral or political concerns (such as being Libertarian or Egalitarian) can imply SWFs, which do not fulfill the

Pareto principle.

Considering this aggregate Social Welfare Function, we then use recent advances in optimal policies in heterogeneous-agent models to derive the policy maximizing the SWF. We use the Lagrangian approach, which allows us to derive first-order conditions of the planner, allowing for a connection with the public finance literature (LeGrand and Ragot 2022, Açıkgöz et al. 2022 or LeGrand and Ragot 2024). In particular, we isolate the Social Marginal Welfare Weights (SMWW) and concept of the Marginal Value of Public Funds (MVPF) in general equilibrium. The first-order conditions have a form close to the GSMWW of Saez and Stantcheva 2016, but as first-order conditions are based on the explicit SWFs, they are immune to the criticisms, which have been made to GSMWW (Sher 2024). We thus aim at contributing to closing the gap between public finance and macroeconomics in the design of optimal policies.

We then apply this methodology to estimate the aggregate SWFs, and the IWFs, of France and the United States to assess the role of Social preferences in the design of the fiscal system and equilibrium inequality. This methodology is called the inverse optimal approach, and it is often used in public finance (Bargain and Keane, 2010; Bourguignon and Amadeo, 2015, Lockwood and Weinzierl (2016) or Hendren (2020)). Our construction extends this strategy to an intertemporal model with general equilibrium effects. More precisely, we consider a heterogeneous-agent model (Bewley-Huggett-Aiyagari type) where agents face income risk and where a rich fiscal system is used to affect income inequality and to finance a public good. The fiscal system is composed of five instruments : a labor tax, which can be progressive, a capital tax, a consumption tax and public debt. These tools are a good redistribution representation of the actual richness of the fiscal system. We indeed show that income and wealth inequality in 2007 can be closely reproduced with a realistic income risk and fiscal system, both in France and in the US. We choose the period 2007 to exclude the financial crisis and the subsequent Covid-19 crisis, which led to significant transitory changes in fiscal structures. Then using the first-order conditions for the planner, we identify the aggregate SWFs for the actual allocation in France and in the US to be a Ramsey steady state.

This estimation generates three sets of results. First, the SWFs in France and in the US are very different from each other. The US SWF is increasing with income and puts the largest weight on rich and high-income agents. The weights for middle-class agents are lower, and low-income agents have the lowest weight. This shape of the SWFs is consistent with decreasing SMWW estimated in the literature, as they include marginal utilities, as we explain. The French SWF is different. It assigns a high weight to low-income agents, with weights decreasing with income and then increasing again for high-income agents. Maybe not surprisingly, the French SWF is more egalitarian at the bottom of the distribution, but its also puts high weights on very productive agents. Second, to understand the role of the SWF, we simulate the optimal US fiscal system if the US were to adopt the French SWF, keeping private preferences constant. We find that the Gini coefficient of wealth would decrease from 78% to 63%, approaching the French Gini coefficient of wealth of 68%. Consequently, social preferences are a primary driver of the fiscal system and household inequality. Third, we decentralize the aggregate SWF in heterogeneous IWFs to assess the heterogeneity of social preferences in each country. To do so, we first use turnout data in the US and in France to estimate political weights, following the

political economy literature surveyed by Cage (2024). We then estimate the set of IWFs, which are the closest to the self-interest one and which are consistent with the aggregate SWF. We find that the middle class is mostly libertarian in the US and egalitarian in France, and that there is a substantial heterogeneity in IWFs within countries.

Related literature Our paper is related to three streams of literature, heterogeneous-agent macroeconomy, public finance and social choice.

First, this paper contributes to the recent literature on optimal policies in heterogeneous-agent models. Early contributions, such as Aiyagari (1995) analyze general properties of capital tax in heterogeneous agent models. Aiyagari and McGrattan (1998) compute the optimal steady-state level of public debt. Dávila et al. (2012) show that the steady-state capital stock can be too low, solving for a constrained-efficient allocation. Some papers compute the optimal path of relevant instruments (Conesa et al. (2009) or Dyrda and Pedroni (2022) more recently). As mentioned above, some papers derive first order conditions of a Ramsey problem to solve for optimal policies (LeGrand and Ragot (2022), Açikgöz et al. (2022), Bhandari et al. (2021)). We use the recent development of the Lagrangian and truncation method, explained in LeGrand and Ragot (2024)) to allow for the inverse optimal approach.

Second, we contribute to the macroeconomics of public finance, expliciting the general equilibrium effects in the MSWW and in the MVPFs (Hendren and Sprung-Keyser (2020) or Ferey et al. (2024) for a recent contribution). Considering the inverse optimal approach, it is now often used in public finance (Bargain and Keane, 2010; Bourguignon and Amadeo, 2015, Lockwood and Weinzierl (2016) or Hendren (2020)). Chang et al. (2018) use an heterogeneous-agent model to estimate inequality aversion across countries, thus avoiding the computation of a Ramsey program. Heathcote and Tsujiyama (2021) also estimate the SWF in a static environment, but they allow for partial private insurance.

Third, this paper contributes to the literature on the Social Welfare Function in heterogeneous-agent models. There is a vast theoretical literature in the Social Choice literature considering relevant restrictions on the SWFs, since Arrow (1951), Harsanyi (1955), Sen (1970), Sen (1977), Fleurbaey and Trannoy (2003) Piacquadio (2017) or Ferey et al. (2024) and Fleurbaey and Maniquet, 2018 for a survey. A more empirical literature identifies relevant aspects of SWFs from experiments (Gaertner and Schokkaert 2012 or Fehr et al., 2013) or from surveys (Stantcheva (2021) for a recent analysis based on large US surveys). Informed by this literature we propose a construction of SWFs as an aggregation of IWFs, which is flexible enough to allow for estimation. We consider our construction to be a possible micro-foundation of weights used to assess optimal policies, such as the Generalized Social Marginal Welfare Weights introduced by Saez and Stantcheva (2016).

The paper proceeds as follows. In Section 2 we present motivating evidence on the French and US fiscal systems. We provide the basics of our Bewley construction of SWFs in Section 3 in the context of a simple model. The construction is generalized to an infinite horizon model in Section 4. Section 5 presents the environment in which we will conduct our estimation of SWF weights, as well as the corresponding Ramsey program. Finally, the quantitative investigation is presented in Section 6, while Section 7 concludes.

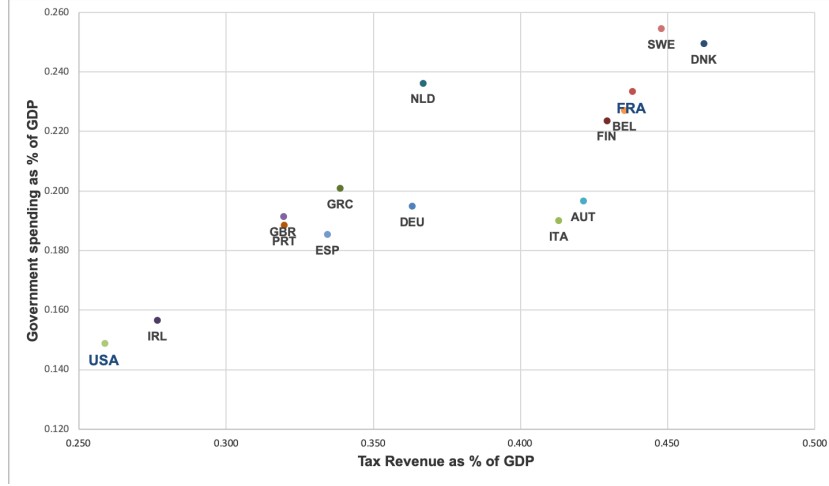


Figure 1: Government spending and tax revenues average from 1995 to 2021 (as a share of GDP). Source: own calculations.

2 The fiscal structure in France and in the US

We report key statistics about the French and US fiscal systems. As explained in the introduction, these two countries have the particularity of being among the most different OECD countries in terms of total taxation. On the one hand, France is one of the countries with the highest mandatory levies, while the US is one of the countries with the lowest levies. This is confirmed in Figure 1, which reports government spending on final goods as a share of GDP and tax revenues as a share of GDP for the two countries. A first observation from Figure 1 is that France and the US drastically differ with respect to the size of their governments. Both government spending and tax revenues are significantly higher in France than in the US. Second, the gap between tax revenues and government spending is larger in France than in the US. This reflects that within-country redistribution – measured as the difference between tax revenues and government spending – is of larger magnitude in France than in the US.

We now turn to the details of the taxation system within each country. We focus on the average tax system from 1995 to 2007, before the 2008-crisis –which we will use as a benchmark for calibrating our initial stationary equilibrium. We acknowledge that the numbers are quantitatively very close if we consider the period from 1995 to 2021, as shown in Figure 1, but consider a period before macroeconomic shocks appears more relevant. We use the results of Trabandt and Uhlig (2011), who provide estimates for the period 1995-2007. Results for France and the US are gathered in Table 1, which also includes some elements related to inequalities.

	Total taxes (%GDP)	τ^K (%)	τ^L (%)	τ^c (%)	B (%GDP)	G (%GDP)	Gini before redist.	Gini after redist.	Gini wealth
France	40	35	46	18	60	24	0.48	0.28	0.68
United States	26	36	28	5	63	15	0.48	0.40	0.77

Table 1: Summary of fiscal systems and inequalities in the US and in France. Total taxes, public debt B and public spending G in percentage of GDP; tax rates τ^K , τ^L and τ^c in percent; Gini indices unitless.

The first column reports the total mandatory levies as a share of GDP for the two countries. Following the literature, we decompose these total levies into three components: capital tax, labor tax, and consumption tax. Since Mendoza et al. (1994), this decomposition is widely used to compare the tax structure across countries (OECD, Eurostat). These three taxes are reported in columns 2–4. The second column shows the implicit capital tax, calculated as tax receipts on capital income divided by the capital tax. The third column provides the same statistic for the labor tax, while the fourth column reports the implicit tax on consumption.

We can observe that overall taxes are 50% higher in France than in the US. Although capital taxes are very close in both countries, the labor and consumption taxes differ significantly. The difference in labor tax partly stems from the financing of the French welfare system, which covers public pensions, unemployment benefits, health care, and family allowances. It mostly relies on social contributions based on the wage bill, which are considered as labor tax. Regarding consumption tax, it is much higher in France compared to the US, although this high value is comparable to those in other European countries.

Tax revenues are used to finance public spending, which includes both public consumption and public investment. Public spending is approximately one-fourth higher in France than in the US. This difference is partly explained by larger investments in public infrastructure in France.

The results in Table 1 considers linear tax for labor, however, progressive labor income tax is commonly used as a fiscal instrument for countries and current evidence suggests that the progressivity rate differ between France and the US. Comparing the progressivity of labor income tax across countries is challenging due to the complex tax schedules and deductions that are specific to each country. One approach to make this comparison tractable is to use a parametric form for the tax function. To compute this, we employ the log-linear tax function currently used in the literature (e.g. Benabou (2002) and Heathcote et al. (2017)).

$$\begin{aligned}\text{Tax:} \quad & T(y_i) = y_i - \kappa y_i^{1-\tau}. \\ \text{Disposable income:} \quad & D(y_i) = \kappa y_i^{1-\tau}. \\ & \log D(y_i) = \log \kappa + (1 - \tau) \log y_i,\end{aligned}$$

where y_i denotes the labor income of the country i , the parameter τ reflect the level of progressivity, and κ the average level of taxation. Notice that the higher the τ , the more progressive the tax system. Using the Luxembourg Income Study (LIS) database for France and the US for 2005, we estimate the tax progressivity for labor income ($\hat{\tau}$). To estimate this parameter, we restrict our attention to the heads of households and their spouses aged between 25 and 60 who were employed. We define *labor income* as the sum of wage income, self-employment income, and private transfers. Using the estimates from Mendoza et al. (1994), we can deduce from the capital income the value of capital income tax and we subtract from the total income tax this last variable. This allow us to obtain an estimate of the labor income tax, and then we define *disposable income* as *labor income* minus the *labor income tax*. Finally, we regress the log of disposable income on the log of labor income.

Table 2 reports our estimates for France and the US. One can notice that France has a much more progressive labor tax than the US. Our estimate of progressivity for the US is 0.16, which closely aligns with values used in the literature. Our value is lower than the 0.181 value estimated

by Heathcote et al. (2017), as we focused solely on estimating the progressivity of labor income and did not consider progressivity estimates for labor and capital income combined.

	$\hat{\tau}$	SE	Obs	R^2
France	0.23	0.0056	5289	0.855
United States	0.16	0.0019	38111	0.942

Table 2: Estimate of the progressivity of the labor income tax in the US and in France for 2005 using the LIS database.

All this heterogeneity in redistribution is also reflected in the evolution of income inequality before and after taxation. We proxy income inequality using the average Gini index between 1995 and 2007 (included), as reported in the OECD Income Inequality Database.¹ Note that the Gini indices barely vary over the period, and the picture would not have been different if we had reported the 2007 data only. The before-tax Gini indices for income are roughly similar in France and in the US. This value for France stems from the accounting of the (high) public pensions in France, which are counted as transfers and not as income. Consequently, this contributes to increasing the before-tax inequalities. However, the after-tax Gini indices are very different in the two countries, which is a consequence in terms of inequalities of the high transfers to households in France. While redistribution diminishes the Gini index for income by less than 10 points in the US, the reduction is twice as large in France and amounts to 20 points.

The second-to-last element we report is the public debt-to-GDP ratio, which appears to be comparable in France and in the US at around 60%.

The last column reports the Gini index for wealth. The data for France come from the Household Finance and Consumption Survey (HFCS) for the 2010 wave, which is the closest wave to our benchmark years. We have checked that the Gini index remains highly similar in the other waves. The wealth Gini index for the US is taken from the PSID in 2006.² As is standard, wealth inequalities in each country are higher than for income. The wealth Gini index in each country is approximately 30 points higher for wealth than for post-tax income. The comparison between the US and France yields a similar result as the income Gini index did. It confirms that inequalities—for wealth here—are sharper in the US than in France.

We will use the elements of Table 1 and the progressivity of labor income estimate in Table 2 to calibrate our heterogeneous-agent model below – and in particular the social weights.

3 A Bewley theory of the SWF: Some initial definitions

We discuss our aggregation theory of the SWF in a simple environment that allows us to abstract from complex algebra and heavy notations. We first explain how we construct individual welfare functions (IWFs), which are aggregated to form the social welfare function (SWF) (Sections 3.1–3.2). We then explain how the SWF and the IWFs can be identified from observed allocations (Section 3.3), and introduce of public finance concepts. Finally, we discuss the

¹See <https://stats.oecd.org/index.aspx?queryid=66670>.

²In the 2007 SCF, the wealth Gini index was found to be 0.78, which is very close to the PSID value.

possible interpretations of these estimations in terms of political and social justice concepts, such as the Utilitarian, Egalitarian, and Libertarian principles.

3.1 The setup

We consider a one-period one-good economy. The unique good is a final consumption good, over which agents have preferences. These preferences are represented by a utility function u , which is assumed to satisfy standard properties: $u' > 0$, $u'' < 0$, and $u'(0) = \infty$. We furthermore assume that $u > 0$. We need this assumption for the combination of weighted utility functions to be well-behaved. The economy is populated by two types of agents, whose population size is normalized to 1. Each type is in equal $1/2$. Agents types only differ along their endowment. We denote by $y_1 > y_2$ the two endowment values.

A benevolent planner has the objective to choose the best allocation subject to a feasibility constraint. Allocations will be ranked according to a SWF whose construction is detailed below. The feasibility constraint reflects the fact that the planner can transfer resources across type 1 and type 2 agents, but with a quadratic redistribution cost. This cost aims at capturing all distortions generated by distribution and is scaled by a parameter $\kappa > 0$. To implement a consumption c_i for an agent of type $i = 1, 2$ receiving the endowment y_i , generates the cost involves a destruction of resources equal to $\frac{\kappa}{2}(c_i - y_i)^2$.

Focusing on the symmetric equilibrium where all agents of the same type i receive the same consumption c_i , the feasibility constraint can be written as:

$$\sum_{i=1}^2 \left(c_i + \frac{\kappa}{2}(c_i - y_i)^2 \right) \leq \sum_{i=1}^2 y_i. \quad (1)$$

To consider meaningful solutions, we assume that the redistribution cost is not too high and verifies $\kappa y_i < 1$, which formally guarantees interior solutions.

3.2 Individual and social welfare functions

Our construction of the SWF from individual ethical preferences (IWFs) proceeds in three steps – that we will replicate in the general case of Section 4: (i) the subjective valuation of each agent for the welfare of others; (ii) the representation of ethical preferences which we will call Individual Welfare Functions (IWFs); (iii) the Social Welfare Function (SWF) representing the planner's preferences.

The Individual Welfare Function (IWF). In our static setup, $u(c_i)$ is the utility of agent i for the consumption c_i . This represents the individual preferences of agent i . However, agent i also has their own view of how their welfare and this of others should be accounted for by the planner. We model this subjective valuation of the utility of others agents by a loading factor that weights the individual utility. Denoting by $\tilde{\omega}_{ij}$ the weight of agent i for agent j , the subjective valuation by agent i of the welfare of agent j is $\tilde{V}_{ij} = \tilde{\omega}_{ij}u(c_j)$.³

³Note that to avoid imposing an implicit normalization, we do not impose $\tilde{\omega}_{ii} = 1$. See also below our discussion about the intensity of preferences.

We then assume that the ethical preferences of agent i are then built as the aggregation of their perception of the welfare of other agents. The IWF of agent i representing their ethical preferences is the weighted sum of subjective valuations by agent i over the two agents' types.: $IWF_i := \frac{1}{2}\tilde{V}_{i1} + \frac{1}{2}\tilde{V}_{i2}$, or:

$$IWF_i = \frac{1}{2}\tilde{\omega}_{i1}u(c_1) + \frac{1}{2}\tilde{\omega}_{i2}u(c_2). \quad (2)$$

The economy thus features heterogeneity in ethical preferences: There are two types of ethical preferences, as there are two income levels. Types thus capture the heterogeneity in both the endowments and in the IWFs. This is in line with empirical studies, that have indeed shown heterogeneity in ethical preferences and that this heterogeneity was partly driven by social position. See Gaertner and Schokkaert (2012) and Stantcheva (2021) for a recent analysis based on large US surveys.

This general case encompasses *self-interested* agents— who only care about their own welfare.⁴ The IWF of self-interested agents will be proportional to their individual utility: $IWF_i = \lambda_i u(c_i)$ for some $\lambda_i > 0$. This will correspond to the weights $\tilde{\omega}_{ij} = \lambda_i \times 1_{i=j}$ — with $1_{i=j} = 1$ if $i = j$ and 0 otherwise.

Two remarks are worth mentioning regarding the generality of the weights $(\tilde{\omega}_{ij})_{i,j=1,2}$

1. *Intensity of preferences.* We do not impose any weight normalization, neither that they sum to 1 ($\sum_j \tilde{\omega}_{ij} = 1$) nor that $\tilde{\omega}_{ii} = 1$. Since we aggregate IWFs together, the IWF weights have indeed both an ordinal and a cardinal meaning. Their normalization is not innocuous, as it would remove the possible heterogeneity in the intensity of ethical preferences (using the wording of Arrow, 1951).
2. *Possibility of spitefulness or discrimination.* We do not restrict the sign of the weights $(\tilde{\omega}_{ij})_{i,j=1,2}$, which are allowed to be negative. In this case, the IWF of an individual may be negatively affected by the utility of other agents: the consumption of some other agents can be perceived as a negative externality by certain agents. Such externalities on individual preferences (and not only on ethical preferences as here) have been modeled in asset pricing or macroeconomics to reflect the idiom that households want “to keep up with the Joneses” (see Abel, 1990 or Campbell and Cochrane, 1999 among others). Such externalities have also received support in experimental studies (see Fehr et al., 2013 among others), where such a behavioral trait has been called spitefulness.⁵ Negative weights in welfare functions could also reflect a possible discrimination of some agents' types (Piacquadio, 2017).

The Social Welfare Function (SWF). Finally, we construct the SWF as the weighted aggregation of the IWFs. Indeed, following the political economy literature, we assume that agents may differ in their political ability to influence the planner in their policy implementation. This heterogeneity in the agents' influence may result from institutional design, lobbying activities,

⁴We avoid the word “selfish” or “rational” agents, as they are too negatively or positively connoted. Sen (1977) uses the word self-seeking for the same idea.

⁵Fehr and Schmidt (2006) describe this behavior as follows: “A spiteful person always values the material payoff of relevant reference agents negatively. Such a person is, therefore, always willing to decrease the material payoff of a reference agent at a personal cost to himself”.

or voting rules, and we will use the political economy literature to estimate these weights. We capture it by political weights $(\omega_{P,i})_i$ which loads IWFs as a function of agent's type. We then define the SWF as the aggregation of IWFs weighted by political weights:

$$SWF = \omega_{P,1}IWF_1 + \omega_{P,2}IWF_2. \quad (3)$$

Using the expression (2) of IWFs, we obtain the following expression for the SWF:

$$SWF = \omega_1 u(c_1) + \omega_2 u(c_2), \quad (4)$$

where we have defined the weights as follows, for some $\omega > 0$:

$$\omega_1 := \frac{\omega}{2}(\omega_{P,1}\tilde{\omega}_{11} + \omega_{P,2}\tilde{\omega}_{21}), \quad (5)$$

$$\omega_2 := \frac{\omega}{2}(\omega_{P,1}\tilde{\omega}_{12} + \omega_{P,2}\tilde{\omega}_{22}). \quad (6)$$

While the IWFs weights have a cardinal interpretation, this is not the case of the SWF weights. We can thus without loss of generality choose the constant ω so as to normalize the sum of weights to 1 ($\omega_1 + \omega_2 = 1$).

Our construction of SWF weights first embeds standard cases, such as the Utilitarian SWF. It corresponds to self-seeking agents: $\tilde{\omega}_{ij} = 1_{i=j}$, with identical preference intensity. With constant political loading factors: $\omega_{P,i} = 1/2$, we obtain that the SWF weights are identical and equal to: $\omega_1 = \omega_2 = \frac{1}{2}$ – where we have set $\omega = 2$ in (5)–(6) for normalization purposes. The resulting SWF is Utilitarian.⁶ Second, SWF weights are not restricted to be positive and the aggregation procedure could be end up with SWF negative. In this simple setup, this implies that the construction does not ensure that the planner only chooses Pareto-optimal allocations. See also Section 3.4 for a lengthier discussion of these aspects.

3.3 Inverse optimal approach: Identifying the Welfare Functions

The identification of SWF weights. Generally speaking, the inverse optimal approach consists in identifying social preferences from the observed allocation and available fiscal tools, assuming that these ones are optimally set by the planner. In our setup, the fiscal system allows the planner to directly choose the allocation (c_1, c_2) that maximizes the aggregate welfare, represented by the SWF of equation (4), subject to the resource constraint of equation (1). For a given pair of SWF weights (ω_1, ω_2) , the optimal allocation is characterized by the following FOC:

$$\frac{\omega_1 u'(c_1)}{1 + \kappa(c_1 - y_1)} = \frac{\omega_2 u'(c_2)}{1 + \kappa(c_2 - y_2)}, \quad (7)$$

together with the constraint (1). Note that the condition $\kappa y_i < 1$ ensures that the FOC is well defined for all consumption levels.

The intuition for this relationship can be provided using concepts of public finance. To clarify this link, we rewrite the planner's program assuming that the planner chooses individual

⁶Even with heterogeneous preference intensity, $\tilde{\omega}_{ij} = \lambda_i 1_{i=j}$, with $\lambda_1 \neq \lambda_2$, we can recover the Utilitarian SWF. Political weights must offset the preference intensity: $\omega_{P,i} = \frac{\lambda_i^{-1}}{\lambda_1^{-1} + \lambda_2^{-1}}$. We then obtain with $\omega = \lambda_1^{-1} + \lambda_2^{-1}$ that SWF weights are identical: $\omega_1 = \omega_2 = \frac{1}{2}$.

lump-sum taxes $(t_i)_i$. The Lagrangian of the planner can be written as $\mathcal{W} + \mu\mathcal{R}$, where $\mathcal{W} := \omega_1 u(y_1 - t_1) + \omega_2 u(y_2 - t_2)$ is the SWF (4) expressed with taxes, $\mathcal{R} := \sum_{i=1}^2 (t_i - \frac{\kappa}{2} t_i^2)$ the resource constraint (1) as a function of taxes, and μ the associated Lagrange multiplier. The FOC associated to the choice of t_i can be written as follows:

$$\mu = \frac{\partial \mathcal{W}}{\partial c_i} \frac{-\frac{\partial c_i}{\partial t_i}}{\frac{\partial \mathcal{R}}{\partial t_i}}. \quad (8)$$

This equation states that the planner equalizes the marginal benefit for the planner's finances of raising t_i to the marginal cost for agent i of financing this marginal resource. The marginal benefit is simply the Lagrange multiplier of the planner's resource constraint. The marginal cost involves two terms.

The first one, $-\frac{\partial c_i}{\partial t_i} \bigg/ \frac{\partial \mathcal{R}}{\partial t_i}$, measure show much the consumption of agent i is affected by the financing of the marginal planner's resource by the tax t_i . The quantity $\frac{\partial \mathcal{R}}{\partial t_i} = 1 - \kappa t_i$ includes the tax base (equal to 1 here) and the financial externality related to the destruction of resources, $-\kappa t_i$. Since $-\frac{\partial c_i}{\partial t_i} = 1$, financing the marginal planner's resource will decrease the consumption of agents i by $-\frac{\partial c_i}{\partial t_i} \bigg/ \frac{\partial \mathcal{R}}{\partial t_i} = 1/(1 + \kappa(c_i - y_i))$. We denote this term $MVPF_i$ and call it the *marginal value of public fund* following Finkelstein and Hendren (2020) and Hendren and Sprung-Keyser (2020).

The second term measures $\frac{\partial \mathcal{W}}{\partial c_i}$ how much social welfare is affected by a variation of the consumption of agents i . In the absence of welfare externality of consumption, only the welfare of agent i is affected and we have $\frac{\partial \mathcal{W}}{\partial c_i} = \omega_i u'(c_i)$. Following the literature (e.g., Ferey et al., 2024 among many others), we call this term the *social marginal welfare weight* attributed by the planner to agents i . We denote it as $SMWW_i$.

Overall, the extra resource of the planner financed by agent i through t_i by the planner implies a consumption cut of $MVPF_i$ units for agents i , and a welfare impact equal to $SMWW_i \times MVPF_i$. This latter quantity can thus be interpreted as the *bang for the buck* of one unit of resources spent by the planner for agents i – following again the denomination of Finkelstein and Hendren (2020) and Hendren and Sprung-Keyser (2020).

The interpretation of equation (7) is then rather simple. The planner sets transfers between the two agents' types up to the point where the planner equalizes the bangs for the buck of the two agent: the planner is indifferent between obtaining one extra unit of resources from agents 1 or from agents 2. Should this not hold, the allocation could not be optimal: Agents with the higher bang for buck should receive resources, at the expense of those with the lower bang for the buck. Furthermore, from (8), the bangs for the buck of the two agents are also equal to the marginal benefit of relaxing the resource constraint (1) Equation (7) can thus be rewritten as:

$$SMWW_1 \times MVPF_1 = SMWW_2 \times MVPF_2 = \mu. \quad (9)$$

Saez and Stantcheva (2016) have generalized this marginal approach by allowing to consider social marginal weights that do not derive an explicit SWF. The weights can adopt general expressions, such as non linear effects or dependencies in endogenous variables other than consumption. These weights are called *generalized social marginal welfare weight*.⁷

⁷The GSMWW approach allows for including a wide variety of political or moral motives. However, when the

The inverse optimal approach still relies on the FOC (7), but takes a different perspective. Instead of deducing the allocation (c_1, c_2) from the weights (ω_1, ω_2) , the weights are computed from the allocation. Formally, for a given allocation (c_1, c_2) satisfying the resource constraint (1), the FOC (7) can be written as:

$$\frac{\omega_1}{\omega_2} = \frac{u'(c_2)}{u'(c_1)} \frac{1 + \kappa(c_1 - y_1)}{1 + \kappa(c_2 - y_2)}, \quad (10)$$

which determines the pair of weights (ω_1, ω_2) with the normalization constraint. Rather than the SWF weights ω_i , we could also compute from (7) the generalized social marginal welfare weight. As we discuss in Section 3.4, focusing on SWF weights is better suited for our analysis, as it allows us to directly qualify SWF and offer possible interpretations in terms of political and ethical terms.

The identification of IWF weights. The most part of the analysis in the quantitative model below only relies on the estimation of the SWF. However to derive insight about IWFs consistent with the estimated SWF, we implement the following strategy. First, we consider measures of the political weights $(\omega_{P,i})_i$ of each group, using insights from the political economy literature (Cage, 2024 for a survey). Second, in the absence of individual-level information, we identify the weights of IWFs as those, which are the closest to the self-interested ones, while being consistent with the estimated SWF weights. See Section 4.4 for a formal presentation.

Other benchmark IWF could easily be used, but the self-interested one seems a natural benchmark in the economic or political economy literature (see Acemoglu, 2010 for various models of this type). The gain of this strategy is to make explicit the identifying assumptions and to allow us to derive closed-form expressions for the IWF weights.

3.4 Interpretations of Welfare Functions

It is known that welfare weights offer a possible interpretation in terms of social choice and moral philosophy, when the allocation is known (Fleurbaey and Maniquet, 2018 or Saez and Stantcheva, 2016). We provide below the definitions that will be used in the quantitative exercise of Section 6.

- *Utilitarian.* Agents of type i will be said to be Utilitarian if they equally weight all agents: $\tilde{\omega}_{i1} = \tilde{\omega}_{i2}$. The Utilitarian planner ($\omega_1 = \omega_2$) will implement the consumption levels $c_1 > c_2$ if $\kappa > 0$ (recall that $y_1 > y_2$). The Utilitarian planner thus accepts some inequality among agents due to the distributional cost.
- *Egalitarian.* Egalitarian agents think that economic inequality isn't justified. Formally, this implies putting a greater weight on poorer agents: $\tilde{\omega}_{i1} < \tilde{\omega}_{i2}$. Compared to a Utilitarian planner, the Egalitarian planner will reduce the consumption of the richest agents (type 1) and increase the consumption of poorest agents (type 2). The planner thus reduces inequality at the cost of a lower total consumption.

Pareto principle is also imposed, Sher (2024) has shown that the GSMWW approach may generate inconsistencies in the ranking of fiscal schedules.

- *Libertarian*. Libertarian agents think that agents deserve what they have. This formally corresponds to a higher weight on richer agents: $\tilde{\omega}_{i1} > \tilde{\omega}_{i2}$. The Libertarian planner will choose an allocation implementing a higher inequality but a higher total consumption than the Utilitarian planner. The resulting allocation is thus closer to the initial income distribution and generates lower distributional costs.

In a two-type economy, these three cases correspond to a partition of the set of welfare weights. Each agent type can only in one of the three above situations. This is also the case for the planner. Note that even if the agent types belong to different categories, the planner will belong to exactly one of those categories – depending on the combination of political weights and preference intensity. However, such a partitioning does not hold with more than two types of agents. We discuss such cases in the general model of Section 4.

What about Pareto deviations? Our SWF construction is not restricted to Pareto-optimal SWF. This is the case in our simple setup with negative SWF weights. In more general settings, this can happen even with positive SWF weights. This is for instance the case in a framework featuring individual risk and a Libertarian planner. In that case, an insurance mechanism could be ex-ante individually optimal for all agents, while not being chosen by the planner. Indeed, a Libertarian planner would be reluctant to favor redistribution. Thus, the SWF allocation may not be Pareto optimal, as it could prevent insurance mechanisms. This is not specific to Libertarian planner. An Egalitarian planner could choose an allocation that reduces so much the inequality that it would come at the expense of some agents. Indeed, since Sen (1970), it is known that the SWF will not necessarily fulfill the Pareto principle if the planner also care for other factors (ethical, moral, or political) than the individuals’ utility. More precisely, Kaplow and Shavell (2001) have shown that the Pareto principle does not hold when the planner departs from welfarism, i.e., when the SWF does not only include the agents’ utility. There is therefore an incompatibility between Pareto principle and the generality of the SWF. One of our paper objective is to propose a rationalization of observed fiscal systems and discuss political implications. As a consequence, we will not relax the assumption of welfarism and require the Pareto principle to be fulfilled by the estimated SWFs. We alternatively consider a weaker restriction, which is that the aggregate welfare cannot decrease when increasing the welfare of any agents. Loosely speaking, this is akin to “everybody counts”. In our setup, this means assuming positive SWF weights: $\omega_1, \omega_2 \geq 0$ (see Definition 1 in the intertemporal case). In general setups, this does not mean Pareto-optimal choices.

4 A Bewley theory of the SWF: The intertemporal case

We extend the previous construction of the SWF to a general intertemporal framework, which is our first contribution. Our approach relies on the sequential representation of the heterogeneous agent model, which is the most suitable for our normative analysis.⁸ We present the construction of the SWF anticipating the model used in the quantitative section below.

⁸Some analyses have considered social weights depending on endogenous variables, such as consumption or wealth. This is a possible source of inconsistencies and multiple equilibria. Indeed, the planner should in that case consider how the social weights change with the allocation.

4.1 The setup

We now consider an infinite-horizon model with incomplete financial markets. Time is discrete, indexed by $t \geq 0$. The economy is populated by a continuum of size 1 of ex-ante identical agents.

Risk structure. Idiosyncratic risk is modeled as an uninsurable idiosyncratic labor productivity shock y_t that can take Y distinct values in the finite set $\mathcal{Y}$. The productivity risk follows a first-order Markov chain with transition matrix $(\Pi_{yy'})_{y,y' \in \mathcal{Y}}$. This matrix is assumed to be irreducible and aperiodic, which ensures that it admits a unique stationary distribution denoted as $(\pi_y)_{y \in \mathcal{Y}}$, normalized such that $\sum_{y \in \mathcal{Y}} \pi_y = 1$. We denote by $y^t = \{\dots, y_{t-1}^t, y_t^t\}$ a one-sided infinite sequence of elements of $\mathcal{Y}$, corresponding to an history of productivity levels up to date t . We denote the set of such histories by $\mathcal{Y}^\infty := \mathcal{Y}^\infty$. Since we will need to consider the evolution of histories from one period to another, we keep time subscripts for histories. To keep notation simple, we will use for an history $y^t \in \mathcal{Y}^\infty$, the following notation: (i) $y_\tau^t \in \mathcal{Y}$ is the productivity level at date $\tau \leq t$ in history y^t ; (ii) y^s is the truncation of y^t at date $s \leq t$ – such that y^t and y^s coincide up to s : $y_\tau^s = y_\tau^t$ for all $\tau \leq s$. We will use decorator to distinguish possible different histories: $\tilde{y}^t$ and y^t can be different at any date.

Initial distribution. To simplify the notation below, we make two assumptions about the initial distribution: (i) all agents start the economy with an initial infinitely-long history belonging to $\mathcal{Y}^\infty$; (ii) agents are initially endowed with a wealth that is only function of their history. The assumption about wealth encompasses, among others, the case where all agents have the same wealth or the steady-state wealth. Since we focus on steady-state distributions, of wealth in particular, this simplification is at no cost in our environment.

The sequential representation. There is first a mathematical subtlety in infinite-horizon models, as the set of histories has the cardinality of the continuum (it is neither finite nor countable).⁹ This explains why the probability space over the set of histories involves a general measure. We thus construct a probability space over the set of all histories denoted by $(\mathcal{Y}^\infty, \mathcal{F}, \mu)$, where $\mathcal{F}$ is a relevant σ -algebra and μ is a measure (see LeGrand and Ragot, 2022, Lemma 3 in Section B.3). In words, for any set of histories $B \in \mathcal{F}$, $\mu(B) \geq 0$ is the measure of agents currently experiencing an history $y^t \in B$. As the population size of agents is normalized to 1, we furthermore have $\int_{y^t \in \mathcal{Y}^\infty} \mu(dy^t) = 1$. We leave the construction of this measure to Appendix A.1.¹⁰

We also need to define transitions across histories. Consider two histories $y^{t+1}, \tilde{y}^t \in \mathcal{Y}^\infty$. The probability to switch from history $\tilde{y}^t$ in the current period to another history y^{t+1} in the next period is simply the probability to switch from state $\tilde{y}_t^t$ to state y_{t+1}^{t+1} if y^t and $\tilde{y}^t$ are equal and 0 otherwise. We denote this conditional probability $\mu_1(\cdot|\tilde{y}^t)$, formally defined as: $\mu_1(dy^{t+1}|\tilde{y}^t) = \Pi_{\tilde{y}_t^t y_{t+1}^{t+1}} \delta_{\tilde{y}^t}(dy^t)$, where $\delta_{\tilde{y}^t}$ is the Dirac delta function in $\tilde{y}^t$.¹¹

⁹An history is an event of measure zero in $\mathcal{F}$. Therefore, every equality that holds for all history y^t should be understood as holding almost surely.

¹⁰In finite time, we would write $\sum_{y^t \in \mathcal{Y}^t} \mu_t(y^t)$ instead of $\int_{y^t \in \mathcal{Y}^\infty} \mu(dy^t)$. All intuitions of the finite time representation are valid.

¹¹The function μ_1 is a measure and verifies the standard properties of a conditional probability: $\mu_1 \geq 0$,

We then define by induction the probability to switch from history $\tilde{y}^t$ to another history y^{t+s} s periods ahead as: $\mu_s(dy^{t+s}|\tilde{y}^t) = \prod_{k=0}^{s-1} \mu_{t+k}(dy^{t+k+1}|\tilde{y}^t)$, or: $\mu_s(dy^{t+s}|\tilde{y}^t) = \prod_{k=0}^{s-1} \mu_{t+k}(dy^{t+k+1}|\tilde{y}^t) \times \delta_{\tilde{y}^t}(dy^t)$. In words, switching from $\tilde{y}^t$ to y^{t+s} imposes that the two histories coincide up to period t and then involves the cumulative probability to successively experience the states from y_{t+1}^{t+s} to y_{t+s}^{t+s} .

Individual intertemporal welfare. For a given allocation, we denote by $U(y^t)$ the period utility of an agent having history y^t . To lighten notation, we choose not to make the dependence in the allocation explicit. For instance in the case of a utility depending on private consumption, and labor supply (as in our quantitative application), $U(y^t) := u(c(y^t)) - v(l(y^t))$, where: $c : \mathcal{Y}^\infty \rightarrow \mathbb{R}^+$ and $l : \mathcal{Y}^\infty \rightarrow \mathbb{R}^+$ are policy functions determining consumption and labor as a function of individual history. We still assume that U is always positive, which ensures, as in the simple case, that the combination of weighted utility functions is well-behaved.

The intertemporal welfare in period t of an agent with history y^t is assumed to be separable in time and of the expected-utility type. It is thus defined as the discounted sum over all future dates of expected period utilities. Formally, the intertemporal utility $V(y^t)$ is:

$$V(y^t) = \sum_{s=0}^{\infty} \beta^s \int_{y^{t+s} \in \mathcal{Y}^\infty} U(y^{t+s}) \mu_s(dy^{t+s}|y^t). \quad (11)$$

As we consider steady-state allocation, the intertemporal welfare can be written recursively as:

$$V(y^t) = U(y^t) + \beta \mathbb{E}_{y^{t+1}} [V(y^{t+1})|y^t], \quad (12)$$

where $\mathbb{E}_{y^{t+1}} [V(y^{t+1})|y^t] = \int_{y^{t+1} \in \mathcal{Y}^\infty} V(y^{t+1}) \mu_1(dy^{t+1}|y^t)$ is a conditional expectation. See Appendix A.3 for a proof.

4.2 Constructing the SWF

We extend the SWF construction of Section 3 to the intertemporal framework. As previously, our construction involves three steps. The first step is slightly different than in the simple case. In the simple framework, agents of type i had their own subjective valuation of how the utility of agents of type j should be valued by the planner. In this general setup, there is no ex ante types. Since histories are the sole source of heterogeneity among agents, the perception of others' situation will thus rely on the perception of other histories. More precisely, an agent with history y^t at date t will form a subjective appreciation of the utility associated to any other history $\tilde{y}^{t+s}$ for any future date $t+s$ – independently of whether the agent is likely experience such an history. This results in a subjective valuation by agent y^t of the utility of any other agent with history $\tilde{y}^t$. The second and third steps are more similar to their counterpart of the simple model. The second step is to aggregate the perception of other histories' over the entire distribution μ of histories, which yields the Individual Welfare Function (IWF) of agents with history y^t . This IWF represents the ethical preferences of the agent y^t , which can be seen as how the agent believes the welfare of others should be valued by the planner. The third and final step is to

$\int_{y^{t+1} \in \mathcal{Y}^\infty} \mu_1(dy^{t+1}|\tilde{y}^t) = 1$ and $\int_{\tilde{y}^t \in \mathcal{Y}^\infty} \mu_1(dy^{t+1}|\tilde{y}^t) \mu(d\tilde{y}^t) = \mu(dy^{t+1})$. See the proofs in Appendix A.2.

construct the Social Welfare Function (SWF) as the weighted sum of individual IWFs over all histories y^t . We now present this aggregation more formally.

Step 1: Constructing the subjective valuation of the utility of another agent. We consider two agents characterized by their histories $y^t \in \mathcal{Y}^\infty$ and $\tilde{y}^t \in \mathcal{Y}^\infty$ at some date t ; the allocation is still considered as given. Given our previous assumption about individual preferences, the tilde agent will value in $s \geq 0$ periods the allocation of any history $\hat{y}^{t+s}$ as $U(\hat{y}^{t+s})$ (whether $\hat{y}^{t+s}$ is a possible successor of $\tilde{y}^t$ or not). However, the non-tilde agent may possibly have a different perception of how the allocation history $\hat{y}^{t+s}$ should be valued. Non-tilde agents assign to the utility $U(\hat{y}^{t+s})$ of the tilde agent a corrective factor, denoted by $\hat{\omega}(y^t, \hat{y}^{t+s})$, such that $\hat{\omega}(y^t, \hat{y}^{t+s})U(\hat{y}^{t+s})$ corresponds to the valuation of history $\hat{y}^{t+s}$ by the non-tilde agent. Summing over all future periods and all future histories the discounted valuations $\hat{\omega}(y^t, \hat{y}^{t+s})U(\hat{y}^{t+s})$ – with the proper conditional probabilities – yields the valuation of the utility of the tilde agent by the non-tilde agent. Denoting by $\hat{V}(y^t, \tilde{y}^t)$ this subjective valuation, we obtain:

$$\hat{V}(y^t, \tilde{y}^t) = \sum_{s=0}^{\infty} \int_{\hat{y}^{t+s} \in \mathcal{Y}^\infty} \beta^s \hat{\omega}(y^t, \hat{y}^{t+s}) U(\hat{y}^{t+s}) \mu_s(d\hat{y}^{t+s} | \tilde{y}^t), \quad (13)$$

which is a direct modification of the utility $V(\tilde{y}^t)$ with the inclusion of the weights $\hat{\omega}(y^t, \hat{y}^{t+s})$.

Step 2: Constructing the Individual Welfare Function (IWF). The IWF of the agent with history y^t is then constructed as the aggregation of their subjective valuation $\hat{V}(y^t, \tilde{y}^t)$ over possible histories $\tilde{y}^t$. Formally:

$$IWF(y^t) = \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \hat{V}(y^t, \tilde{y}^t) \mu(d\tilde{y}^t). \quad (14)$$

The IWF is a representation of the ethical preferences of agents with history y^t . It represents how the agent y^t thinks the welfare of all other agents should be accounted for by the planner. As in the simple model, we do impose any weight normalization at this stage.

Step 3: Aggregating IWFs to obtain the Social Welfare Function. We assume that the planner observes the IWFs in the population and aggregates them all depending on the weights assigned to each agent. Not all agents have the same importance for the planner and they differ along what we will call their political weights – as in the simple model. Formally, the IWF of agents with history y^t will be assigned by the planner the weight $\omega_P(y^t)$. This weight is a shortcut for the importance of agents with history y^t in the political process and hence in their ability to have their own IWF accounted for by the planner. Formally:

$$SWF = \int_{y^t \in \mathcal{Y}^\infty} \omega_P(y^t) IWF(y^t) \mu(dy^t). \quad (15)$$

Special cases. To illustrate our construction, we now consider special cases.

The first case is when agents identically value other histories. Formally, the weights of agent y^t are the same for all histories $\hat{y}^{t+s}$: $\hat{\omega}(y^t) := \hat{\omega}(y^t, \hat{y}^{t+s})$, with a slight abuse of notation. In that case, the ethical preferences can be shown to be represented by an IWF that is proportional to

the utilitarian SWF: $IWF(y^t) = \hat{\omega}(y^t) \int_{\tilde{y}^t \in \mathcal{Y}^\infty} V(\tilde{y}^t) \mu(d\tilde{y}^t)$. All agents have the same weights in the agents' ethical preferences. Consequently, there is no disagreement in the population for the ordering of allocation. The SWF reflects this absence of disagreement and is also proportional to the utilitarian SWF: $SWF = (\int_{\tilde{y}^t \in \mathcal{Y}^\infty} \omega_P(\tilde{y}^t) \hat{\omega}(\tilde{y}^t) \mu(d\tilde{y}^t)) \int_{y^t \in \mathcal{Y}^\infty} V(y^t) \mu(dy^t)$.¹²

Second, we consider the so-called self-interested agents, who only care about the histories they can possibly experience. Formally, the weights of an agent with history y^t will be zero for histories that are not possible continuations of y^t : $\hat{\omega}(y^t, \tilde{y}^{t+s}) := \delta_{y^t}(\tilde{y}^t) \hat{\omega}(y^t)$, with a slight abuse of notation again. In that case, the ethical preferences of agents with history y^t are identical to their individual preferences and their IWF is proportional to their intertemporal utility: $IWF(y^t) = \hat{\omega}(y^t) V(y^t)$. This illustrates that these agents only care about themselves, which justifies our denomination of “self-seeking”. In that case, the SWF is equal to a weighted sum of individual intertemporal utilities: $SWF = \int_{y^t \in \mathcal{Y}^\infty} \omega_P(y^t) \hat{\omega}(y^t) V(y^t) \mu(dy^t)$, which is a weighted additive SWF. It reduces to a utilitarian SWF when the weight product $\omega_P(\cdot) \hat{\omega}(\cdot)$ is constant.

4.3 Properties of the SWF

An explicit expression of the SWF. We state the following proposition.

Proposition 1 *The SWF (15) admits the following expression:*

$$SWF = \sum_{t=0}^{\infty} \int_{y^t \in \mathcal{Y}^\infty} \beta^t \omega(y^t) U(y^t) \mu(dy^t). \quad (16)$$

where the weights ω are given by:

$$\omega(y^{t+s}) = \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \omega_P(\tilde{y}^t) \hat{\omega}(\tilde{y}^t, y^{t+s}) \mu(d\tilde{y}^t). \quad (17)$$

The proof can be found in Appendix A.4. Proposition 1 provides a simple expression for the SWF. It states that we can find period weights ω depending on the current history such that the SWF expresses as the discounted sum over all dates and histories of the utility of that date and history, weighted by the factor ω . In other words, this twists the standard utilitarian SWF by weighting period utilities by a factor depending on the period history – the utilitarian SWF corresponding to a constant ω . The SWF weight $\omega(y^{t+s})$ in (17) can be interpreted as the “average” weight given to history y^{t+s} by all agents in the economy, where agents are weighted by their political leverage ω_P .

The sequential representation (16) of the SWF can also be written a recursive representation: $SWF = \int_{y^t \in \mathcal{Y}^\infty} \omega(y^t) U(y^t, A) \mu(dy^t) + \beta \cdot SWF$, which can be seen as the extension of the recursive representation of the utilitarian SWF to history-dependent weights. Note that this recursive representation is very simple because of our stationarity assumption. When one considers the whole dynamics of the economy, the representation would be: $SWF_t = \int_{y^t \in \mathcal{Y}^\infty} \omega(y^t) U_t(y^t, A) \mu(dy^t) + \beta \cdot SWF_{t+1}$. We have use the latter representation when solving the Ramsey program.

¹²This property is known at least since Aiyagari (1995), to justify the use of the Utilitarian under the veil of ignorance.

Weight restriction. As discussed in the simple framework, we will not restrict the SWF to satisfy the Pareto principle. We do, however, impose a weaker restriction. To formally express this restriction, we need to make the dependence in the allocation explicit. We now denote the period utility $U : \mathcal{Y}^\infty \times \mathcal{A} \rightarrow \mathbb{R}$ and the SWF: $SWF : \mathcal{A} \rightarrow \mathbb{R}$, where $\mathcal{A}$ is the set of allocations. The period utility for an history y^t and an allocation A will be denoted by $U(y^t, A)$. For instance, in the case of our quantitative application, we denote $U(y^t, A) := u(c(y^t)) - v(l(y^t)) + u_G(G)$, where A is the pair of policy functions (c, l) . We can state our result using this notation.

Definition 1 *A SWF $SWF : \mathcal{A} \rightarrow \mathbb{R}$, associated with a period utility $U : \mathcal{Y}^\infty \times \mathcal{A} \rightarrow \mathbb{R}$, is said to be element-wise monotone if for any two allocations A and A' such that $U(y^t, A) \geq U(y^t, A')$ for all y^t , we have $SWF(A) \geq SWF(A')$.*

This definition states that with a monotone SWF, if an allocation is in every period better (in the sense of the period utility) than another one, the former will always be preferred, in the sense of the SWF, to the latter. This property is similar to element-wise monotonicity for utility functions. Obviously, this is weaker than the Pareto principle, which would require A to be preferred to A' in the sense of the intertemporal utility, and not only of period utility.¹³

Proposition 2 *A SWF fulfills element-wise monotonicity iff the weights ω defined in (17) are nonnegative.*

The proof can be found in Appendix A.5. Our quantitative estimations may impose the positivity of weights, which correspond to an element-wise monotone SWF. This means that the SWF cannot increase if the welfare of one agent is reduced: everybody (positively) counts.

4.4 Identification of the weights

The SWF expression in Proposition 1 is very general and doesn't easily lend itself to estimation. We thus introduce a tractability assumption that will allow us to compute the weights in the SWF and the IWF. We will assume that agents with the same current productivity level will all value identically future histories and that this valuation will actually only depend on the current value of productivity of the valued history¹⁴. We make a similar assumption for political weights that are also supposed to only depend on the current productivity level. More formally:

Assumption A *There exist two functions, denoted with a slight abuse of notation $\tilde{\omega} : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$ and a function $\omega_P : \mathcal{Y} \rightarrow \mathbb{R}$ such that for all histories $y^t, \tilde{y}^{t+s} \in \mathcal{Y}^\infty$, we have:*

$$\begin{cases} \tilde{\omega}(y^t, \tilde{y}^{t+s}) := \hat{\omega}_{y_t^t, \tilde{y}_{t+s}^{t+s}}, \\ \omega_P(y^t) := \omega_{P, y_t^t}, \end{cases}$$

¹³This explains why the planner may choose not to implement an insurance mechanism, even if this mechanism is individually desirable. Indeed, the mechanism implies higher utility in some states (typically the “bad” ones) and lower utility in others (typically the “good” ones). This does not verify our element-wise monotonicity of Definition 1.

¹⁴In a first version of the paper, we extended the representation to allow weight to depend on the truncated history of each agent. The estimated results are very similar as the current productivity is almost a sufficient statistics. For this reason, we directly consider the restriction that weights depend on current productivity only. We report the results of the extended estimation in ??

where we recall that y_t^t is the current productivity level in history y^t .

Assumption A reduces the dimensionality of loading factors, which are now defined on finite sets. Weights can now be interpreted as loading factors on productivity levels. To avoid heavy notation, we keep the same notation but use subscripts to denote the dependence in productivity level. The weights ω of equation (17) can also now be shown to verify for all $y \in \mathcal{Y}$: $\omega_y = \sum_{\tilde{y} \in \mathcal{Y}} \pi_{\tilde{y}} \omega_{P, \tilde{y}} \tilde{\omega}_{\tilde{y}, y}$, where $\pi_{\tilde{y}}$ is recalled to be the share of agents with productivity $\tilde{y}$.

The identification of the weights proceeds in two steps: (i) the SWF weights $(\omega_y)_{y \in \mathcal{Y}}$ from the data; and (ii) the IWF weights $(\omega_{\tilde{y}y})_{y, \tilde{y} \in \mathcal{Y}}$ from the SWF weights and the data. Without loss of generality, we now impose for the identification a normalization constraint of the weights that are assumed to sum to 1: $\sum_y \pi_y \omega_y = 1$.

The SWF weights. Our identification strategy for the SWF weights consists in finding the weights for which a given fiscal system (typically the observed one) can be seen as the outcome of a Ramsey program. Formally, the FOCs of the Ramsey planner imply some linear constraints for the SWF weights $(\omega_y)_{y \in \mathcal{Y}}$. The number of constraints, n , depends on the number of instruments of the planner. In the general case, the system of constraints (the n linear ones and the normalization) is weakly underdetermined, which means that the number of productivity state $|\mathcal{Y}|$ is greater than the number of constraints, $|\mathcal{Y}| \geq n + 1$ (for any set X , $|X|$ is the cardinality of X).¹⁵ There are thus $p = |\mathcal{Y}| - n - 1$ degree of freedom.

Our solution to handle the underdetermination is to follow Heathcote and Tsujiyama (2021) and to assume that the weights are a parametric function of y , with exactly $n + 1$ parameters. There is no degree of freedom and the weights are defined as the solution of the non-linear system of constraints. We summarize it in the following definition.

Definition 2 *We consider as given a set of $1 \leq n \leq |\mathcal{Y}| - 1$ linear constraints represented by the matrix $(L_{k,y})_{k=1, \dots, n, y \in \mathcal{Y}}$ and a set of parametric functions $f_y : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ ($y \in \mathcal{Y}$) characterizing weights. The estimated SWF weights are characterized by the vector $(\omega_y)_{y \in \mathcal{Y}} = (f_y(\theta))_y$, where $\theta \in \mathbb{R}^p$ solves the following system:*

$$0 = \sum_{y \in \mathcal{Y}} L_{k,y} f_y(\theta) \text{ for all } k = 1, \dots, n, \quad (18)$$

$$1 = \sum_{y \in \mathcal{Y}} \pi_y f_y(\theta). \quad (19)$$

As a robustness check, we also consider a non-parametric estimation of the weights. In this case, the system (18)–(19) is underdetermined and the weights are chosen as the solution of (18)–(19) with the lowest variance across productivity levels. Both solutions imply weights that are quantitatively very similar, which we see as positive for the identification strategy. See Appendix H for definitions and results in the non-parametric case.

The IWF weights. The identification of the IWF weights is of higher dimensionality and requires to compute the $|\mathcal{Y}| \times |\mathcal{Y}|$ parameters $\hat{\omega}_{\tilde{y}y}$. As explained in Section 3.3 for the simple

¹⁵Typically, in the quantitative exercise, there are 10 idiosyncratic states and 4 instruments of the planner.

setup, we choose the weights $\tilde{\omega}$ that are the closest to the self-seeking ones, while being consistent with the SWF weights. The following definition formalizes it.

Definition 3 We consider as given SWF weights $(\omega_y)_y$ and policy weights $(\omega_{P,y})_y$. The estimated IWF weights are given by the matrix $(\tilde{\omega}_{\tilde{y}y})_{\tilde{y},y}$ that solves the following program:

$$\begin{aligned} (\tilde{\omega}_{\tilde{y}y})_{\tilde{y},y} = \underset{(\tilde{\omega}_{\tilde{y}y})_{\tilde{y},y}}{\operatorname{argmin}} \quad & \sum_{(y,\tilde{y}) \in \mathcal{Y}^{\infty 2}} \pi_{\tilde{y}} \left(\hat{\omega}_{\tilde{y}y} - \frac{1_{y=\tilde{y}}}{\omega_{P,y}\pi_y} \right)^2, \\ \text{s.t. } \omega_y = \quad & \sum_{\tilde{y} \in \mathcal{Y}^{\infty}} \pi_{\tilde{y}} \omega_{P,\tilde{y}} \hat{\omega}_{\tilde{y}y}. \end{aligned}$$

The solution to this program is for $(y, \tilde{y}) \in \mathcal{Y}^{\infty 2}$:

$$\hat{\omega}_{\tilde{y}y} = \frac{1_{y=\tilde{y}}}{\omega_{P,y}\pi_y} + \frac{\omega_{P,\tilde{y}}}{\sum_{\tilde{y} \in \mathcal{Y}^{\infty}} \pi_{\tilde{y}} (\omega_{P,\tilde{y}})^2} (\omega_y - 1). \quad (20)$$

The proof for the derivation of the weight expression can be found in Appendix A.6. The IWF weights in equation (20) are quite straightforward to interpret. They are separable in two terms, which are the self-seeking weights and the distance of the SWF weights to utilitarian ones, weighted by a factor proportional to the political weight $\omega_{P,\tilde{y}}$. Formally:

$$\begin{aligned} \tilde{\omega}_{\tilde{y}y} = \quad & \underbrace{\frac{1_{y=\tilde{y}}}{\omega_{P,y}\pi_y}}_{=\text{self-seeking weights}} + \frac{\omega_{P,\tilde{y}}}{\sum_{\tilde{y} \in \mathcal{Y}^{\infty}} \pi_{\tilde{y}} (\omega_{P,\tilde{y}})^2} \times \underbrace{(\omega_y - 1)}_{=\text{distance to utilitarian SWF}}. \end{aligned} \quad (21)$$

This simple decomposition mostly comes from choosing $(\frac{1_{y=\tilde{y}}}{\omega_{P,y}\pi_y})_{y,\tilde{y}}$ for the benchmark weights to which the distance should be computed. Indeed, it implies that the IWF weights reduce to self-seeking ones ($\tilde{\omega}_{\tilde{y}y} = 0$ if $y \neq \tilde{y}$) for utilitarian SWF weights ($\omega_y = 1$ for all y). It can be shown that they are the only benchmark for which this property holds.

To push the interpretation further, we consider the quantity $\pi_{\tilde{y}} \omega_{P,\tilde{y}} \hat{\omega}_{\tilde{y}y}$, which measures how much the perception of agents $\tilde{y}$ contribute to the SWF ω_y of agent y . We have: $\pi_{\tilde{y}} \omega_{P,\tilde{y}} \hat{\omega}_{\tilde{y}y} = 1_{y=\tilde{y}} + \frac{\pi_{\tilde{y}} (\omega_{P,\tilde{y}})^2}{\sum_{\tilde{y} \in \mathcal{Y}^{\infty}} \pi_{\tilde{y}} (\omega_{P,\tilde{y}})^2} (\omega_y - 1)$. It includes a visible self-seeking component: $1_{y=\tilde{y}}$, showing that along this dimension, the perception of agents $\tilde{y}$ to the SWF matter only when they perceive themselves. The second component can be perceived as an altruistic dimension when it is positive, or a spiteful component when it is negative. All agents $\tilde{y}$ will perceive agents y proportionally to the distance of the SWF weights of these agents to utilitarian. The loading factor put by agents $\tilde{y}$ is proportional to $\pi_{\tilde{y}} (\omega_{P,\tilde{y}})^2$, which is increasing in the population share and political weights of agents $\tilde{y}$.

5 The general model and the Ramsey program

We now construct the macroeconomic model allowing for estimation of the SWF and IWFs. We consider a mass 1 of ex-ante identical agents that is affected by a productivity risk denoted by y . We further assume two goods in the economy: a final consumption good, whose consumption is denoted by c and labor, whose supply is denoted by l . The rest of the specification involves: the planner's fiscal structure in Section 5.1 and the households' program and the competitive

equilibrium in Section 5.2. The corresponding Ramsey program and its FOCs are described in Section 5.3. We finally discuss the identification of weights in Section 5.5.

5.1 Production and government

Production. In any period t , a production technology with constant returns to scale transforms capital K_{t-1} and labor L_t into $F(K_{t-1}, L_t)$ units of output. The production function is smooth in K and L and satisfies the standard Inada conditions. This formulation allows for capital depreciation, which is subsumed by the production function F . Labor L_t is the total labor supply measured in efficient units. The good is produced by a unique profit-maximizing representative firm. We denote by $\tilde{w}_t$ the real before-tax wage rate in period t and by $\tilde{r}_t$ the real before-tax rental rate of capital in period t . Profit maximization yields in each period $t \geq 1$:

$$\tilde{r}_t = F_K(K_{t-1}, L_t) \quad \text{and} \quad \tilde{w}_t = F_L(K_{t-1}, L_t). \quad (22)$$

Government. A benevolent government must finance a path of public spending, (G_t) using several instruments. First, the government can levy one-period public debt B_t , assumed to be default-free. As there is no aggregate risk, public debt and capital are perfectly substitute and they payoff the same pre-tax interest rate $\tilde{r}_t$. Second, the government can raise a number of distortionary taxes, which concern consumption, labor income, and capital revenues. Consumption and capital taxes are linear and are denoted by τ_t^c , and τ_t^K at date t . Regarding the tax on labor income, note that the pre-tax labor income of an agent with productivity y and labor supply l is $\tilde{w}_t y l$. The associated labor income tax, denoted by $\mathcal{T}_t(\tilde{w}_t y l)$, is assumed to be non-linear and possibly time-varying, as in Heathcote et al. (2017) (henceforth, HSV):

$$\mathcal{T}_t(\tilde{w}_t y l) := \tilde{w}_t y l - \kappa_t (\tilde{w}_t y l)^{1-\tau_t}, \quad (23)$$

where κ captures the level of labor taxation and τ the progressivity. Both parameters will be planner's instruments. When $\tau_t = 0$, labor tax is linear with a rate $1 - \kappa_t$. Oppositely, the case $\tau_t = 1$ corresponds to full income redistribution, where all agents earn the same post-tax income κ_t . Functional form (23), combined with a linear capital tax, allows one to realistically reproduce the actual US system and its progressivity.¹⁶

These three taxes imply a total governmental revenue equal to $\tau_t^c C_t + \int_i \mathcal{T}_t(\tilde{w}_t y_{i,t} l_{i,t}) \ell(di) + \tau_t^K \tilde{r}_t (K_{t-1} + B_{t-1})$, where C_t is the aggregate consumption, and $A_{t-1} := K_{t-1} + B_{t-1}$ is the aggregate savings in period $t - 1$ and $\tilde{r}_t A_{t-1}$ the capital revenues in period t .

With these elements, the governmental budget constraint can be written as follows:

$$G + (1 + \tilde{r}_t) B_{t-1} = \tau_t^c C_t + \int_i \mathcal{T}_t(\tilde{w}_t y_{i,t} l_{i,t}) \ell(di) + \tau_t^K \tilde{r}_t A_{t-1} + B_t. \quad (24)$$

We define the post-tax rates r_t and w_t , as follows:

$$r_t := (1 - \tau_t^K) \tilde{r}_t, \quad w_t := \kappa_t (\tilde{w}_t)^{1-\tau_t}. \quad (25)$$

¹⁶The literature uses either the combination of a linear tax and of a lump-sum transfer (e.g., Dyrda and Pedroni, 2022, Açıkgöz et al., 2022) or the HSV structure (see Ferriere and Navarro, 2023). Heathcote and Tsujiyama (2021) show that the HSV structure is quantitatively more relevant.

Using the property of constant-return-to-scale for F and the definition of post-tax rates (25), the governmental budget constraint can be written as:

$$G + (1 + r_t)B_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + r_t K_{t-1} = \tau_t^c C_t + F(K_{t-1}, L_t) + B_t. \quad (26)$$

5.2 Households program

Period utility. We specify the period utility function U of agents. It is defined over private consumption c and labor supply l , and is assumed to be separable. Formally:

$$U(c, l) := u(c) - v(l). \quad (27)$$

The function $u : \mathbb{R}_+ \rightarrow \mathbb{R}$ is twice continuously differentiable, strictly increasing, and strictly concave, with $u'(0) = \infty$, while $v : \mathbb{R}_+ \rightarrow \mathbb{R}$ is twice continuously differentiable, strictly increasing, and strictly convex, with $v'(0) = 0$.¹⁷

Agents' program. Agents' resources consist of labor income and saving payoffs. The post-tax labor income of an agent with productivity $y_{i,t}$ and supplying labor effort $l_{i,t}$ amounts to $\tilde{w}_t y_{i,t} l_{i,t} - \mathcal{T}_t(\tilde{w}_t y_{i,t} l_{i,t}) = w_t (y_{i,t} l_{i,t})^{1-\tau_t}$. Since public debt and capital shares are perfect substitutes, savings payoffs are equal to $(1 + r_t)a_{i,t-1}$, where $a_{i,t-1}$ is the end-of-period- $t-1$ saving of agent i . Agents use these resources to save and to consume. Formally:

$$\max_{\{c_{i,t}, l_{i,t}, a_{i,t}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t (u(c_{i,t}) - v(l_{i,t})), \quad (28)$$

$$(1 + \tau_t^c)c_{i,t} + a_{i,t} \leq w_t (y_{i,t} l_{i,t})^{1-\tau_t} + (1 + r_t)a_{i,t-1}, \quad (29)$$

$$a_{i,t} \geq -\bar{a}, c_{i,t} > 0, l_{i,t} > 0, \quad (30)$$

where $\mathbb{E}_0$ an expectation operator (with respect to idiosyncratic risk), and where the initial state $(y_{i,0}, a_{i,-1})$ is given.

At date 0, agents decide their consumption $(c_{i,t})_{t \geq 0}$, their labor supply $(l_{i,t})_{t \geq 0}$, and their saving plans $(a_{i,t})_{t \geq 0}$ that maximize their intertemporal utility of equation (28), subject to a budget constraint (29) and a previous borrowing limit (30), while prices are assumed to be exogenous. These decisions are functions of the initial endowment $a_{i,-1}$, and the history of idiosyncratic shocks y_i^t . However, to simplify notation, instead of writing the agents' optimal decision as a function of these variables (as was done in Section 4), we simply denote it with the subscripts i and t . For instance for a generic variable X , instead of using the dependence in the history y_i^t , we simply write it $X_{i,t}$. Similarly, instead of summing over all histories in period t , we simply sum over all agents in a given period: $\int_i X_{i,t} \ell(di)$, where ℓ is the distribution of agents on the population interval J .

¹⁷Without loss of generality we can assume that U is positive for all choices actually made by agents. We can indeed shift u or v by a harmless constant. Note that this constant has no effect on our estimation of the SWF as the strategy of Definition 2 only involves Ramsey FOCs: marginal utilities matter, but utilities in level do not. See Algorithm 1 below.

The FOCs associated with the agents' program (28)–(30) can be written as follows:

$$u'(c_{i,t}) = \beta \mathbb{E}_t \left[(1 + r_{t+1}) \frac{1 + \tau_t^c}{1 + \tau_{t+1}^c} u'(c_{i,t+1}) \right] + \nu_{i,t}, \quad (31)$$

$$v'(l_{i,t}) = \frac{1 - \tau_t}{1 + \tau_t^c} w_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}), \quad (32)$$

where the quantity $\beta^t \nu_{i,t}$ denotes the Lagrange multiplier on agent i 's credit constraint.

Market clearing and resources constraints. The clearing conditions for capital and labor markets can be written as follows:

$$\int_i a_{i,t} \ell(di) = A_t = B_t + K_t, \quad \int_i y_{i,t} l_{i,t} \ell(di) = L_t. \quad (33)$$

Equilibrium definition. We provide a formal definition in Appendix B. Intuitively, for a given fiscal policy $(\tau_t^c, \tau_t^K, \tau_t, \kappa_t, B_t)_t$, the competitive equilibrium is a collection of individual decisions $(c_{i,t}, l_{i,t}, a_{i,t}, \nu_{i,t})_{t,i}$, of aggregate quantities $(K_t, L_t, Y_t)_t$, and prices $(w_t, r_t, \tilde{w}_t, \tilde{r}_t)_t$ that are consistent with the agents' optimization program (28)–(30), the clearing equation (33) of financial and labor markets, as well as the definition of pre- and post-tax factor prices (22) and (25). The competitive equilibrium is at the steady state when all quantities are time-invariant.

5.3 The Ramsey problem and the identification of weights

We follow the construction of Proposition 1 for the SWF. We also require Assumption A to hold for identification purposes. With our period utility separable in consumption and labor, the period 0 SWF is:

$$SWF_0 := \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \int_i \omega_{y_{i,t}} (u(c_{i,t}) - v(l_{i,t})) \ell(di) \right], \quad (34)$$

where the weights $(\omega_y)_{y \in \mathcal{Y}}$ solely depend on the current productivity state due to Assumption A.

In the Ramsey program, the planner aims to determine the fiscal policy corresponding to the competitive equilibrium that maximizes aggregate welfare according to the criterion in equation (34), while satisfying the government's budget constraint. A Ramsey equilibrium is a fiscal policy, prices, individual allocations and aggregate quantities solving the Ramsey program. A Ramsey steady state equilibrium is a time-invariant Ramsey equilibrium. Formally, the Ramsey program

can be stated as follows.

$$\max_{(w_t, r_t, \tilde{w}_t, \tilde{r}_t, \tau_t^c, \tau_t^K, \tau_t, \kappa_t, B_t, G_t, K_t, L_t, (c_{i,t}, l_{i,t}, a_{i,t}, \nu_{i,t})_i)_{t \geq 0}} W_0, \quad (35)$$

$$G_t + (1 + r_t)B_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + r_t K_{t-1} = \tau_t^c C_t + F(K_{t-1}, L_t, s_t) + B_t, \quad (36)$$

$$\text{for all } i \in \mathcal{I}: (1 + \tau_t^c) c_{i,t} + a_{i,t} = (1 + r_t) a_{i,t-1} + w_t (y_{i,t} l_{i,t})^{1-\tau_t}, \quad (37)$$

$$a_{i,t} \geq -\bar{a}, \quad \nu_{i,t}(a_{i,t} + \bar{a}) = 0, \quad \nu_{i,t} \geq 0, \quad (38)$$

$$u'(c_{i,t}) = \beta \mathbb{E}_t \left[(1 + r_{t+1}) \frac{1 + \tau_t^c}{1 + \tau_{t+1}^c} u'(c_{i,t+1}) \right] + \nu_{i,t}, \quad (39)$$

$$v'(l_{i,t}) = \frac{(1 - \tau_t)}{1 + \tau_t^c} w_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}), \quad (40)$$

$$K_t + B_t = \int_i a_{i,t} \ell(di), \quad L_t = \int_i y_{i,t} l_{i,t} \ell(di), \quad (41)$$

and subject to the definition (22) of pre-tax wage and interest rates $\tilde{w}_t$ and $\tilde{r}_t$, the definition (25) of post-tax rates, the positivity of labor and consumption choices, and initial conditions.

Since the Ramsey program involves selecting a competitive equilibrium, its constraints include the equations characterizing this equilibrium: individual budget constraints (37), individual credit constraints (and related constraints on $\nu_{i,t}$) (38), Euler equations for consumption and labor - (39) and (40) - and market clearing conditions for financial and labor markets (41). Moreover, the fiscal policy selected by the Ramsey equilibrium should also fulfill the governmental budget constraint (36) – which is also a constraint.

The general Ramsey program can be simplified. First, in our setup with a linear tax on capital, a progressive tax on labor, and one-period public debt, the consumption tax is redundant with other fiscal instruments. Second, we can follow Chamley (1986) and express the program using post-tax prices only. Combining the two simplifications implies that the planner's fiscal instruments are: post-tax wage and interest rates W_t and R_t , labor tax progressivity and public debt. They need to be chosen such that the governmental budget constraint (26) holds. The other choice variables of the planner also include individual and aggregate allocations that have to be chosen so as to correspond to a competitive equilibrium. This means that individual budget constraints (29), borrowing limits (30), and FOCs (31)–(32) are constraints of the Ramsey program – as well as market clearing conditions (33). The reformulated Ramsey program is formally stated in Proposition 3 of Appendix C.2.

5.4 Interpreting the Ramsey FOCs in the light of public finance

The economics trade-offs faced by the planner can be identified by the FOCs of the Ramsey program, which can be found in Appendix C.3. Solving such program through a Lagrangian raises a number of technical questions that have been discussed in LeGrand and Ragot (2022) and in detail studied in a simpler setup in LeGrand and Ragot (2024) – with no capital and GHH utility function. We here discuss the economic interpretation of the FOCs of the Ramsey program using the concepts of public finance and extending our discussion of Section 3.3.

We mostly focus here on the capital tax – or equivalently the post-tax rate R_t in our setting. We consider that the planner raises resources through the capital tax, which decreases the

consumption of all agents. As a preliminary, observe that setting the capital tax at a given date only affects welfare and constraints of the same date. The Lagrangian associated to problem (35) can be written as

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t (\mathcal{W}_t + \mu_t \mathcal{R}_t)$$

First, $\mathcal{W}_t$ is the “augmented welfare” of date t that includes the pure welfare component $\omega_{y_{i,t}}(u(c_{i,t}) - v(l_{i,t}))$, as well as the general-equilibrium effects implied by individual decisions about savings (i.e., Euler equation) and labor supply (i.e., FOC labor supply). This term depends on the capital tax τ_t^K , and on all consumption levels $(c_{i,t})_i$. Second, $\mathcal{R}_t$ is the governmental budget constraint at date t and μ_t the associated Lagrange multiplier – we keep the same notation as in the simple case. The quantity $\mathcal{R}_t$ depends on τ_t^K but not on $(c_{i,t})_i$. With this notation, the FOC with respect to τ_t^K can be written as:¹⁸

$$\mu_t = \int_i \underbrace{\frac{\partial \mathcal{W}_t}{\partial c_{i,t}} \Big|_{\tau_t^K}}_{=SVL_{i,t}} \underbrace{\frac{-\frac{\partial c_{i,t}}{\partial \tau_t^K}}{\frac{\partial \mathcal{R}_t}{\partial \tau_t^K} + \frac{1}{\mu_t} \frac{\partial \mathcal{W}_t}{\partial \tau_t^K} \Big|_{c_{i,t}}}}_{=MVPF_{i,t}^K} \ell(di), \quad (42)$$

or

$$\mu_t = \int_i SVL_{i,t} \times MVPF_{i,t}^K \ell(di). \quad (43)$$

As in the simple case, the planner sets the capital tax to the point where the shadow price of the governmental budget constraint (μ_t) equals its marginal cost, which equals to the sum over all agents of the individual (negative) bangs for the buck of a cut in agents’ consumption due to a higher capital tax. The cost covers all agents as the capital tax is a non-specific tax. Similarly to the simple case (9), the individual bangs for the buck of one unit of resources raised through capital tax equals the product of SVL (which generalizes the notion of social marginal weight) and the MVPF.

Similarly to as in the simple model, $SVL_{i,t}$ (which we denote by $\psi_{i,t}$ in Appendix C) quantifies the welfare reduction (increase) associated to a one-unit reduction (increase) in the consumption of agents i . This gain includes the fiscal externalities generated by an increase on consumption. Indeed, changing the consumption of agent i affects its saving and its labor supply. The total welfare effect thus includes three terms:

$$\begin{aligned} SVL_{i,t} &= \underbrace{\omega_{i,t} u'(c_{i,t})}_{=direct\ effect} - \underbrace{(\lambda_{c,i,t} - R_t \lambda_{c,i,t-1}) u''(c_{i,t})}_{=externality\ on\ savings\ incentives} \\ &\quad + \underbrace{\lambda_{l,i,t} (1 - \tau_t) W_t(y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t} u''(c_{i,t})}_{=externality\ on\ labor\ supply\ incentives}, \end{aligned}$$

where $\beta^t \lambda_{c,i,t}$ and $\beta^t \lambda_{l,i,t}$ are the Lagrange multipliers on agent i ’s Euler equation and labor supply FOC, respectively. The first term, $\omega_{i,t} u'(c_{i,t})$, reflects the direct welfare effect of a consumption

¹⁸ $\frac{\partial \mathcal{W}_t}{\partial c_{i,t}} \Big|_{\tau_t^K}$ is the derivative of $\mathcal{W}_t$ with respect to $c_{i,t}$ while keeping τ_t^K constant; $\frac{\partial \mathcal{W}_t}{\partial \tau_t^K} \Big|_{c_{i,t}}$ is the derivative of $\mathcal{W}_t$ with respect to τ_t^K while keeping all $c_{i,t}$ constant.

variation. This is identical to the social marginal welfare weight in equation (9). However, in our setting, the welfare is also affected through the externality of the consumption variation on savings and labor supply. The welfare externality on savings channels through the Euler equation, while the one labor supply channels through the FOC on labor supply. This explains why the welfare externalities are proportional to the Lagrange multipliers. The notion of SVL thus generalizes the notion of MSWW to welfare externalities.¹⁹

The MVPF measures the variation (here, a decrease) in consumption implied by a one resource unit taken from agent i by a marginal increase in capital tax, including all fiscal externalities which are independent of the change in consumption of agent i . In the term $MVPF_{i,t}^K$ of equation (42), in the absence of direct pecuniary externality, $\frac{\partial \mathcal{R}_t}{\partial \tau_t^K}$ measures the direct effect of capital tax on planner's resources and is equal to the capital tax base (i.e., all payoffs of interest-bearing assets). Conversely, $\frac{1}{\mu_t} \frac{\partial \mathcal{W}_t}{\partial \tau_t^K} \Big|_{c_{i,t}}$ is the fiscal externality of the capital tax and channels through the modification of the savings incentives and the Euler equation. This fiscal instrument reflects that capital tax is a distortionary instrument.

In the case of labor tax (level and progressivity), we obtain a FOC, which can be rewritten similarly to equations (42)–(43). The main difference is in the MVPF term, which depends on the instrument: In particular, the tax base $\frac{\partial \mathcal{R}_t}{\partial \text{tax instrument}}$ and the welfare externality $-\frac{1}{\mu_t} \frac{\partial \mathcal{W}_t}{\partial \text{tax instrument}} \Big|_{c_{i,t}}$. Are obviously not the same. In the case of the labor tax level, the tax base is the aggregate labor supply (multiplied by the common wage rate), while the externality operates through the labor supply FOC. In the case of the labor tax progressivity, the tax base is more complex to describe and its externality also channels via individual labor supply.²⁰

Note that if the planner would have access to a standard aggregate lump-sum transfer T , the MVPF associated to that tax instrument would simply be 1. Indeed, the lump-sum transfer involves no externality and is a flow of resources from the transfer to agents', such that: $\frac{\partial \mathcal{W}_t}{\partial T_t} \Big|_{c_{i,t}} = 0$ and $-\frac{\partial c_{i,t}}{\partial T_t} = \frac{\partial \mathcal{R}_t}{\partial T_t}$. Therefore, the planner would set T_t such that $\mu_t = \int_i SVL_{i,t} \ell(di)$: the marginal cost for governmental budget equals the marginal benefit for all agents. Should the planner would further have access to individual-specific lump-sum transfers T^i , each would be set such that would $\mu_t = SVL_{i,t}$. As explained in LeGrand and Ragot (2024), the difference $SVL_{i,t} - \mu_t$ captures the cost of distortionary fiscal instruments for the planner, concerning agent i .

5.5 Identification of weights

Our objective is to estimate a SWF such that the observed allocation and fiscal system of a given country is the solution of a Ramsey steady state. We follow the identification strategy of Definition 2 to estimate the social weights $(\omega_y)_{y \in \mathcal{Y}}$ of the SWF SWF_0 .

Fiscal policy is composed of five instruments $(\tau^K, \tau^c, B, \kappa, \tau)$, but these five instruments actually impose only two constraints on social weights. Indeed, consumption taxes τ^c are redundant, as explained above (see Appendix C.2 for the details). Second, the public debt FOC, provided in equation (77) of Appendix C.3, imposes a steady-state value on the before-tax real

¹⁹This SVL is similar to the quantity $\hat{g}$ defined in Ferey et al. (2024) and that they describe as “the social marginal welfare weights augmented with the fiscal impact of income effects” and which represent “the full social value of marginally increasing the disposable income of [an] individual”.

²⁰The consumption tax is redundant and as such does not imply any specific FOC.

interest rate $1 + \tilde{r} = 1/\beta$, but does not restrict the social weights. Therefore, this means that the instruments $(\tau^K, \tau^c, \kappa, \tau)$ actually imply three FOCs. Third the budget constraint of the government implies a restrictions of the instruments. As a consequence, the five instruments actually impose two linear constraints on the SWF weights.

The identification strategy of Definition 2 can thus be readily applied with three constraints: the two linear constraints coming from the Ramsey FOCs and the normalization constraint. We thus consider a parametric estimation, with three degrees of freedom that will be exactly identified by the three constraints. We adopt the following functional form, which naturally extend the one in Heathcote and Tsujiyama (2021):

$$\forall y \in \mathcal{Y}, \log \omega_y := \bar{\omega}_0 + \bar{\omega}_1 \log(y) + \bar{\omega}_2 (\log(y))^2, \quad (44)$$

where $(\bar{\omega}_i)_{i=1,\dots,3}$ are the three free parameters.

We can now provide our algorithm to estimate the SWF and IWF weights.

Algorithm 1 (Computing the SWF weights) *The SWF weights $(\omega_y)_{y \in \mathcal{Y}}$ can be computed using the following steps.*

1. *Calibrate the model parameters and the fiscal system for the steady-state competitive equilibrium of Definition 4 to be as close as possible to the empirical allocation and the fiscal system under consideration.*
2. *Compute the FOCs of the Ramsey program (68)–(74) and deduce the two linear constraints on the SWF weights. Use them as inputs in (18) for the identification strategy of Definition 2.*
3. *We parametrize the weights as $\omega_y := e^{\bar{\omega}_0 + \bar{\omega}_1 \log(y) + \bar{\omega}_2 (\log(y))^2}$ with $(\bar{\omega}_i)_{i=1,\dots,3}$ being the parameters. We apply the identification strategy of Definition 2.*
4. *Using some measures of political weights $\omega_{P,y}$, determine the IWFs using the expression (20) of the identification strategy of Definition 3.*

Let us comment some computational aspects of Algorithm 1. Step 1 is standard in heterogeneous-agent models, as well-calibrated models can effectively replicate observed inequality, as demonstrated below. Step 2 is more challenging, as the Ramsey FOCs imply a joint distribution of various Lagrange multipliers, which are very hard to compute. To address this difficulty, we employ a so-called truncation method to reduce the state space, initially developed in LeGrand and Ragot (2022) and subsequently refined in LeGrand and Ragot (2023) and LeGrand and Ragot (2024). Basically, this truncation method allows us to compute the steady-state optimal allocation and characterize it with a finite number of equations. The idea is to build an aggregation of the Bewley model (under a specific policy and no aggregate shock), where the agents sharing the same history over last N (where N is a fixed horizon) periods are aggregated into a single “agent”. Each of the “aggregate” agents is endowed with the average wealth and average allocation of all individuals with this N -period history. This finite state-space representation is particularly well-suited for the resolution of the Ramsey program. In particular, the linear constraints of Algorithm 1 can be computed using matrix algebra. Steps 3 and 4 are thus quite straightforward

to implement us with this finite state-space representation. We provide all the construction and derivation in Appendix D. We further discuss the choice of the measures of the political weights in Section 3.4.

6 Quantitative investigation

We first provide the calibrations reproducing the tax system and the wealth distribution in both the US and France for the period 1995-2007. We then estimate the SWFs in both countries.

6.1 Calibration

The US calibration

The estimation parameters are gathered in Table 3, and we detail below our calibration strategy.

Preference parameters. The period is a quarter. The discount factor is set to $\beta = 0.992$ to match a realistic capital-to-output ratio. The period utility is specified as $U(\cdot) = \frac{c^{1-\gamma} - 1}{1-\gamma} + \frac{1}{\chi} l^{\frac{1}{\varphi}+1}$ with $\gamma = 1.8$. This risk-aversion is set to match a realistic wealth inequality for the targeted capital-to-output ratio. Furthermore, the Frisch elasticity for labor is set to $\varphi = 0.5$, which is recommended by Chetty et al. (2011) for the intensive margin. We set the labor-scaling parameter to $\chi = 0.0477$, which implies normalizing the aggregate labor supply to 0.33.

Technology. The production function is of the Cobb-Douglas form and subsumes capital depreciation: $F(K, L, s) = sK^\alpha L^{1-\alpha} - \delta K$. The capital share is set to the standard value, $\alpha = 36\%$, while the depreciation rate is set to $\delta = 2.5\%$.

Idiosyncratic labor risk. Various estimations of the idiosyncratic process can be found in the literature. The productivity follows an AR(1) process: $\log y_t = \rho_y \log y_{t-1} + \varepsilon_t^y$, with $\varepsilon_t^y \sim_{\text{IID}} \mathcal{N}(0, \sigma_y^2)$. The calibration features an autocorrelation $\rho_y = 0.99$ and a standard deviation $\sigma_y = 0.0995$, which is close to the estimates of Krueger et al. (2018). We discretize this AR(1) process using the Tauchen (1986) procedure, with 10 states. This calibration implies a Gini index of post-tax and transfers of 0.40, which is the same as that reported in Table 1. We obtain a Gini for wealth of 0.78. The model does a good job in matching the income and wealth distribution in the US.

Taxes and government budget constraint. Fiscal parameters are calibrated based on the computations by Trabandt and Uhlig (2011) reported in Table 1, with the exception of the progressivity of the labor tax, which was computed using our own calculations based on the Luxembourg Income Study Database (LIS) and summarized in Table 2. We recall that their estimations for the US in the period 1995-2007 yield a capital tax of $\tau^K = 36\%$ and a consumption tax of $\tau^c = 5\%$. In our estimation for the progressivity parameter, we obtain

$\tau = 0.16$, which is close to the estimates in the literature.²¹ We consider the years 1995-2007 to examine a period unaffected by major macroeconomic shocks.

Finally, we estimate the parameter κ such that it matches the public spending over GDP ratio. By doing this, we use a value of $\kappa = 0.85$, which is close to the estimates in Ferriere and Navarro (2023). With this fiscal system, the model generates a public-debt-to-GDP ratio equal to 63%. The model also implies a public-spending-to-GDP ratio of 15%. These two values are reported in Table 1.

Additionally, the model performs well in replicating the ratios of consumption over GDP and investment over GDP. The model predicts a consumption-to-GDP ratio of 58%, very close to its empirical counterpart of 60% for the period 1995-2007. The investment-to-GDP ratio generated by the model amounts to 27%, close to the empirical value of 25%. Finally, regarding inequalities, the model generates a Gini index for post-tax income equal to 0.40, identical to its empirical counterpart in Table 1. The Gini index for wealth is found to be 0.78, very close to its empirical value of 0.77 in Table 1.

We gather the model implications in Table 4. These implications show that our tax system provides a good approximation of the redistributive effects of the actual tax system. This confirms the results of Heathcote et al. (2017) and Dyrda and Pedroni (2022).

French calibration

The calibration for France shares a number of similarities with the one for the US. We use the same period and the same functional forms. For the sake of clarity, we mimic the structure of the US calibration, even though our presentation is more streamlined. The calibration parameters can be found, as those for the US, in Table 3.

Preference parameters. The discount factor is set to $\beta = 0.996$ and the Frisch elasticity for the labor supply is still equal to $\varphi = 0.5$. We fix the scaling parameter to $\chi = 0.0228$, which implies an aggregate labor supply normalized to 0.29. It happens that the same risk-aversion parameter $\gamma = 1.8$ is consistent with French statistics.

Technology and TFP shock. The production function is of the Cobb-Douglas form and subsumes capital depreciation: $F(K, L, s) = sK^\alpha L^{1-\alpha} - \delta K$. The capital share is set to the standard value, $\alpha = 36\%$, while the depreciation rate is set to $\delta = 2.5\%$.²²

Idiosyncratic risk. The AR(1) productivity process is calibrated using $\rho_y = 0.99$ and $\sigma_y = 0.0646$. These values are in line with the estimates of Fonseca et al. (2023). We discretize this

²¹To estimate this parameter, we use the dataset for the US in 2005 and restrict our attention to the heads of households and their spouses aged between 25 and 60 who were employed. We define labor income as the sum of wage income, self-employment income, and private transfers. We then define a variable net tax as the income tax and contributions minus public transfers and occupational pensions. Capital income includes interest, dividends, rental income, and private pensions. We apply the estimate of Trabandt and Uhlig (2011) of $\tau^K = 36\%$ to the capital income and obtain the capital income tax. Finally, the labor income tax is simply the variable net tax minus the capital income tax. We then proceed by defining disposable income as labor income minus the labor income tax, and we regress the log of disposable income on the log of labor income. The results are presented in Table 2.

²²We are keeping the same values as in the United States to emphasize more the differences in the SWFs due to differences in the fiscal systems of both countries.

			US	France	
Parameter	Description	Value	Target or ref.	Value	Target or ref.
<i>Preference parameters</i>					
β	discount factor	0.992	$K/Y = 2.7$	0.996	$K/Y = 3.1$
u	utility function	.	$\gamma = 1.8$	.	$\gamma = 1.8$
φ	Frisch elasticity	0.5	Chetty et al. (2011)	0.5	Chetty et al. (2011)
χ	hours worked	0.33	Penn World Table	0.29	Penn World Table
α	capital share	36%	Profit Share, NIPA	36%	Profit Share, INSEE
δ	depreciation rate	2.5%	Chetty et al. (2011)	2.5%	Own calc., INSEE
<i>Productivity parameters</i>					
σ^y	std. err. prod.	0.10	Gini for income	0.06	Fonseca et al. (2023)
ρ^y	autocorr. prod.	0.99	Gini for income	0.99	Fonseca et al. (2023)

Table 3: Parameter values.

AR(1) process using the Tauchen (1986) procedure, with 10 states. With the process, the model is also able to replicate a realistic level of wealth inequalities. The Gini coefficient of income after taxes and transfers in the model is 0.28, matching the value reported in Table 1, while the Gini coefficient of wealth is estimated to be 0.68.

Taxes and government budget constraint. We use the values summarized in Table 1 for the French taxes, with exception for the labor tax that in our model is progressive. We consider a capital tax of $\tau^K = 35\%$, a parameter $\tau = 0.23$ (estimated in the same way as estimated for the US using the dataset of France for the year 2005), and a consumption tax of $\tau^c = 18\%$. This tax system has realistic implications for the model. In terms of public finance, we use $\kappa = 0.728$ to match the empirical public-spending-to-GDP of 24 %. The public-debt-to-GDP ratio amounts to 60%, as the value of Table 1. Regarding private consumption and investment, the model generates aggregate private consumption equal to 44% of GDP, which is close to the empirical counterpart of 45% estimated by Trabandt and Uhlig (2011) for the period 1995-2007, while investment amounts to 31% of GDP, compared to 31% for its empirical counterpart. Finally, in terms of inequalities, the model implies a Gini coefficient for post-tax income equal to 0.28, the same as the empirical value reported in Table 1. The Gini index for wealth generated by the model amounts to 0.68, which is also the same as the empirical counterpart. These elements confirm that the tax system, comprising a linear tax for capital and progressive tax for labor, is very relevant from an empirical point of view.

Model outcomes

We gather the model implications for both France and the US in Table 4.

6.2 Estimation of the SWFs

The estimation procedure follows the algorithm presented in Section 5.3 and the algebra of Appendix D. For the exercises below, we use a truncation length of $N = 5$, although the main characteristic of the results do not change when we consider a longer truncation length. As we

		US		France	
		Model	Data	Model	Data
Parameter		Description			
<i>Public finance aspects</i>					
B/Y	Public debt (%GDP)	63%	63%	60%	60%
G/Y	Public spending (%GDP)	15%	15%	25%	24%
	Total tax revenues (%GDP)	16%	26%	25%	40%
<i>Aggregate quantities</i>					
C/Y	Aggregate consumption (%GDP)	58%	60%	44%	45%
I/Y	Aggregate investment (%GDP)	27%	25%	31%	31%
<i>Inequality measures</i>					
	Gini for post-tax income	40%	40%	28%	28%
	Gini for wealth	78%	77%	68%	68%

Table 4: Model implications for key variables. Empirical values are discussed in Section 2 and summarized in Table 1.

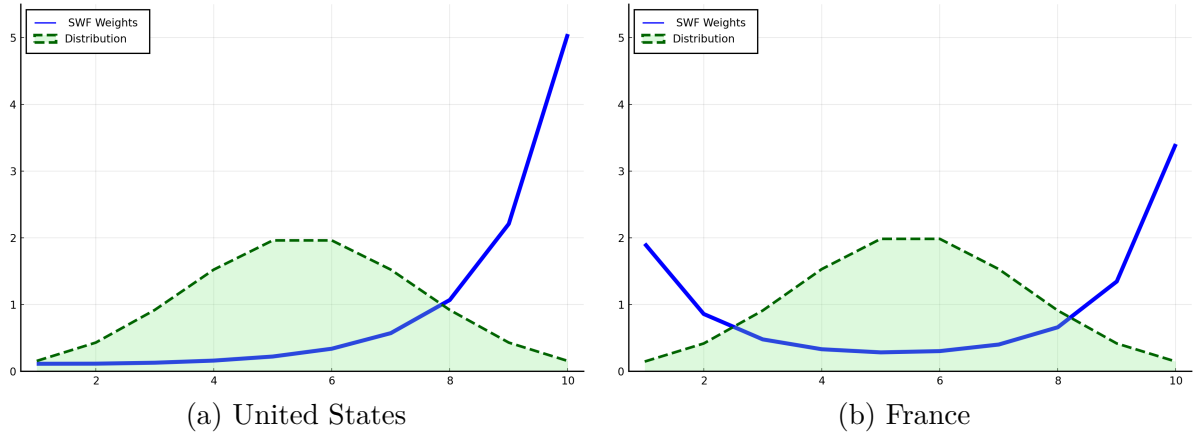


Figure 2: Parametric period weights as a function of productivity for the US and France.

consider 10 idiosyncratic productivity levels, this implies $N^{tot} = 10^5 = 100000$ different truncated histories.

As discussed in Section 5.5, the weights are obtained such that the FOCs of the planner are exactly identified. We consider the same functional form as in equation (44). We obtain the following parametric function for the US and France, respectively:

$$\begin{aligned}\log \omega(y)^{us} &= -0.25 + 1.06 \log(y) + 0.22(\log(y))^2, \\ \log \omega(y)^{fr} &= -0.51 + 0.62 \log(y) + 1.44(\log(y))^2.\end{aligned}$$

In Figure 2, we plot the weights of the SWF as a function of the ten productivity indices of agents. We observe that in the US, the period weights increase with productivity level, whereas for France they exhibit a U-shaped pattern, assigning higher weights to low-productivity agents compared to those at the top of the productivity distribution.

In the US, agents with the highest weight in the population are those with high productivity. In France, low-productivity agents have a higher weight than those with medium productivity. The high productivity agents have the highest weights.

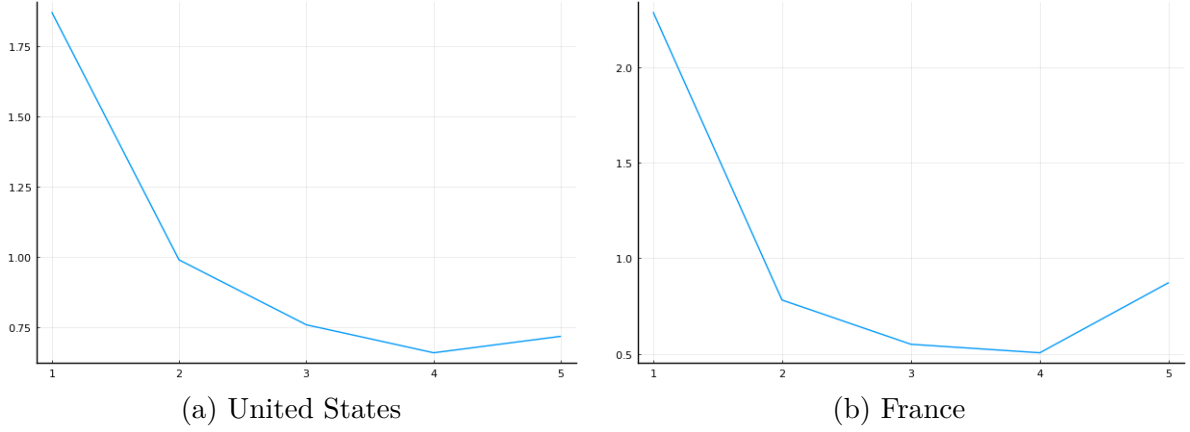


Figure 3: Normlized GSMWW as a function quintile of labor income, defined as $\omega_i \bar{u}'_i$, in the United States and in France.. See text for comments.

Implied partial equilibrium GSMWW

As discussed in Section 5.4, the public finance literature often considers the GSMWWs following Saez and Stantcheva (2016), which is the product of the social weights by the average marginal utility for each productivity level : $\omega_i \bar{u}'_i$. Although the relevant concept for the planner in our environment is the SVL, it is useful to represent implied GSMWW as it has been estimated in the US. Figure 3 represents the social weights for increasing productivity levels, normalized for the mean to be 1, as a function quintile of labor income for each country.

The shape in both countries are similar. The GSMWWs are decreasing, expect for the last quintile for which they are increasing. Interestingly, the shape for the US is similar to the one estimated by Hendren (2020) using data on fiscal externalities. The GSMWW is found to be decreasing except at the end of distribution, where it increases slightly. There are however some differences, as Hendren (2020) finds that the initial point for the first decile, with a similar normalization is closer to 1.2. This difference comes from the fact that low productivity agents have a very high marginal utility, as we don't model additional form of money transfers, which are captured in Hendren (2020) estimates. In addition, our fiscal system, although relevant for the macroeconomy is very simple compared to actual transfer to agents by type in the US. However, we consider that the similarity in the general shape is promising for heterogeneous-agent models to be consistent with the empirical public finance literature. To our knowledge, there is no such estimation for France. The French GSMWW shows however that the same general shape of GSMWWs can be consistent with very different social weights, as can be seen in Figure 2.

6.3 Investigating the drivers of weight differences between the United States and France

Before interpreting these weights, we use the previous methodology to investigate the drivers behind the differing weights assigned to agents in the United States and France. Our objective is to determine if we can replicate the weights of one country using fundamentals or the tax system of the other. Initially, we examine the role of preferences by observing the behavior of the US economy when it adopts the French discount factor, β .

In panel (a) of Figure 4, the red dashed line represents the weights as a function of productivity for the US, adopting the French discount factor. Compared to the original weights, there is a slight increase in the weights for lower productivity agents, but the overall trend remains: higher weights are given to agents with higher productivity levels. In panel (b), the red dashed line shows the same case for France adopting the US preference parameter. We observe that the weights for low-productivity agents in France decrease, while those for high-productivity agents increase. However, the discount factor alone does not fully account for the differences in weights between the two countries.

Next, we analyze the impact of fiscal systems. We examine how the US economy behaves under the French fiscal system, characterized by higher labor tax progressivity. This system, which is more aligned with reducing inequality, contrasts with the US system, which is more libertarian as we will discuss below. Panel (a) of Figure 4 shows the US weights with the French tax system (orange dashed line). The weights for lower productivity agents increase at the expense of those for higher productivity agents. Conversely, for France adopting the US tax system, the weights for low-productivity agents decrease, while those for high-productivity agents increase (panel (b) of Figure 4).

Figure 9 in Appendix F illustrates the changes in period weights, utility, labor, and capital income when the US adopts the French tax system and vice versa. Adopting the French tax system in the US reduces labor income for high-productivity agents due to decreased labor supply incentives. Heavier taxation reduces their utility and weights assigned by the social planner. Conversely, the progressive tax system boosts consumption and utility for low-productivity agents, increasing their weights.

This experiment demonstrates that consistent changes in the tax system can rationalize changes in weights. For the US to increase weights for low-productivity agents and decrease weights for high-productivity agents, adopting a more progressive labor tax is effective. The relatively low labor tax in the US favors high-income/high-productivity agents.

Finally, to fully uncover the differences in weights, we incorporate the French income process into the US economy. This allows us to completely replicate the French weights (blue line in panel (a) of Figure 4). Similarly, incorporating the US income process into the French economy allows us to replicate the US weights (panel (b)). This highlights that weights depend on preference parameters, taxes, and income processes, underscoring the significant role of taxes and income processes in explaining weight differences.

6.4 A world where the US have the SWF of France

We now compute the US fiscal system that makes social weights as close as possible to those of France. The goal is to find a fiscal system in the US where the distance between the weights in the modified US economy and the initial France economy is minimized. This exercise aims to understand the role of social preferences in shaping the actual tax system, distinct from the influence of technology and individual preferences.

We conduct the experiment as follows. Once the capital-to-output ratio has been set to its steady-state value, we iterate over every combination of the capital tax rate and the progressivity of the labor tax, adjusting κ to keep the government spending-to-output ratio constant. This

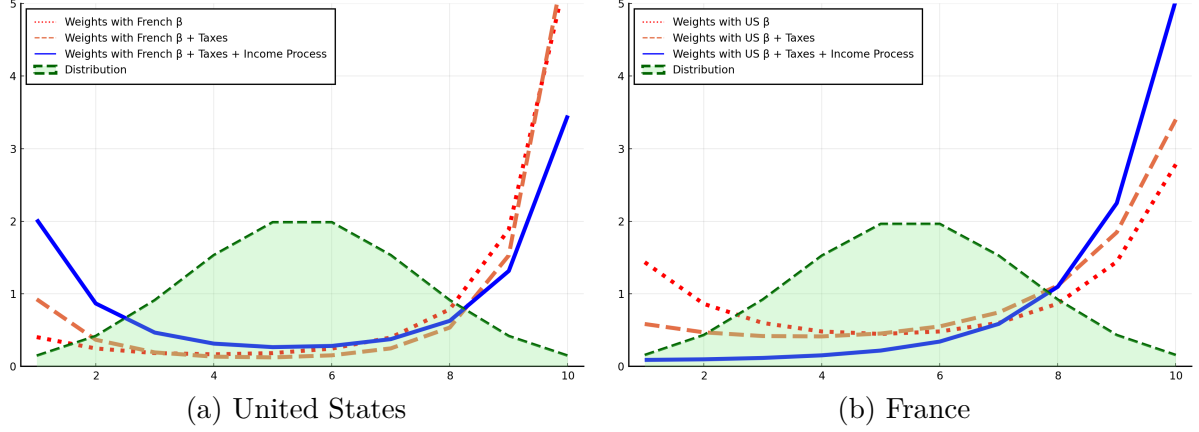


Figure 4: Change in the period weights for the United States and France due to different preference parameters, tax systems, and income processes.

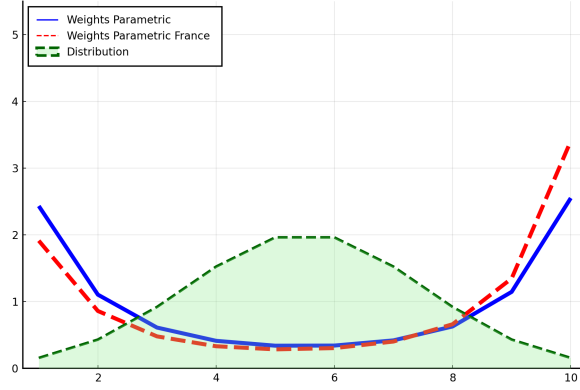


Figure 5: SWF weights for France (red dashed line) and for the United States with the tax system that minimizes the distance to the French weights (blue line).

allows us to match the main macroeconomic ratios. Specifically, all model parameters and primary macroeconomic ratios remain unchanged. The only variable adjusted in each case is the vector representing the fiscal system. $(\tau_K, \tau_c, \tau, \kappa, B)$. For each combination, we calculate the weights and, by using interpolation on the productivity states of the target economy (i.e., in this case, France), we can obtain the weights on the union of the grids that represent the whole set of productivity states. Once we obtain these weights, we can calculate the distance between the vectors using standard measures such as the Euclidean distance. We then consider, as in our benchmark case, that this new fiscal system, which represents the minimum distance between the weights, results from the optimal planner's decisions.

In this experiment, we are considering the distance between the weights on the union of the grids in the productivity states.

Figure 5 plots the weights as a function of productivity for the US with the new fiscal system and for France. As can be seen, our procedure allows us to find a fiscal system, such the weights in the US with this modified fiscal system are quite close to the French weights.

We report in Table 5 the values of the new fiscal system that allows the weights in the US with this modified fiscal system to minimize the Euclidean distance to the French weights. We also report the fiscal system values for the US benchmark economy and the French benchmark

economy. The column Gini post-tax income represents the Gini of income after taxes and transfers, and the last one represents the Gini for wealth.

	Public debt (%GDP)	τ_k (%)	τ (%)	κ (%)	Gini post-tax income	Gini wealth
US	63	36	16	85	40	78
France	60	35	23	72.8	28	68
US with French SWF	299	9	57	71	27	63

Table 5: Comparison between the benchmark economies and the US economy with the French SWF.

The first observation is that the distribution of income and wealth for the United States is now much closer to their French counterparts, and therefore less unequal. This mostly comes from a much higher progressivity that is needed to increase the weights on low-productivity/low-income agents. The progressivity indeed increases from 16% to 57%. We recall that the other policy parameters, as well as the main macro ratios (e.g., consumption-to-GDP, government spending-to-GDP, investment-to-GDP), remain the same as in the benchmark economy. In particular, the pre-tax interest rate is kept at its optimal value, which is the inverse of the discount factor. Because of the labor taxation, the labor supply falls, which means that the capital also falls to keep the capital-to-labor ratio constant. However, agents are still patient, and more than in the French economy, which means that they overall save more. This requires an increase in public debt to absorb the excess savings. The decrease in capital tax from 36% to 9% is needed for low productivity agent to save more, as their weight in the SWF is now higher.

This increase in public debt and the decrease in capital taxes requires an increase in the labor tax to compensate for the loss in tax returns. Moreover, the increase in the labor tax was expected since the relatively low labor tax in the United States favors high-income/high-productivity agents. This decreases the estimated weights for high-labor-income agents.

6.5 Identification of IWFs

We now derive the estimated IWFs from the estimated SWF for each country, as presented in Section 4.4. The first step is to estimate political weights of various groups of agents. We use the participation rates of main elections as a function of the average income as an estimation of the political weights of each productivity level. We are here following the literature using turnout inequality as a proxy of the evolution of political inequality (Cage, 2024 for a discussion). Other strategies are used in the political economy literature, notably relying on donations in the US. As political donations are very small in France, considering turnout and participation rates allow for comparison between France and the US.

Figure 6 plots the participation rate in percent as a function of the average income in the US (panel a) and in France (panel b). For each country, the income is expressed in percent of the average income of the country.²³ In the US, average income is 51726\$, and it is 31093€ for

²³Us data are taken from Table 8 the Current Population Survey of November 2008 of the U.S. Census Bureau. French data are taken from IPSOS data for participation rates as a function of occupations and DADS 2007 to obtain the annual income of each occupation.

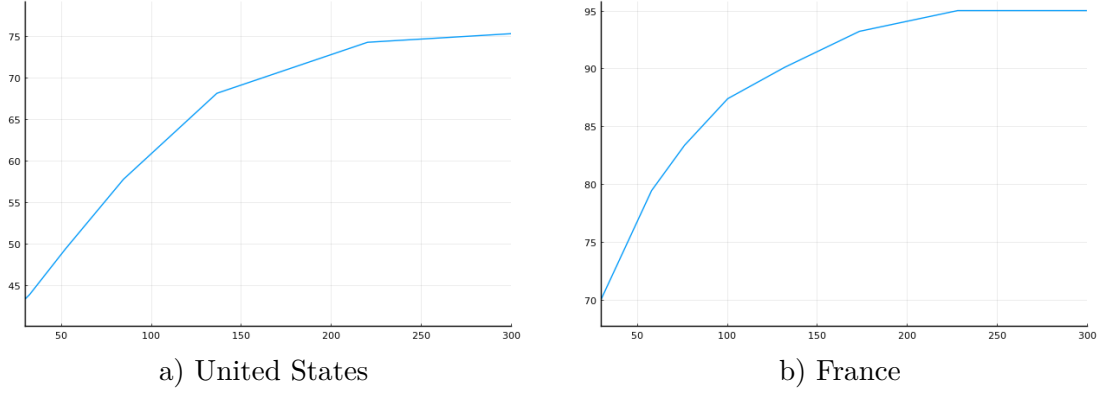


Figure 6: Participation rates in percent as a function of annual income, expressed in percent of average income. The data for the US are for the presidential election of 2008. The data for France are for the presidential election of 2007. The 100 on the x-axis is the average income in each country.

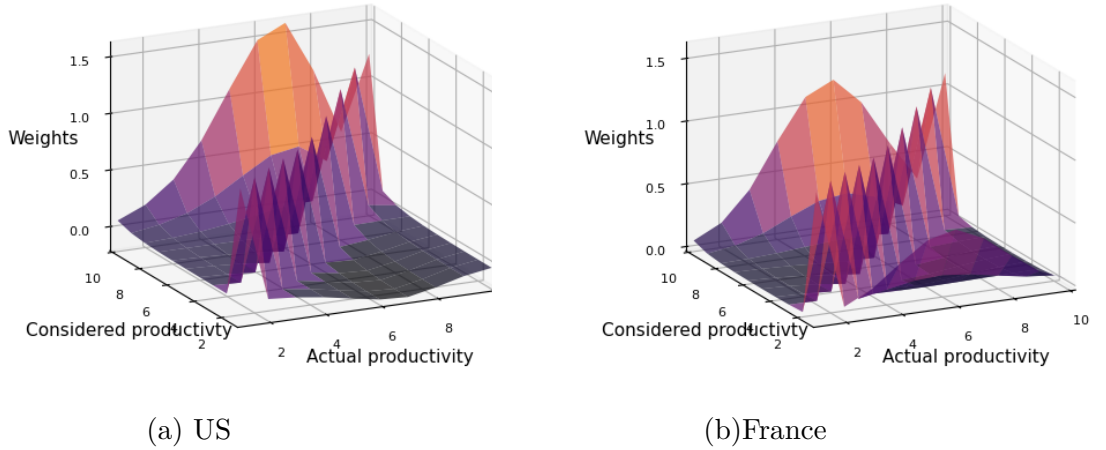


Figure 7: Total weights for each productivity group. Actual productivity is the productivity y of the agents, Considered productivity is the productivity $\tilde{y}$ of the agents under consideration.

France. We observe that the participation rate is increasing and concave in income for each country. The average participation rate is higher for France than for the US, but it is the ratio of participation rates between income groups, which is identifying the political weights. Indeed we assume that $\frac{\omega_{P,y}}{\omega_{P,\tilde{y}}} = \frac{Part_y}{Part_{\tilde{y}}}$, where $Part_y$ is the participation rate of income group y , which is interpolated from figure 6. As a consequence, the shape of political weights follow the shape of participation rates.

Considering these political weights, it is now possible to estimate the IWFs from the SWFs, minimizing the quadratic distance presented in Definition 3. The results are reported in Figure 7.

We report the politically-weighted weights given by the mass of agents of productivity y (Actual productivity) to agents of productivity $\tilde{y}$ (Considered productivity). The z-axis thus reports the value of $\omega_{P,y}\pi_y\omega_{y,\tilde{y}}$ for each $y = 1, \dots, 10$ and $\tilde{y} = 1, \dots, 10$. The higher the weights the more the actual agents y affect the valuations of agents $\tilde{y}$ by the planner. The left panel a) is the result for the US, the right panel b) is the results for France.

First, one can observe that in each country the diagonal has a high weights, implying that

agents are self-seeking, putting a high weight to their own productivity. In the US, the diagonal is more increasing as a function of productivity, and it reaches higher levels than in France. This implies that the most productive agents have higher weights in the SWF, especially in the US, what is consistent with the results of Section 6.2.

Second, not considering the diagonal, the US weights exhibit an increasing pattern in $\tilde{y}$ for each productivity level y , notably for intermediate value of y , who are the most numerous. This shape has been qualified of Libertarian shape for the welfare functions in Section 3.4, i.e. higher weights are attributed to the most productive agents. Third, not considering the diagonal, the French weights exhibit a U-shape pattern, consistent with the finding of 6.2 for the French SWF. Hence, the French welfare functions are Egalitarian for low level of productivity and Libertarian for higher productivity levels.

7 Conclusion

We derive a methodology to identify the Social Welfare Function (SWF) and Individual Welfare Functions (IWF), which are consistent with the empirical wealth and income distributions given the actual tax structure in both France and the US. Using four fiscal instruments—consumption, capital and progressive labor taxes, and public debt—we estimated the SWFs of the two countries and showed that they differ from each other. The SWF for France gives a higher weight to low-productivity agents and is less heterogeneous than that of the US, while the US SWF has an increasing shape in income with more weight to high-income agents. The US thus appear more Libertarian than France, which is more Egalitarian, especially for low income levels.

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Appendix

A Proofs related to the SWF construction

A.1 Construction of the measure on the set of idiosyncratic histories $\mathcal{Y}^\infty$

We can construct a probability space related to the set of infinite idiosyncratic histories, $\mathcal{Y}^\infty$. We summarize here the construction and further details can be found in LeGrand and Ragot (2022, Appendix B.3.1). An element $y^\infty \in \mathcal{Y}^\infty$ can be described as a left-infinite sequence that can be written as follows:

$$y^\infty = (\dots, y_{-k}(y^\infty), \dots, y_{-1}(y^\infty), y_0(y^\infty)),$$

where each $y_{-k} : \mathcal{Y}^\infty \rightarrow \mathcal{Y}$ is a coordinate function returning the idiosyncratic state k periods ago. For the sake of simplicity, since we need to mark the passage of time, we keep the time index t . We thus typically denote by y^t an element of $\mathcal{Y}^\infty$ (at date t) and by $y_s^t := y_{s-t}(y^t)$ for any $s \leq t$ the state in date s , which is $t - s$ periods ahead of date t .

We define $L(y^\infty)$ the past of history y^∞ – which discards the current state $y_0(y^\infty)$:

$$L(y^\infty) = (\dots, y_{-k}(y^\infty), \dots, y_{-1}(y^\infty)).$$

Consistently with our previous notation, we denote in the paper by y^{t-1} the past history of y^t : $y^{t-1} = L(y^t)$.

We can then define the cylinder sets $C_k(A)$ for any $k \geq 1$ and any $A \subset \mathcal{Y}^k$ as:

$$C_k(A) = \{y^\infty \in \mathcal{Y}^\infty : (y_{-k+1}(y^\infty), \dots, y_0(y^\infty)) \in A\}.$$

The cylinder set $C_k(A)$ is the subset of $\mathcal{Y}^\infty$ containing all idiosyncratic histories whose truncation of length k belongs to A . We then define $\mathcal{C}_0$ as the set of all cylinder sets, which can be shown to be an algebra (LeGrand and Ragot, 2022, Lemma 1). We denote by $\mathcal{F} := \sigma(\mathcal{C}_0)$ the cylindrical σ -algebra generated by $\mathcal{C}_0$, and we define the set function $\mu : \mathcal{C}_0 \rightarrow \mathbb{R}$ from the transition matrix Π and the stationary vector π , such that for any $k \geq 2$ and any $A \subset \mathcal{Y}^k$:

$$\begin{cases} \mu(C_k(A)) = \sum_{(y_{-k+1}, \dots, y_0) \in A} \pi_{y_{-k+1}} \Pi_{y_{-k+1} y_{-k+2}} \dots \Pi_{y_{-1} y_0} & \text{for any } k \geq 2 \text{ and } A \subset \mathcal{Y}^k, \\ \mu(C_1(A)) = \sum_{y_0 \in A} \pi_{y_0} & \text{for any } A \subset \mathcal{Y}. \end{cases} \quad (45)$$

Finally, we can state the following lemma.

Lemma 1 *The triplet $(\mathcal{Y}^\infty, \mathcal{F}, \mu)$ is a probability space.*

Proof. The proof can be found in LeGrand and Ragot (2022, Lemma 3 in Section B.3). The key part of the proof is to extend the measure μ defined on $\mathcal{C}_0$ to a measure defined on $\sigma(\mathcal{C}_0) = \mathcal{F}$.

■ A consequence of Lemma 1 is that $\mu(\mathcal{Y}^\infty) = 1$, or $\int_{y^\infty \in \mathcal{Y}^\infty} \mu(dy^\infty) = 1$.

A.2 The conditional measure

To lighten formulas, for any $(y_{-k+1}, \dots, y_0) \in \mathcal{Y}^k$, we define $y_{-k+1:0} := (y_{-k+1}, \dots, y_0)$ the vector of length k , containing elements whose indices range from $-k+1$ to 0.

We define μ_1 by:

$$\mu_1(C_1(Y_1)|\tilde{y}^\infty) = \sum_{y_0 \in Y_1} \Pi_{y_0(\tilde{y}^\infty)y_0}, \text{ for any } Y_1 \subset \mathcal{Y}, \quad (46)$$

$$\forall k \geq 2, \mu_1(C_k(Y_k)|\tilde{y}^\infty) = \sum_{y_{(-k+1):0} \in Y_k} \Pi_{y_0(\tilde{y}^\infty)y_0} 1_{y_{(-k+1):-1}=(y_{-k+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))}, \text{ for any } Y_k \subset \mathcal{Y}^k, \quad (47)$$

where for any elements $x, \tilde{x}$ of the same set, $1_{x=\tilde{x}} = 1$ if $x = \tilde{x}$ and $1_{x=\tilde{x}} = 0$ otherwise. Intuitively, the expression in (47) sums over all possible vectors $y_{(-k+1):0}$ of length k , the probability to switch from $\tilde{y}^\infty$ to an history ending up in $y_{(-k+1):0}$. The latter probability is equal to the probability to switch from $y_0(\tilde{y}^\infty)$ to y_0 , provided that $\tilde{y}^\infty$ and $y_{(-k+1):1}$ are compatible (i.e., the $k-1$ last realization of $\tilde{y}^\infty$ equals $y_{(-k+1):1}$). Note that we could also write: $1_{y_{(-k+1):-1}=(y_{-k+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))} = 1_{y_{-k+1}=y_{-k+2}(\tilde{y}^\infty)} \times \dots \times 1_{y_{-1}=y_0(\tilde{y}^\infty)} = \prod_{j=1}^{k-1} 1_{y_{-j}=y_{-j+1}(\tilde{y}^\infty)}$.

Lemma 2 For any $\tilde{y}^\infty \in \mathcal{Y}^\infty$, the set function $C \in \mathcal{C}_0 \mapsto \mu_1(C|\tilde{y}^\infty)$ is a pre-measure.

Proof.

In the remainder of the proof, we set $\tilde{y}^\infty \in \mathcal{Y}^\infty$.

As a preliminary, we state two properties that we will use extensively below:

- For all $k' \geq k \geq 1$, for all $Y_k \subset \mathcal{Y}^k$, for all $Y'_{k'-k} \subset \mathcal{Y}^{k'-k}$:

$$\sum_{y_{(-k'+1):0} \in Y'_{k'-k} \times Y_k} \Pi_{y_0(\tilde{y}^\infty)y_0} 1_{y_{(-k'+1):-1}=(y_{-k'+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))} = \quad (48)$$

$$\sum_{y_{(-k+1):0} \in Y_k} \left(\Pi_{y_0(\tilde{y}^\infty)y_0} 1_{y_{(-k+1):-1}=(y_{-k+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))} \times \right. \quad (49)$$

$$\left. \sum_{y_{(-k'+1):-k} \in Y'_{k'-k}} 1_{y_{(-k'+1):-k}=(y_{-k'+2}(\tilde{y}^\infty), \dots, y_{-k+1}(\tilde{y}^\infty))} \right)$$

- For all $k' \geq k \geq 0$:

$$\sum_{y_{-k':-k} \in \mathcal{Y}^{k'-k+1}} 1_{y_{(-k':-k)}=(y_{-k'+1}(\tilde{y}^\infty), \dots, y_{-k+1}(\tilde{y}^\infty))} = 1. \quad (50)$$

In the remainder they will be referred to by their equation numbering. The proof of (50) is straightforward and comes from the fact that $(y_{-k'+1}(\tilde{y}^\infty), \dots, y_{-k+1}(\tilde{y}^\infty))$ is a unique element of the set $\mathcal{Y}^{k'-k+1}$. For the proof of (50), we denote by $S_{y,k,k'}$ the left hand side. We have:

$$S_{y,k,k'} = \sum_{y_{(-k+1):0} \in Y_k} \sum_{y_{(-k'+1):-k} \in Y'_{k'-k}} \left(\Pi_{y_0(\tilde{y}^\infty)y_0} \times \right. \quad (51)$$

$$\left. 1_{y_{(-k+1):-1}=(y_{-k+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))} 1_{y_{(-k'+1):-k}=(y_{-k'+2}(\tilde{y}^\infty), \dots, y_{-k+1}(\tilde{y}^\infty))} \right),$$

where we have used the properties of a sum on a product space and the fact that $(x, y) = (x', y')$ iff $x = x'$ and $y = y'$ (where (x, x') and (y, y') are two pairs of vectors of the same length). Then we can factorize $\Pi_{y_0(\tilde{y}^\infty)y_0}$ and $1_{y_{(-k+1):-1}=(y_{-k+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))}$ in (51), as they are independent from the sum over $y_{(-k'+1):-k} \in Y'_{k'-k}$. We then readily obtain (48).

We now go back to the proof of Lemma 2. Three properties need to hold: (i) well-defined, (ii) (countably) additive, (iii) $\mu_1(\mathcal{Y}^\infty|\tilde{y}^\infty) = 1$ for all $\tilde{y}^\infty \in \mathcal{Y}^\infty$.

For Point (i), we need to check that $\mu_1(C_k(Y_k)|\tilde{y}^\infty) = \mu_1(C_{k'}(\mathcal{Y}^{k'-k} \times Y_k)|\tilde{y}^\infty)$ for all $k' \geq k$. We have for $k \geq 2$:

$$\begin{aligned} \mu_1(C_{k'}(\mathcal{Y}^{k'-k} \times Y_k)|\tilde{y}^\infty) &= \sum_{y_{(-k'+1):0} \in Y_k} \Pi_{y_0(\tilde{y}^\infty)y_0} 1_{y_{(-k'+1):-1}=(y_{-k'+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))}, \\ &= \sum_{y_{(-k+1):0} \in Y_k} \Pi_{y_0(\tilde{y}^\infty)y_0} 1_{y_{(-k+1):-1}=(y_{-k+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))} = \mu_1(C_k(Y_k)|\tilde{y}^\infty), \end{aligned} \quad (52)$$

where the first and last equalities use the definition of μ_1 , and the second the combination of properties (48) and (50).

We need to prove the result for $k = 1$. We have for $k' \geq 1$:

$$\begin{aligned} \mu_1(C_{k'}(\mathcal{Y}^{k'-1} \times Y_1)|\tilde{y}^\infty) &= \sum_{y_0 \in Y_1} \Pi_{y_0(\tilde{y}^\infty)y_0} \left(\sum_{y_{(-k'+1):-1} \in \mathcal{Y}^{k'-1}} 1_{y_{(-k'+1):-1}=(y_{-k'+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))} \right), \\ &= \sum_{y_0 \in Y_1} \Pi_{y_0(\tilde{y}^\infty)y_0} = \mu_1(C_1(Y_1)|\tilde{y}^\infty), \end{aligned}$$

where the first equality combines the definition of μ_1 and (48), the second uses the property (50), and the last the definition of μ_1 .

For point (ii), we consider two disjoint cylinders, $C_k(Y_k)$ and $C_{k'}(Y'_{k'})$ ($k' \geq k$, $Y_k \subset \mathcal{Y}^k$ and $Y'_{k'} \subset \mathcal{Y}^{k'}$). Since both cylinders are disjoint, then $\mathcal{Y}^{k'-k} \times Y_k$ and $Y'_{k'}$ are disjoint too. We deduce that:

$$\begin{aligned} \mu_1(C_k(Y_k) \cup C_{k'}(Y'_{k'})|\tilde{y}^\infty) &= \mu_1(C_{k'}((\mathcal{Y}^{k'-k} \times Y_k) \cup Y'_{k'})|\tilde{y}^\infty), \\ &= \sum_{y_{(-k'+1):0} \in (\mathcal{Y}^{k'-k} \times Y_k) \cup Y'_{k'}} \Pi_{y_0(\tilde{y}^\infty)y_0} 1_{y_{(-k'+1):-1}=(y_{-k'+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))}, \\ &= \sum_{y_{(-k'+1):0} \in (\mathcal{Y}^{k'-k} \times Y_k)} \Pi_{y_0(\tilde{y}^\infty)y_0} 1_{y_{(-k'+1):-1}=(y_{-k'+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))} \\ &\quad + \sum_{y_{(-k'+1):0} \in Y'_{k'}} \Pi_{y_0(\tilde{y}^\infty)y_0} 1_{y_{(-k'+1):-1}=(y_{-k'+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))}, \\ &= \sum_{y_{(-k+1):0} \in Y_k} \Pi_{y_0(\tilde{y}^\infty)y_0} 1_{y_{(-k+1):-1}=(y_{-k+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))} \\ &\quad + \sum_{y_{(-k'+1):0} \in Y'_{k'}} \Pi_{y_0(\tilde{y}^\infty)y_0} 1_{y_{(-k'+1):-1}=(y_{-k'+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))}, \\ &= \mu_1(C_k(Y_k)|\tilde{y}^\infty) + \mu_1(C_{k'}(Y'_{k'})|\tilde{y}^\infty), \end{aligned}$$

where the first equality uses the algebra property of cylinder sets, the second the definition of μ_1 , the third the property that $\mathcal{Y}^{k'-k} \times Y_k$ and $Y'_{k'}$ are disjoint, the fourth the combination of

properties (48) and (50), and the last the definition of μ_1 twice. We have thus proved that $\mu_1(\cdot|\tilde{y}^\infty)$ is finitely additive.

For point (iii), let $k \geq 2$. We have:

$$\begin{aligned}\mu_1(C_k(\mathcal{Y}^k)|\tilde{y}^\infty) &= \sum_{y_0 \in \mathcal{Y}} \Pi_{y_0(\tilde{y}^\infty)y_0} \sum_{y_{(-k+1):1} \in \mathcal{Y}^{k-1}} 1_{y_{(-k+1):-1}=(y_{-k+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))}, \\ &= \sum_{y_0 \in \mathcal{Y}} \Pi_{y_0(\tilde{y}^\infty)y_0} = 1,\end{aligned}$$

where the first equality uses the definition of μ_1 and (48), the second property (50), and the third the property of the transition matrix Π . We thus deduce that $\mu_1(\mathcal{Y}^\infty|\tilde{y}^\infty) = 1$.

We have proven that $\mu_1(\cdot|\tilde{y}^\infty)$ is a finitely additive probability measure on the algebra $\mathcal{C}_0$. Billingsley (2012, Theorem 2.3) states that any finitely additive probability measure on the cylinder algebra is countably additive. We thus conclude that μ_1 is a countably additive probability measure on $\mathcal{C}_0$ and is thus a pre-measure on $\mathcal{C}_0$. ■

We then prove the following lemma. We recall that $\mathcal{F}$ the cylindrical σ -algebra, $\sigma(\mathcal{C}_0)$, generated by $\mathcal{C}_0$.

Lemma 3 *For all $\tilde{y}^\infty \in \mathcal{Y}^\infty$, the function $\mu_1(\cdot|\tilde{y}^\infty)$ uniquely extends to a measure on $\mathcal{F}$.*

The proof is similar to the one showing the extension of μ as a pre-measure on $\mathcal{C}_0$ to a measure on $\mathcal{F}$. It relies on the Hahn-Kolmogorov theorem (Billingsley 2012, Theorem 3.1). See LeGrand and Ragot (2022, Lemma 3 in Section B.3).

Finally we can state the following lemma, showing that μ_1 is a conditional measure.

Lemma 4 *For all $\tilde{y}^\infty \in \mathcal{Y}^\infty$ and for all $F \in \mathcal{F}$:*

$$\int_{\tilde{y}^\infty \in \mathcal{Y}^\infty} \mu_1(F|\tilde{y}^\infty) \mu(d\tilde{y}^\infty) = \mu(F) \quad (53)$$

Proof. We first prove (53) for F being a cylinder sets.

First, let $Y_1 \subset \mathcal{Y}$ and consider $C_1(Y_1)$. We have:

$$\begin{aligned}\int_{\tilde{y}^\infty \in \mathcal{Y}^\infty} \mu_1(C_1(Y_1)|\tilde{y}^\infty) \mu(d\tilde{y}^\infty) &= \int_{\tilde{y}^\infty \in \mathcal{Y}^\infty} \sum_{y_0 \in Y_1} \Pi_{y_0(\tilde{y}^\infty)y_0} \mu(d\tilde{y}^\infty), \\ &= \sum_{\tilde{y}_0 \in \mathcal{Y}} \pi_{\tilde{y}_0} \sum_{y_0 \in Y_1} \Pi_{\tilde{y}_0 y_0}, \\ &= \sum_{y_0 \in Y_1} \sum_{\tilde{y}_0 \in \mathcal{Y}} \pi_{\tilde{y}_0} \Pi_{\tilde{y}_0 y_0}, \\ &= \sum_{y_0 \in Y_1} \pi_{y_0} = \mu(C_1(Y_1)),\end{aligned}$$

where the first equality comes from the definition (46) of μ_1 , the second from the fact that the integral is actually carried over a cylinder set of the form $C_1(\mathcal{Y})$ and from the definition (45) of μ , the third from the permutation of finite sums, the fourth from the fact that π is stationary ($\sum_{y \in \mathcal{Y}} \pi_y \Pi_{yy'} = \pi_{y'}$), and the last from the definition (45) of μ .

Second, let $Y_k \subset \mathcal{Y}^k$ and consider $C_k(Y_k)$. We have:

$$\begin{aligned}
& \int_{\tilde{y}^\infty \in \mathcal{Y}^\infty} \mu_1(C_k(Y_k)|\tilde{y}^\infty) \mu(d\tilde{y}^\infty) \\
&= \int_{\tilde{y}^\infty \in \mathcal{Y}^\infty} \sum_{y_{(-k+1):0} \in Y_k} \Pi_{y_0(\tilde{y}^\infty)y_0} 1_{y_{(-k+1):-1}=(y_{-k+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))} \mu(d\tilde{y}^\infty), \\
&= \sum_{\tilde{y}_{(-k+2):0} \in \mathcal{Y}^{k-1}} \pi_{\tilde{y}_{-k+2}} \Pi_{\tilde{y}_{-k+2}\tilde{y}_{-k+3}} \cdots \Pi_{\tilde{y}_{-1}\tilde{y}_0} \sum_{y_{(-k+1):0} \in Y_k} \Pi_{\tilde{y}_0 y_0} 1_{y_{(-k+1):-1}=\tilde{y}_{(-k+2:0)}}, \\
&= \sum_{y_{(-k+1):0} \in Y_k} \sum_{\tilde{y}_{(-k+2):0} \in \mathcal{Y}^{k-1}} \pi_{\tilde{y}_{-k+2}} \Pi_{\tilde{y}_{-k+2}\tilde{y}_{-k+3}} \cdots \Pi_{\tilde{y}_{-1}\tilde{y}_0} \Pi_{\tilde{y}_0 y_0} 1_{y_{(-k+1):-1}=\tilde{y}_{(-k+2:0)}}, \\
&= \sum_{y_{(-k+1):0} \in Y_k} \pi_{y_{-k+1}} \Pi_{y_{-k+1}y_{-k+2}} \cdots \Pi_{y_{-2}y_{-1}} \Pi_{y_{-1}y_0} = \mu(C_k(Y_k)),
\end{aligned}$$

where the first equality comes from the definition (47) of μ_1 , the second from the fact that the integral is actually carried over a cylinder set of the form $C_{k-1}(\mathcal{Y}^{k-1})$ and from the definition (45) of μ , the third from the permutation of finite sums, the fourth from the fact that all terms in the sum over $\tilde{y}_{(-k+2):0} \in \mathcal{Y}^{k-1}$ are zero but the one for $\tilde{y}_{(-k+2:0)} = y_{(-k+1):-1}$, and the last from the definition (45) of μ . This proves (53) on $\mathcal{C}_0$. Since cylinder sets are a generating family of $\mathcal{F}$ and a π -system, the equality (53) also holds on $\sigma(\mathcal{C}_0) = \mathcal{F}$. ■

We finally prove the following lemma. It justifies that in the main text, we define μ_1 as $\mu_1(dy^{t+1}|\tilde{y}^t) = \Pi_{\tilde{y}_t^t y_{t+1}^{t+1}} \delta_{\tilde{y}^t}(dy^t)$.

Lemma 5 *For all $\tilde{y}^\infty \in \mathcal{Y}^\infty$ and for all $C \in \mathcal{C}_0$, we have:*

$$\mu_1(C|\tilde{y}^\infty) = \int_{y^\infty \in C} \Pi_{y_0(\tilde{y}^\infty)y_0(y^\infty)} \delta_{\tilde{y}^\infty}(L(dy^\infty)),$$

where $\delta_{\tilde{y}^\infty}$ is the Dirac mass in $\tilde{y}^\infty$.

Proof. Let $Y_k \subset \mathcal{Y}^k$ for some $k \geq 1$.

For $k = 1$, we have:

$$\begin{aligned}
\int_{y^\infty \in C_1(Y_1)} \Pi_{y_0(\tilde{y}^\infty)y_0(y^\infty)} \delta_{\tilde{y}^\infty}(L(dy^\infty)) &= \int_{(\hat{y}^\infty, y_0) \in \mathcal{Y}^\infty \times Y_1} \Pi_{y_0(\tilde{y}^\infty)y_0} \delta_{\tilde{y}^\infty}(d\hat{y}^\infty), \\
&= \sum_{y_0 \in Y_1} \Pi_{y_0(\tilde{y}^\infty)y_0} \int_{\hat{y}^\infty \in \mathcal{Y}^\infty} \delta_{\tilde{y}^\infty}(d\hat{y}^\infty), \\
&= \sum_{y_0 \in Y_1} \Pi_{y_0(\tilde{y}^\infty)y_0} = \mu_1(C_1(Y_1)|\tilde{y}^\infty),
\end{aligned}$$

where the first equality comes from using $C_1(Y_1) = \mathcal{Y}^\infty \times Y_1$, the second from using the Fubini theorem (we consider σ -finite measure spaces and $(\hat{y}^\infty, y_0) \in \mathcal{Y}^\infty \times Y_1 \mapsto \Pi_{y_0(\tilde{y}^\infty)y_0} \delta_{\tilde{y}^\infty}(d\hat{y}^\infty)$ is integrable), the third the property of Dirac mass, and fourth the definition of μ_1 .

For $k \geq 2$, we have:

$$\int_{y^\infty \in C_k(Y_k)} \Pi_{y_0(\tilde{y}^\infty)y_0(y^\infty)} \delta_{\tilde{y}^\infty}(L(dy^\infty)) \quad (54)$$

$$= \int_{(\hat{y}^\infty, y_{(-k+1):0}) \in \mathcal{Y}^\infty \times Y_k} \Pi_{y_0(\tilde{y}^\infty)y_0} 1_{y_{(-k+1):-1}=(y_{-k+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))} \delta_{L^{(k-1)}(\tilde{y}^\infty)}(d\hat{y}^\infty), \quad (55)$$

where $L^{(k-1)}(\cdot)$ is k -iterate of the shift operator L : $L^{(k-1)}(\tilde{y}^\infty) = (\dots, \tilde{y}_{-k-1}^\infty, \tilde{y}_{-k}^\infty, \tilde{y}_{-k+1}^\infty)$. To write equality (54), we have used that $C_k(Y_k) = \mathcal{Y}^\infty \times Y_k$ and the fact that $\delta_{\tilde{y}^\infty}(Y_{k-1} \times C) = \sum_{y_{(-k+1):-1} \in Y_{k-1}} 1_{y_{(-k+1):-1} = (y_{-k+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))} \delta_{L^{(k-1)}(\tilde{y}^\infty)}(C)$ for all $C \in \mathcal{C}_0$ and $Y_{k-1} \subset \mathcal{Y}^{k-1}$. Hence the two measures coincide on $\mathcal{C}_0$ and are also σ -finite. Using Fubini and the property of a Dirac mass, we obtain from (54):

$$\begin{aligned} \int_{y^\infty \in C_k(Y_k)} \Pi_{y_0(\tilde{y}^\infty)y_0(y^\infty)} \delta_{\tilde{y}^\infty}(L(dy^\infty)) &= \sum_{y_{(-k+1):0} \in Y_k} \Pi_{y_0(\tilde{y}^\infty)y_0} 1_{y_{(-k+1):-1} = (y_{-k+2}(\tilde{y}^\infty), \dots, y_0(\tilde{y}^\infty))}, \\ &= \mu_1(C_k(Y_k)|\tilde{y}^\infty), \end{aligned}$$

where the last equality comes from the definition of μ_1 . This concludes the proof. ■

A.3 Proof of Equality 12

We consider an agent whose individual preferences are represented by a utility function $V : \mathcal{Y}^\infty \rightarrow \mathbb{R}$. The allocation is subsumed. Let $y^t \in \mathcal{Y}^\infty$. The representation result of equation (11) becomes, after splitting the sum for $s = 0$ and $s \geq 1$:

$$V(y^t) = U(y^t) + \beta \sum_{s=1}^{\infty} \int_{y^{t+s} \in \mathcal{Y}^\infty} \beta^{s-1} U(y^{t+s}) \mu_s(dy^{t+s}|y^t). \quad (56)$$

Using the definition of μ_s and Bayes rule, we have, for all $s \geq 1$:

$$\mu_s(dy^{t+s}|y^t) = \int_{y^{t+1} \in \mathcal{Y}^\infty} \mu_{s-1}(dy^{t+s}|y^{t+1}) \mu_1(dy^{t+1}|y_0^\infty). \quad (57)$$

In words, it means that the probability of transitioning from y^t to y^{t+s} in s periods is equal to the product of the probabilities of transitioning from y^t to y^{t+1} (in 1 period) and from y^{t+1} to y^{t+s} (in $s-1$ periods), summed over all possible histories y^{t+1} . Using (57), the expression of V in (56) becomes:

$$\begin{aligned} V(y^t) &= U(y^t) + \beta \sum_{s=1}^{\infty} \int_{y^{t+s} \in \mathcal{Y}^\infty} \beta^{s-1} U(y^{t+s}) \\ &\quad \times \int_{y^{t+1} \in \mathcal{Y}^\infty} \mu_{s-1}(dy^{t+s}|y^{t+1}) \mu_1(dy^{t+1}|y_0^\infty). \end{aligned}$$

Since all function are integrable, we can use the Fubini theorem twice, to swap the order of integrals and then of the sum and of the integral on $y^{t+1} \in \mathcal{Y}^\infty$. We obtain:

$$\begin{aligned} V(y^t) &= U(y^t) + \\ &+ \beta \int_{y^{t+1} \in \mathcal{Y}^\infty} \left\{ \sum_{s=1}^{\infty} \int_{y^{t+s} \in \mathcal{Y}^\infty} \beta^{s-1} U(y^{t+s}) \mu_{s-1}(dy^{t+s}|y^{t+1}) \right\} \mu_1(dy^{t+1}|y_0^\infty). \end{aligned} \quad (58)$$

The term between curly braces is $V(y^{t+1})$ using (11) applied to $t+1$. The previous equality becomes:

$$V(y^t) = U(y^t) + \beta \int_{y^{t+1} \in \mathcal{Y}^\infty} V(y^{t+1}) \mu_1(dy^{t+1}|y_0^\infty).$$

which gives the representation (12) using the notation $\mathbb{E}_{y^{t+1}} [V(y^{t+1})|y^t] = \int_{y^{t+1} \in \mathcal{Y}^\infty} V(y^{t+1}) \mu_1(dy^{t+1}|y^t)$ for the conditional expectation.

A.4 Proof of Proposition 1

Using the definition (14), the expression (15) of the SWF becomes:

$$SWF = \int_{y^t \in \mathcal{Y}^\infty} \omega_P(y^t) \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \hat{V}(y^t, \tilde{y}^t) \mu(d\tilde{y}^t) \mu(dy^t),$$

which can be further simplified using the expression (13) of $\hat{V}$:

$$\begin{aligned} SWF &= \int_{y^t \in \mathcal{Y}^\infty} \omega_P(y^t) \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \sum_{s=0}^{\infty} \int_{\hat{y}^{t+s} \in \mathcal{Y}^\infty} \beta^s \hat{\omega}(y^t, \hat{y}^{t+s}) U(\hat{y}^{t+s}) \mu_s(d\hat{y}^{t+s}|\tilde{y}^t) \mu(d\tilde{y}^t) \mu(dy^t). \\ &= \int_{y^t \in \mathcal{Y}^\infty} \omega_P(y^t) \sum_{s=0}^{\infty} \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \int_{\hat{y}^{t+s} \in \mathcal{Y}^\infty} \beta^s \hat{\omega}(y^t, \hat{y}^{t+s}) U(\hat{y}^{t+s}) \mu_s(d\hat{y}^{t+s}|\tilde{y}^t) \mu(d\tilde{y}^t) \mu(dy^t). \end{aligned}$$

Since all functions under consideration are positive and measurable and since it is assumed that $SWF < \infty$, we can use the Fubini theorem to permute the order of integrals and obtain:

$$\begin{aligned} SWF &= \sum_{s=0}^{\infty} \int_{\hat{y}^{t+s} \in \mathcal{Y}^\infty} \beta^s \left[\int_{y^t \in \mathcal{Y}^\infty} \omega_P(y^t) \hat{\omega}(y^t, \hat{y}^{t+s}) \mu(dy^t) \right] U(\hat{y}^{t+s}) \\ &\quad \times \left[\int_{\tilde{y}^t \in \mathcal{Y}^\infty} \mu_s(d\hat{y}^{t+s}|\tilde{y}^t) \mu(d\tilde{y}^t) \right]. \end{aligned}$$

A straightforward extension of Lemma 4 for μ_s (for any $s \geq 1$) yields: $\int_{\tilde{y}^t \in \mathcal{Y}^\infty} \mu_s(d\hat{y}^{t+s}|\tilde{y}^t) \mu(d\tilde{y}^t) = \mu(d\hat{y}^{t+s})$. Using the definition (17) of the weights ω , we deduce:

$$SWF = \sum_{s=0}^{\infty} \int_{\hat{y}^{t+s} \in \mathcal{Y}^\infty} \beta^s \omega(\hat{y}^{t+s}) U(\hat{y}^{t+s}) \mu(d\hat{y}^{t+s}),$$

which proves Proposition 1.

As a final remark, observe that by splitting the sum over s into $s = 0$ and a sum for $s \geq 1$, we also obtain the following expression for SWF :

$$SWF = \int_{y^t \in \mathcal{Y}^\infty} \omega(y^t) U(y^t, A) \mu(dy^t) + \beta SWF.$$

A.5 Proof of Proposition 2

We recall the expression of the SWF when the allocation is explicit:

$$SWF(A) = \sum_{t=0}^{\infty} \beta^t \int_{y^t \in \mathcal{Y}^\infty} \omega(y^t) U(y^t, A) \mu(dy^t).$$

Let assume that the weights are non-negative and consider two allocations A and A' such that A element-wise dominates A' . We thus have $U(y^t, A) \geq U(y^t, A')$ for all y^t . The non-negativity of weights implies that: $\beta^t \omega(y^t) U(y^t, A) \geq \beta^t \omega(y^t) U(y^t, A')$ for all y^t , which after integration and sum yields $SWF(A) \geq SWF(A')$.

Let us assume that $SWF(A) \geq SWF(A')$ for any pair of allocations A and A' such that A element-wise dominates A' . Let us assume that the weights are strictly negative on a subset $\mathcal{X} \subset \mathcal{Y}^\infty$ of positive measure. We consider an allocation A' . We construct the allocation A such that A and A' coincide on $\mathcal{Y}^\infty \setminus \mathcal{X}$ and A strictly dominates A' on $\mathcal{X}$. We thus have $U(y^t, A) = U(y^t, A')$ for all $y^t \in \mathcal{Y}^\infty \setminus \mathcal{X}$ and $U(y^t, A) > U(y^t, A')$ for all $y^t \in \mathcal{X}$: A element-wise dominates A' . We thus deduce that:

$$\begin{aligned} \int_{y^t \in \mathcal{Y}^\infty} \omega(y^t)(U(y^t, A) - U(y^t, A'))\mu(dy^t) &= \int_{y^t \in \mathcal{Y}^\infty \setminus \mathcal{X}} \omega(y^t)(U(y^t, A) - U(y^t, A'))\mu(y^t) \\ &\quad + \int_{y^t \in \mathcal{X}} \omega(y^t)(U(y^t, A) - U(y^t, A'))\mu(y^t), \\ &= \int_{y^t \in \mathcal{X}} \omega(y^t)(U(y^t, A) - U(y^t, A'))\mu(y^t), \\ &< 0, \end{aligned}$$

where the first equality is a split of the integral over two disjoint sets, the second comes from $U(y^t, A) = U(y^t, A')$ on $\mathcal{Y}^\infty \setminus \mathcal{X}$, and the third from $U(y^t, A) > U(y^t, A')$ and $\omega(y^t) < 0$ on $\mathcal{X}$.

Summing the previous discounted inequality implies $SWF(A) < SWF(A')$, which is a contradiction. We must thus have positive weights. This concludes the proof.

A.6 Proof of equation (20) in Definition 3

The program in Definition 3 is:

$$\begin{aligned} (\tilde{\omega}_{\tilde{y}y})_{\tilde{y},y} = \operatorname{argmin}_{(\hat{\omega}_{\tilde{y}y})_{\tilde{y},y}} \quad & \sum_{(y,\tilde{y}) \in \mathcal{Y}^{\infty 2}} \pi_{\tilde{y}} \left(\hat{\omega}_{\tilde{y}y} - \frac{1_{y=\tilde{y}}}{\omega_{P,y}\pi_y} \right)^2, \\ \text{s.t. } \omega_y = \quad & \sum_{\tilde{y} \in \mathcal{Y}^\infty} \pi_{\tilde{y}} \omega_{P,\tilde{y}} \hat{\omega}_{\tilde{y}y} \quad (y \in \mathcal{Y}). \end{aligned} \tag{59}$$

We denote by $2\lambda_y$ the Lagrange multiplier to the constraint of equation (59) for $y \in \mathcal{Y}$. We obtain the following Lagrangian:

$$\mathcal{L} = \frac{1}{2} \sum_{(y,\tilde{y}) \in \mathcal{Y}^{\infty 2}} \pi_{\tilde{y}} \left(\hat{\omega}_{\tilde{y}y} - \frac{1_{y=\tilde{y}}}{\omega_{P,y}\pi_y} \right)^2 - \sum_{y \in \mathcal{Y}} \lambda_y \left(\sum_{\tilde{y} \in \mathcal{Y}^\infty} \pi_{\tilde{y}} \omega_{P,\tilde{y}} \hat{\omega}_{\tilde{y}y} - \omega_y \right).$$

Computing the derivative with respect to $\hat{\omega}_{\tilde{y}y}$ yields the following FOC:

$$\hat{\omega}_{\tilde{y}y} = \frac{1_{y=\tilde{y}}}{\omega_{P,y}\pi_y} + \lambda_y \omega_{P,\tilde{y}}.$$

Using the constraint of equation (59), we deduce:

$$\lambda_y = \frac{\omega_{P,\tilde{y}}}{\sum_{\tilde{y} \in \mathcal{Y}^\infty} \pi_{\tilde{y}} (\omega_{P,\tilde{y}})^2} (\omega_y - 1),$$

which finally implies equation (A.6).

B Competitive equilibrium

We provide a formal definition of a competitive equilibrium.

Definition 4 (Competitive equilibrium) *A sequential competitive equilibrium is a collection of individual allocations $(c_{i,t}, l_{i,t}, a_{i,t}, \nu_{i,t})_{t \geq 0, i \in \mathcal{I}}$, of aggregate quantities $(K_t, L_t, Y_t)_{t \geq 0}$, of price processes $(w_t, r_t, \tilde{w}_t, \tilde{r}_t)_{t \geq 0}$, and of fiscal policies $(\tau_t^c, \tau_t^K, \tau_t, \kappa_t, B_t)_{t \geq 0}$, such that, for initial conditions and initial values of capital stock and public debt verifying $K_{-1} + B_{-1} = \int_i a_{i,-1} \ell(di)$, we have:*

1. *given prices, the functions $(c_{i,t}, l_{i,t}, a_{i,t}, \nu_{i,t})_{t \geq 0, i \in \mathcal{I}}$ solve the agent's optimization program in equations (28)–(30);*
2. *financial, labor, and goods markets clear at all dates: for any $t \geq 0$, equation (33) holds;*
3. *the government budget is balanced at all dates: equation (26) holds for all $t \geq 0$;*
4. *factor prices $(w_t, r_t, \tilde{w}_t, \tilde{r}_t)_{t \geq 0}$ are consistent with condition (22) and post-tax definitions (25).*

A steady-state competitive equilibrium is a competitive equilibrium for which the joint distribution of agents' decisions (c, l, a, ν) , aggregate quantities K, L, Y , prices $w, r, \tilde{w}, \tilde{r}$, and fiscal policy $(\tau^c, \tau^K, \tau, \kappa, B)$ are time-invariant.

C The Ramsey program 3

C.1

C.2 Reformulating the Ramsey program

We now provide a reformulation of the Ramsey problem, using the change of variables:

$$\tilde{a}_{i,t} := \frac{a_{i,t}}{1 + \tau_t^c}, \quad (60)$$

$$W_t := \frac{w_t}{1 + \tau_t^c}, \quad (61)$$

$$R_t := \frac{(1 + r_t)(1 + \tau_{t-1}^c)}{1 + \tau_t^c}, \quad (62)$$

which represents the asset choices in (60), the wage rate in (61), and the interest rate in (62). With this notation, the agent's budget and credit constraints become:

$$c_{i,t} + \tilde{a}_{i,t} = W_t(y_{i,t} l_{i,t})^{1-\tau_t} + R_t \tilde{a}_{i,t-1}, \quad (63)$$

$$\tilde{a}_{i,t} \geq -\frac{\bar{a}}{1 + \tau_t^c} := -\tilde{\bar{a}}. \quad (64)$$

Since taxes and prices are considered as given by agents, we can equivalently state their optimization program using the notation (60)–(62) and the constraints (63) and (64), rather than

the original notation and the constraints (29) and (30). This modifies Euler equations (31)–(32) as follows:

$$\begin{aligned} u'(c_{i,t}) &= \beta \mathbb{E}_t \left[R_{t+1} u'(c_{i,t+1}) \right] + \nu_{i,t}, \\ v'(l_{i,t}) &= (1 - \tau_t) W_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}). \end{aligned}$$

We now turn to the governmental budget constraint. We further define:

$$\tilde{B}_t := \frac{B_t}{(1 + \tau_t^c)}, \quad (65)$$

$$\tilde{A}_t := \frac{A_t}{1 + \tau_t^c}. \quad (66)$$

and

$$\hat{B}_t := (1 + \tau_t^c) \tilde{B}_t - \tau_t^c \tilde{A}_t, \quad (67)$$

With these new definitions, the financial market equilibrium given by (41) holds, as we have $\tilde{A}_t = \int_i \tilde{a}(i) l(di)$.

Using the government budget constraint defined in (26), we have:

$$G_t + (1 + r_t) B_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + r_t K_{t-1} = \tau_t^c C_t + F(K_{t-1}, L_t, s_t) + B_t.$$

Now replace (??) into the equation above and obtain:

$$\begin{aligned} G_t + (1 + r_t) B_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + r_t K_{t-1} = \\ \tau_t^c (F(K_{t-1}, L_t, s_t) - G_t - (K_t - K_{t-1})) + F(K_{t-1}, L_t, s_t) + B_t. \end{aligned}$$

Divide both sides of the equation above by $(1 + \tau_t^c)$ and obtain:

$$\begin{aligned} G_t + \frac{(1 + r_t)}{1 + \tau_t^c} B_{t-1} + \frac{w_t}{1 + \tau_t^c} \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + \frac{r_t}{1 + \tau_t^c} K_{t-1} = \\ - \frac{\tau_t^c}{1 + \tau_t^c} (K_t - K_{t-1}) + F(K_{t-1}, L_t, s_t) + \frac{B_t}{1 + \tau_t^c}. \end{aligned}$$

Using the definitions (61), (62), and (65):

$$G_t + R_t \tilde{B}_{t-1} + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + \frac{r_t}{1 + \tau_t^c} K_{t-1} = - \frac{\tau_t^c}{1 + \tau_t^c} (K_t - K_{t-1}) + F(K_{t-1}, L_t, s_t) + \tilde{B}_t.$$

Now using definitions in (65) and (66) we have $K_{t-1} = A_{t-1} - B_{t-1} = (1 + \tau_{t-1}^c)(\tilde{A}_{t-1} - \tilde{B}_{t-1})$ and:

$$\begin{aligned} G_t + R_t \tilde{B}_{t-1} + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + \frac{r_t(1 + \tau_{t-1}^c)}{1 + \tau_t^c} (\tilde{A}_{t-1} - \tilde{B}_{t-1}) = \\ \frac{\tau_t^c(1 + \tau_{t-1}^c)}{1 + \tau_t^c} (\tilde{A}_{t-1} - \tilde{B}_{t-1}) + F(K_{t-1}, L_t, s_t) - \tau_t^c (\tilde{A}_t - \tilde{B}_t) + \tilde{B}_t. \end{aligned}$$

Observe $\frac{r_t(1 + \tau_{t-1}^c)}{1 + \tau_t^c} = R_t - \frac{1 + \tau_{t-1}^c}{1 + \tau_t^c}$ and that $-\tau_t^c (\tilde{A}_t - \tilde{B}_t) + \tilde{B}_t = \hat{B}_t$ given by equation (67),

which leads us:

$$G_t + R_t \tilde{B}_{t-1} + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + \left(R_t - \frac{1 + \tau_{t-1}^c}{1 + \tau_t^c} \right) (\tilde{A}_{t-1} - \tilde{B}_{t-1}) = \frac{\tau_t^c (1 + \tau_{t-1}^c)}{1 + \tau_t^c} (\tilde{A}_{t-1} - \tilde{B}_{t-1}) + F(K_{t-1}, L_t, s_t) + \hat{B}_t.$$

Hence, we have:

$$G_t + R_t \tilde{B}_{t-1} + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + (R_t - (1 + \tau_{t-1}^c)) (\tilde{A}_{t-1} - \tilde{B}_{t-1}) = F(K_{t-1}, L_t, s_t) + \hat{B}_t.$$

Finally, using (67) in period $t - 1$ (i.e., $\hat{B}_{t-1} = (1 + \tau_{t-1}^c) \tilde{B}_{t-1} - \tau_{t-1}^c \tilde{A}_{t-1}$) we get:

$$G_t + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + (R_t - 1) \tilde{A}_{t-1} + \hat{B}_{t-1} = F(K_{t-1}, L_t, s_t) + \hat{B}_t.$$

Since the public debt can be freely chosen by the planner, it is equivalent for the planner to choose $\hat{B}_t$ rather than B_t .

The reformulated Ramsey program. We reformulate the the Ramsey program (68)–(74) using the new variables $\tilde{a}_{i,t}$, W_t , R_t , $\tilde{A}_t$, $\hat{B}_t$ introduced in (60)–(62) and (66)–(67). The program can be expressed in post-tax prices R_t and W_t – taxes and pre-tax factor prices can be deduced from the allocation and the post-tax price definitions. The following proposition summarizes the reformulation of the Ramsey program.

Proposition 3 *The Ramsey program (68)–(74) can be rewritten as:*

$$\max_{(W_t, R_t, \tau_t, \hat{B}_t, \tilde{A}_t, K_t, L_t, (c_{i,t}, l_{i,t}, \tilde{a}_{i,t}, \nu_{i,t})_i)_{t \geq 0}} SWF_0, \quad (68)$$

$$G + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + (R_t - 1) \tilde{A}_{t-1} + \hat{B}_{t-1} = F(K_{t-1}, L_t, s_t) + \hat{B}_t, \quad (69)$$

$$\text{for all } i \in \mathcal{I}: c_{i,t} + \tilde{a}_{i,t} = W_t (y_{i,t} l_{i,t})^{1-\tau_t} + R_t \tilde{a}_{i,t-1}, \quad (70)$$

$$\tilde{a}_{i,t} \geq -\tilde{a}, \quad \nu_{i,t} (\tilde{a}_{i,t} + \tilde{a}) = 0, \quad \nu_{i,t} \geq 0, \quad (71)$$

$$u'(c_{i,t}) = \beta \mathbb{E}_t \left[R_{t+1} u'(c_{i,t+1}) \right] + \nu_{i,t}, \quad (72)$$

$$v'(l_{i,t}) = (1 - \tau_t) W_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}), \quad (73)$$

$$K_t + \hat{B}_t = \tilde{A}_t = \int_i \tilde{a}_{i,t} \ell(di), \quad L_t = \int_i y_{i,t} l_{i,t} \ell(di). \quad (74)$$

C.3 Lagrangian and the FOCs of the Ramsey program

The Lagrangian associated to the Ramsey program (68)–(74) can be written as:

$$\begin{aligned}\mathcal{L} = & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int_i \omega_{i,t} (u(c_{i,t}) - v(l_{i,t})) \ell(di) \\ & - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int_i (\lambda_{c,i,t} - R_t \lambda_{c,i,t-1}) u'(c_{i,t}) \ell(di) \\ & - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int_i \lambda_{l,i,t} (v'(l_{i,t}) - (1 - \tau_t) W_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t})) \ell(di) \\ & - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \mu_t \left(G_t + (1 - \delta) \hat{B}_{t-1} + (R_t - 1 + \delta) \int_i \tilde{a}_{i,t-1} \ell(di) + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) - Y_t - \hat{B}_t \right).\end{aligned}$$

where the instruments are: $\tilde{a}_{i,t}$, $l_{i,t}$, W_t , R_t , τ_t , and $\hat{B}_t$, and we have:

$$c_{i,t} = -\tilde{a}_{i,t} + R_t \tilde{a}_{i,t-1} + W_t (y_{i,t} l_{i,t})^{1-\tau_t}, \quad (75)$$

$$Y_t = \left(\int_i \tilde{a}_{i,t-1} \ell(di) - \hat{B}_{t-1} \right)^\alpha \left(\int_i y_{i,t} l_{i,t} \ell(di) \right)^{1-\alpha}. \quad (76)$$

The Lagrangian can be written as

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t (\mathcal{W}_t + \mu_t \mathcal{R}_t)$$

with

$$\begin{aligned}\mathcal{W}_t := & \int_i (\omega_{i,t} (u(c_{i,t}) - v(l_{i,t})) - (\lambda_{c,i,t} - R_t \lambda_{c,i,t-1}) u'(c_{i,t}) \\ & - \lambda_{l,i,t} (v'(l_{i,t}) - (1 - \tau_t) W_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}))) \ell(di) \\ \mathcal{R}_t := & G_t + (1 - \delta) \hat{B}_{t-1} + (R_t - 1 + \delta) \int_i \tilde{a}_{i,t-1} \ell(di) + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) - Y_t - \hat{B}_t\end{aligned}$$

The value $\mathcal{W}_t$ aggregates all individual variables, whereas $\mathcal{R}_t$ is the aggregated resource constraint of the government.

FOC with respect to public debt $\hat{B}_t$.

$$\mu_t = \beta \mathbb{E}_t [(1 + \tilde{r}_{t+1}) \mu_{t+1}]. \quad (77)$$

FOC with respect to savings choices $\tilde{a}_{i,t}$. We define the marginal social value of liquidity for agent i at date t as:

$$\psi_{i,t} := \omega_{i,t} u'(c_{i,t}) - \left(\lambda_{c,i,t} - R_t \lambda_{c,i,t-1} - \lambda_{l,i,t} (1 - \tau_t) W_t (y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t} \right) u''(c_{i,t}), \quad (78)$$

and $\hat{\psi}_{i,t} = \psi_{i,t} - \mu_t$ as the marginal social value of liquidity net of the cost for planner's resources. We obtain using (77):

$$\hat{\psi}_{i,t} = \beta \mathbb{E}_t [R_{t+1} \hat{\psi}_{i,t+1}]. \quad (79)$$

FOC with respect to labor supply $l_{i,t}$. We define:

$$\psi_{l,i,t} := \omega_{i,t} v'(l_{i,t}) + \lambda_{l,i,t} v''(l_{i,t}).$$

The FOC with respect to labor supply $l_{i,t}$ is:

$$\psi_{l,i,t} = (1 - \tau_t) W_t(y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t} \hat{\psi}_{i,t} + \mu_t F_{L,t} y_{i,t} - (1 - \tau_t) W_t(y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t} \lambda_{l,i,t} \tau_t \frac{u'(c_{i,t})}{l_{i,t}}.$$

FOC with respect to the wage rate W_t .

$$0 = \int_j (y_{j,t} l_{j,t})^{1-\tau_t} \left(\hat{\psi}_{j,t} + \lambda_{l,j,t} (1 - \tau_t) u'(c_{j,t}) / l_{j,t} \right) \ell(dj).$$

FOC with respect to the interest rate R_t .

$$0 = \int_j (\hat{\psi}_{j,t} \tilde{a}_{t-1}^j + \lambda_{c,j,t-1} u'(c_{j,t})) \ell(dj).$$

FOC with respect to progressivity τ_t .

$$\begin{aligned} 0 &= \int_j (y_{j,t} l_{j,t})^{1-\tau_t} (\hat{\psi}_{j,t} + \lambda_{l,j,t} (1 - \tau_t) (u'(c_{j,t}) / l_{j,t})) \ln(y_{j,t} l_{j,t}) \ell(dj) \\ &\quad + \int_j \lambda_{l,j,t} (y_{j,t} l_{j,t})^{1-\tau_t} (u'(c_{j,t}) / l_{j,t}) \ell(dj). \end{aligned}$$

D Truncating the model and identification of Pareto Weights

The Ramsey problem discussed in Section 5.3 involves a joint distribution across wealth and Lagrange multipliers. This high-dimensional object raises a number of difficulties for the program resolution. For instance, this joint distribution influences the planner's instruments in a non-obvious way, which renders methods based on perturbation of a well-identified steady-state not usable for solving such problems (as Reiter, 2009, Boppart et al., 2018, Bayer et al., 2019 or Auclert et al., 2021).

The solution we provide here builds on LeGrand and Ragot (2022). It allows us to compute the steady-state allocation and to derive a finite number of equations that can simulate by perturbation the dynamics of the Ramsey program under small aggregate shocks. The intuition of the method can be summarized as follows. We construct an aggregation of the Bewley model (thus for a given fiscal policy and no aggregate shock) where agents with the same history over the last N periods (where N is a fixed horizon) are aggregated into a single “agent”. The method implies that the “aggregate” agent is endowed with the average wealth of all individual agents with this N -period history. The wealth heterogeneity among these individual agents is captured in our aggregate model through additional parameters – that will be referred as “ ξ s”.²⁴

The aggregation method yields a so-called truncated model, which, thanks to the “ ξ s”, is an exact aggregation of the underlying Bewley model in the absence of aggregate shocks. In the

²⁴Werning (2015) also proposes an aggregation method where the heterogeneity is captured by a change in the discount factor, while our method involves the introduction of explicit correction factors, denoted as ξ s. Using the ξ s enables us to solve for the Ramsey program, as all agents are endowed with the same discount factor.

presence of aggregate shocks, we can simulate the truncated model using standard perturbation techniques. This truncated model also enables us to solve for the Ramsey program.

D.1 The truncated model

Let $N \geq 0$ be a truncation length. The key step of the aggregation consists of assigning the same wealth and allocation to all agents sharing the same idiosyncratic history over the last $N \geq 0$ periods. Such an N -period history will be referred to as a truncated history. For a history $y^t = \{y_0, \dots, y_{t-N}, y_{t-N+1}, \dots, y_{t-1}, y_t\}$, this corresponds to the N -length vector denoted $y^N = \{y_{-N+1}^N, \dots, y_{-1}^N, y_0^N\}$. To sum up, we can represent the truncated history of an agent i whose idiosyncratic history is y^t as:

$$\begin{aligned} y^t &= \underbrace{\{y_0, \dots, y_{t-N-2}, y_{t-N-1}, y_{t-N}\}}_{\xi_{y^N}} \underbrace{\{y_{t-N+1}, \dots, y_{t-1}, y_t\}}_{=y^N} \\ &:= \underbrace{\{\dots, y_{t-N-2}^t, y_{t-N-1}^t, y_{t-N}^t\}}_{\xi_{y^N}} \underbrace{\{y_{t-N+1}^t, \dots, y_{t-1}^t, y_t^t\}}_{=y^N}, \end{aligned}$$

where the parameter ξ_{y^N} captures the residual heterogeneity for the truncated history y^N , and y_{-k}^t represents the idiosyncratic variable (at date t) k periods in the past. The method to compute this parameter will be discussed further below. In what follows, we will discuss the various elements needed to apply the aggregation procedure.

Firstly, we need to compute the measure of agents with the same history y^N . An agent with history $\tilde{y}^N$ at $t-1$ will have a different truncated history in period t depending on the realization of the idiosyncratic variable at date t . The probability to transit from truncated history $\tilde{y}^N$ to truncated history y^N will be denoted by $\Pi_{t, \tilde{y}^N y^N}$ (with $\sum_{y^N} \Pi_{t, \tilde{y}^N y^N} = 1$) and can be computed from the transition probabilities for the productivity process as:

$$\Pi_{t, \tilde{y}^N y^N} = 1_{y^N \succeq \tilde{y}^N} \Pi_{t, \tilde{y}_0^N y_0^N} \geq 0.$$

With those elements, we can now compute the share of agents with truncated history y^N as S_{t, y^N} . This element will be

$$S_{t, y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^{\infty N}} S_{t-1, \tilde{y}^N} \Pi_{t, \tilde{y}^N y^N}, \quad (80)$$

where the initial shares $(S_{-1, y^N})_{y^N \in \mathcal{Y}^{\infty N}}$ are given with $\sum_{y^N \in \mathcal{Y}^{\infty N}} S_{-1, y^N} = 1$.

The model aggregation then assigns to each truncated history the average choices (whether for consumption, savings, or labor supply) of the group of agents sharing the same truncated history. Let's consider a generic variable denoted by $X_t(y^t, s^t)$, and denote by X_{t, y^N} the average quantity of X assigned to truncated history y^N . Formally:

$$X_{t, y^N} = \frac{1}{S_{t, y^N}} \sum_{y^t \in \mathcal{Y}^{\infty t+1} | (y_{-N+1}^t, \dots, y_{-1}^t, y_0^t) = y^N} X_t(y^t, s^t) \theta_t(y^t), \quad (81)$$

where we remind that $\theta_t(y^t)$ is the measure of agents with history y^t . Definition (81) can be applied to consumption, savings, labor supply, and credit-constraint Lagrange multiplier. This

leads to the quantities c_{t,y^N} , $\tilde{a}_{t,y^N}$, l_{t,y^N} , and ν_{t,y^N} , respectively. Note that applying (81) to beginning-of-period wealth involves accounting for the fact that agents with truncated history y^N at t may have come from various truncated history at $t-1$. Specifically, this variable consists of the wealth of all agents having history y^N in period t and any other possible history in $t-1$. Formally, the beginning-of-period wealth $\tilde{a}_{t,y^N}$ for truncated history y^N is:

$$\tilde{a}_{t,y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^{\infty N}} \frac{S_{t-1,\tilde{y}^N}}{S_{t,y^N}} \Pi_{t,\tilde{y}^N y^N} \tilde{a}_{t-1,\tilde{y}^N}. \quad (82)$$

Now, in what follows, we will define the various “ ξ s”. Observe:

$$\sum_{y^t \in \mathcal{Y}^{\infty t+1} | (y_{-N+1}^t, \dots, y_{-1}^t, y_0^t) = y^N} u(c_t(y^t)) = \xi_{y^N}^{u,0} u\left(\sum_{y^t \in \mathcal{Y}^{\infty t+1} | (y_{-N+1}^t, \dots, y_{-1}^t, y_0^t) = y^N} c_t(y^t)\right),$$

which can be written compactly as:

$$\sum_{y^N \in \mathcal{Y}^{\infty N}} u(c_{i,t}) = \xi_{y^N}^{u,0} u(c_{t,y^N}). \quad (83)$$

The same procedure applied to the other variables for the Ramsey problem (68)–(74) yields:

$$\sum_{y^N \in \mathcal{Y}^{\infty N}} v(l_{i,t}) := \xi_{y^N}^{v,0} v(l_{t,y^N}), \quad (84)$$

$$\sum_{y^N \in \mathcal{Y}^{\infty N}} u'(c_{i,t}) := \xi_{y^N}^{u,1} u'(c_{t,y^N}), \quad (85)$$

$$\sum_{y^N \in \mathcal{Y}^{\infty N}} (y_{i,t} l_{i,t})^{1-\tau_t} := \xi_{y^N}^y (y_0^N l_{t,y^N})^{1-\tau_t}. \quad (86)$$

We can now proceed with the aggregation of the full-fledged model. First, the aggregation of individual budget constraints (63) yields the following equation:

$$c_{t,y^N} + \tilde{a}_{t,y^N} = W_t \xi_{y^N}^y (l_{t,y^N} y_0^N)^{1-\tau_t} + R_t \tilde{a}_{t,y^N}, \text{ for } y^N \in \mathcal{Y}^{\infty N}. \quad (87)$$

The aggregation of Euler equations for consumption (72) and labor (73) yields:

$$\xi_{y^N}^{u,E} u'(c_{t,y^N}) = \beta \mathbb{E}_t \left[R_{t+1} \sum_{\tilde{y}^N \in \mathcal{Y}^{\infty N}} \Pi_{t+1,y^N \tilde{y}^N} \xi_{\tilde{y}^N}^{u,E} u'(c_{t+1,\tilde{y}^N}) \right] + \nu_{t,y^N}, \quad (88)$$

$$\xi_{y^N}^{v,1} v'(l_{t,y^N}) := (1 - \tau_t) W_t \xi_{y^N}^y (l_{t,y^N} y_0^N)^{1-\tau_t} \xi_{y^N}^{u,1} (u'(c_{t,y^N})/l_{t,y^N}), \quad (89)$$

where the coefficients $(\xi_{y^N}^{u,E})_{y^N}$ for the consumption Euler equations ensure that the aggregate Euler equations yield Euler equations with aggregate consumption levels. In other words, the $(\xi_{y^N}^{u,E})_{y^N}$ are determined such that the aggregated consumption levels (for each history) satisfy the steady state consumption Euler equation (88). These coefficients are necessary because Euler equations involve non-linear marginal utilities. The same idea for the coefficients $(\xi_{y^N}^{v,1})_{y^N}$ for the Euler equation for labor.

Finally, market clearing conditions can be expressed as:

$$K_t + \hat{B}_t = \sum_{y^N \in \mathcal{Y}^{\infty N}} S_{t,y^N} \tilde{a}_{t,y^N}, \quad L_t = \sum_{y^N \in \mathcal{Y}^{\infty N}} S_{t,y^N} y_{y^N} l_{t,y^N}. \quad (90)$$

Equations (87)–(90) exactly characterize the dynamics of the aggregated variables c_{t,y^N} , $\tilde{a}_{t,y^N}$, l_{t,y^N} , and ν_{t,y^N} , as well as aggregate quantities K_t , $\hat{B}_t$, and L_t . In the Appendix D.3, we have the derivation of the program formulation in the Projected model as well as the equations that characterize the social weights.

Steady-state and computation of the ξ s. Steady-state allocations allow us to compute the parameters ξ s as follows. We compute policy functions and wealth distribution of the Bewley model, as well as identify credit-constrained histories. Aggregation equations (81) and (82) can then be used to aggregate (steady-state) allocations c_{y^N} , $\tilde{a}_{y^N}$, l_{y^N} , and ν_{y^N} . We then invert the consumption Euler equations (88) to deduce the preference parameters $(\xi_{y^N}^{u,E})_{y^N}$. The other ξ s are computed explicitly by equations (83), (84), (85), (86), and (89).

The truncated model in the presence of aggregate shocks. We state two further assumptions that enable us to use our truncation method in the presence of aggregate shocks, resulting in the so-called truncated model.

Assumption B *We make the following two assumptions.*

1. *The preference parameters $(\xi_{y^N})_{y^N}$ remain constant and equal to their steady-state values.*
2. *The set of credit-constrained histories, denoted by $\mathcal{C} \subset \mathcal{Y}^{\infty N}$, is time-invariant.*

Two properties are finally worth mentioning. First, a straightforward consequence of the construction of the ξ s is that the steady-state allocations of the initial and truncated models are identical. Second, as the truncation length N becomes increasingly long, truncated allocations (in the presence of aggregate shocks) can be shown to converge to those of the full-fledged equilibrium. Section 6 shows that from a quantitative standpoint, the ξ s efficiently capture the heterogeneity within truncated histories, even when the truncation length remains limited.

D.2 Ramsey program

Program formulation. The finite state-space representation of the truncated model allows us to solve for the Ramsey program in the presence of aggregate shocks.²⁵ Let $(\omega_y)_{y \in \mathcal{Y}^\infty}$ denote the period weights associated with each productivity level. The Ramsey program in the truncated economy can be written as follows:

$$\max_{(W_t, R_t, \tilde{w}_t, \tilde{r}_t, \tau_t^C, \tau_t^K, \tau_t, \kappa_t, \hat{B}_t, G_t, K_t, L_t, (c_{t,y^N}, l_{t,y^N}, \tilde{a}_{t,y^N}, \nu_{t,y^N})_{y^N})_{t \geq 0}} W_0, \quad (91)$$

where $W_0 := \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \sum_{y^N \in \mathcal{Y}^{\infty N}} S_{t,y^N} \omega_{y^N} (\xi_{y^N}^{u,0} u(c_{t,y^N}) - \xi_{y^N}^{v,0} v(l_{t,y^N}) + u_G(G_t)) \right]$ and subject to aggregate Euler equations (88) and (89), aggregate budget constraint (87), aggregate market clearing conditions (90), credit constraints $\tilde{a}_{t,y^N} \geq -\tilde{a}$, as well as the governmental budget constraint (69), which is already present in the full-fledged Ramsey program.

²⁵Our method involves deriving the first-order conditions of the truncated model, rather than truncating the first-order conditions of the full-fledged Ramsey model. This ensures numerical stability, as the truncated model is "well-defined" for the fiscal policy under consideration by construction.

First-order conditions. We define the net social value of liquidity of history y^N as in (78):

$$\begin{aligned} \hat{\psi}_{t,y^N} &= \omega_{y^N} \xi_{y^N}^{u,0} u'(c_{t,y^N}) - \mu_t \\ &\quad - \left(\lambda_{c,t,y^N} \xi_{y^N}^{u,E} - R_t \tilde{\lambda}_{c,t,y^N} \xi_{y^N}^{u,E} - \lambda_{l,t,y^N} \xi_{y^N}^y (1 - \tau_t) W_t (y_0^N)^{1-\tau_t} l_{t,y^N}^{-\tau_t} \xi_{y^N}^{u,1} \right) u''(c_{t,y^N}). \end{aligned} \quad (92)$$

FOC with respect to $\tilde{a}_{t,y^N}$:

$$\hat{\psi}_{t,y^N} = \beta \mathbb{E}_t \left[R_{t+1} \sum_{\tilde{y}^N \in \mathcal{Y}^{\infty N}} \Pi_{t,y^N \tilde{y}^N} \hat{\psi}_{t+1,\tilde{y}^N} \right] \text{ if } \nu_{y^N} = 0 \text{ and } \lambda_{c,t,y^N} = 0 \text{ otherwise.} \quad (93)$$

FOC with respect to l_{t,y^N} :

$$\begin{aligned} \frac{\omega_{y^N} \xi_{y^N}^{v,0} v'(l_{t,y^N}) + \lambda_{l,t,y^N} \xi_{y^N}^{v,1} v''(l_{t,y^N})}{(1 - \tau_t) W_t \xi_{y^N}^y (y_0^N)^{1-\tau_t} l_{t,y^N}^{-\tau_t}} &= \hat{\psi}_{t,y^N} - \lambda_{l,t,y^N} \tau_t \xi_{y^N}^{u,1} (u'(c_{t,y^N})/l_{t,y^N}) \\ &\quad + \mu_t (1 - \alpha) \frac{Y_t}{(1 - \tau_t) W_t \xi_{y^N}^y (y_0^N)^{-\tau_t} l_{t,y^N}^{-\tau_t} L_t}. \end{aligned} \quad (94)$$

FOC with respect to W_t :

$$\sum_{y^N \in \mathcal{Y}^{\infty N}} S_{t,y^N} \xi_{y^N}^y (l_{t,y^N} y_{y^N})^{1-\tau_t} \left(\hat{\psi}_{t,y^N} + \lambda_{l,t,y^N} (1 - \tau_t) \xi_{y^N}^{u,1} (u'(c_{t,y^N})/l_{t,y^N}) \right) = 0. \quad (95)$$

FOC with respect to R_t :

$$\sum_{y^N \in \mathcal{Y}^{\infty N}} S_{t,y^N} \left(\hat{\psi}_{t,y^N} \tilde{a}_{t,y^N} + \tilde{\lambda}_{c,t,y^N} \xi_{y^N}^{u,E} u'(c_{t,y^N}) \right) = 0. \quad (96)$$

FOC with respect to τ_t :

$$\begin{aligned} \sum_{y^N \in \mathcal{Y}^{\infty N}} S_{t,y^N} \left(\hat{\psi}_{t,y^N} + \lambda_{l,t,y^N} (1 - \tau_t) \xi_{y^N}^{u,1} (u'(c_{t,y^N})/l_{t,y^N}) \right) \ln \left(l_{t,y^N} y_{y^N} \right) \xi_{y^N}^y (l_{t,y^N} y_{y^N})^{1-\tau_t} \\ = - \sum_{y^N \in \mathcal{Y}^{\infty N}} S_{t,y^N} \lambda_{l,t,y^N} \xi_{y^N}^y (l_{t,y^N} y_{y^N})^{1-\tau_t} \xi_{y^N}^{u,1} (u'(c_{t,y^N})/l_{t,y^N}). \end{aligned} \quad (97)$$

FOC with respect to $\hat{B}_t$:

$$\mu_t = \beta \mathbb{E} \left[\mu_{t+1} \left(1 + \alpha \frac{Y_{t+1}}{K_t} - \delta \right) \right]. \quad (98)$$

We must furthermore have:

$$\tilde{\lambda}_{c,t,y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^{\infty N}} \frac{S_{t-1,\tilde{y}^N}}{S_{t,y^N}} \Pi_{t,\tilde{y}^N y^N} \lambda_{c,t-1,\tilde{y}^N}, \quad (99)$$

$$\tilde{a}_{t,y^N} \geq 0 \text{ and } \tilde{a}_{t,y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^{\infty N}} \frac{S_{t-1,\tilde{y}^N}}{S_{t,y^N}} \Pi_{t,\tilde{y}^N y^N} \tilde{a}_{t-1,\tilde{y}^N}. \quad (100)$$

D.3 Matrix expression

In this section, we provide closed-form formulas for preference multipliers ξ s (Section D.1) and the weights ω s. We start with some notation:

$\circ$ is the Hadamard product, $\otimes$ is the Kronecker product, $\times$ is the usual matrix product.

For any vector V , we denote by $\text{diag}(V)$ the diagonal matrix with V on the diagonal.

The matrix representation consists in stacking together the equations characterizing the steady-state, so as to provide a convenient matrix notation for solving the steady state. It also provides an efficient solution to compute the values for the coefficients (ξ_{y^N}) and (ω_{y^N}) . The starting point is to observe that a history y^N can be seen as an N -length vector $\{y_{-N+1}^N, \dots, y_0^N\}$ of elements of $\mathcal{Y}^\infty$. The number of histories is $N_{tot} = Y^N$. We can identify each history by an integer $k_{y^N} = 1, \dots, N_{tot}$:

$$k_{y^N} = \sum_{k=0}^{N-1} N_{tot}^{-N+1-k} (y_k - 1) + 1, \quad (101)$$

which corresponds to an enumeration in base Y .

D.3.1 A closed-form formula for the ξ s

Let $\mathbf{S}$ be the N_{tot} -vector of steady-state history sizes, defined as $\mathbf{S} = (S_{k_{y^N}})_{k_{y^N}=1, \dots, N_{tot}}$, by stacking history sizes for all histories using the enumeration given by (101). Similarly, let $\tilde{\mathbf{a}}, \mathbf{c}, \ell, \boldsymbol{\nu}, \mathbf{u}'(\mathbf{c}), \mathbf{v}'(\mathbf{l}), \mathbf{u}''(\mathbf{c}), \mathbf{v}''(\mathbf{l})$ be the N_{tot} -vectors of end-of-period wealth, consumption, labor supply, Lagrange multipliers, marginal utilities, and derivatives of the marginal utility, respectively. These vectors are known from the steady-state equilibrium of the Bewley model. Each element is defined as the truncation of the relevant variable computed using equation (81). We also define:

$$\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_Y \end{bmatrix} \otimes \mathbf{1}_B,$$

where $\mathbf{1}_B$ is a vector of 1s of length Y^{N-1} . We define $\mathbb{P}$ as the diagonal matrix having 1 on the diagonal at y^N if and only if the history y^N is not credit constrained (i.e., $\nu_{y^N} = 0$), and 0 otherwise. We similarly define $\mathbb{P}^c = \mathbf{I} - \mathbb{P}$, where $\mathbf{I}$ is the $(N_{tot} \times N_{tot})$ -identity matrix. Noting Π as the transition matrix across histories, we obtain the following steady-state relationships.

D.3.2 Matrix expressions for the definition of ξ s

Writing $S_{y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^{\infty N}} S_{\tilde{y}^N} \Pi_{\tilde{y}^N y^N}$, (87) and the credit constraint yield, respectively:

$$\mathbf{S} = \Pi \mathbf{S}, \quad (102)$$

$$\mathbf{S} \circ \mathbf{c} + \mathbf{S} \circ \tilde{\mathbf{a}} = R \Pi (\mathbf{S} \circ \tilde{\mathbf{a}}) + W \mathbf{S} \circ \boldsymbol{\xi}^y \circ (\mathbf{y} \circ \mathbf{l})^{1-\tau}, \quad (103)$$

$$(\mathbf{I} - \mathbf{P}) \tilde{\mathbf{a}} = \mathbf{0}_{N_{tot} \times 1}. \quad (104)$$

The Euler equation for consumption in (88) becomes:

$$\xi^{u,E} \circ u'(c) = \beta R \Pi^\top (\xi^{u,E} \circ u'(c)) + \nu,$$

where the matrix $\Pi^\top$ (the transpose of Π) is used to make expectations about next period histories. Or equivalently:

$$D_{u'(c)} \xi^{u,E} = \beta R \Pi^\top D_{u'(c)} \xi^{u,E} + \nu,$$

where D stands for the diagonal matrix. Finally:

$$\xi^{u,E} = \left[(I - \beta R \Pi^\top) D_{u'(c)} \right]^{-1} \nu. \quad (105)$$

From the FOC on labor in (89), we obtain:

$$\xi^{v,1} = (1 - \tau) W(y \circ l)^{1-\tau} \circ \xi^y \circ \xi^{u,1} \circ u'(c) / (l \circ v'(l)). \quad (106)$$

Now, from the aggregation equations in (83)–(86), we get:

$$\xi^{u,0} = \frac{\sum_{y^N \in \mathcal{Y}^{\infty N}} u(c_{i,t})}{u(c_{t,y^N})}, \quad \xi^{u,1} = \frac{\sum_{y^N \in \mathcal{Y}^{\infty N}} u'(c_{i,t})}{u'(c_{t,y^N})}, \quad (107)$$

$$\xi^{v,0} = \frac{\sum_{y^N \in \mathcal{Y}^{\infty N}} v(l_{i,t})}{v(l_{t,y^N})}, \quad \xi^y = \frac{\sum_{y^N \in \mathcal{Y}^{\infty N}} (y_{i,t} l_{i,t})^{1-\tau}}{(y_0^N l_{t,y^N})^{1-\tau}}. \quad (108)$$

Finally, we define the following variables: $\tilde{\xi}^{u,1} := \xi^{u,1} / l$, $\tilde{\xi}^{v,1} := \xi^{v,1} / ((1 - \tau) W \xi^y \circ y^{1-\tau} \circ l^{-\tau})$, and $\tilde{\xi}^{v,0} := \xi^{v,0} / ((1 - \tau) W \xi^y \circ y^{1-\tau} \circ l^{-\tau})$.

D.3.3 Matrix expressions for the FOCs

We define the following variables: $\bar{\lambda}^l := S \circ \lambda^l$, $\bar{\psi} := S \circ \hat{\psi}$, $\bar{\Pi} := S \circ \Pi^\top \circ (1/S)$, $\bar{\omega} := S \circ \omega$, $\bar{\lambda}_c := S \circ \lambda_c$, and notice that $S \circ \bar{\lambda}_c = \bar{\Pi} \bar{\lambda}_c$. With this notation, the FOCs (92)–(98) become:

$$\bar{\psi} = \bar{\omega} \circ \xi^{u,0} \circ u'(c) - \mu S \quad (109)$$

$$- \left(\bar{\lambda}_c \circ \xi^{u,E} - R \Pi \bar{\lambda}_c \circ \xi^{u,E} - (1 - \tau) W \bar{\lambda}_l \circ \xi^y \circ (y \circ l)^{1-\tau} \circ \tilde{\xi}^{u,1} \right) \circ u''(c),$$

$$P \bar{\psi} = \beta R P \bar{\Pi} \bar{\psi}, \quad (110)$$

$$(I - P) \bar{\lambda}_c = 0, \quad (111)$$

$$\left(\xi^y \circ (y \circ l)^{1-\tau} \right)^\top \bar{\psi} = -(1 - \tau) \left(\xi^y \circ (y \circ l)^{1-\tau} \circ \tilde{\xi}^{u,1} \circ u'(c) \right)^\top \bar{\lambda}_l, \quad (112)$$

$$\tilde{a}^\top \bar{\psi} = - \left(\xi^{u,E} \circ u'(c) \right)^\top \Pi \bar{\lambda}_c, \quad (113)$$

$$\bar{\omega} \circ \tilde{\xi}^{v,0} \circ v'(l) + \bar{\lambda}_l \circ \tilde{\xi}^{v,1} \circ v''(l) = \bar{\psi} - \tau \tilde{\xi}^{u,1} \circ u'(c) \circ \bar{\lambda}_l + \mu F_L S / ((1 - \tau) W \xi^y \circ y^{-\tau} \circ l^{-\tau}), \quad (114)$$

$$\left(\ln(y \circ l) \circ \xi^y \circ (y \circ l)^{1-\tau} \right)^\top \bar{\psi} = - \left((1 + (1 - \tau) \ln(y \circ l)) \circ \xi^y \circ (y \circ l)^{1-\tau} \circ \tilde{\xi}^{u,1} \circ u'(c) \right)^\top \bar{\lambda}_l. \quad (115)$$

D.3.4 Solving the system

Equation (114) yields:

$$\begin{aligned} D_{\tilde{\xi}^{v,1} \circ v''(l) + \tau \tilde{\xi}^{u,1} \circ u'(c)} \bar{\lambda}_l &= \mu F_L S. / ((1 - \tau) W \xi^y \circ \mathbf{y}^{-\tau} \circ l^{-\tau}) + \bar{\psi} - D_{\tilde{\xi}^{v,0} \circ v'(l)} \bar{\omega}, \\ \bar{\lambda}_l &= M_0 \bar{\omega} + M_1 \bar{\psi} + \mu V_0. \end{aligned} \quad (116)$$

with: $M_0 := -M_1 D_{\tilde{\xi}^{v,0} \circ v'(l)}$, $M_1 := D_{\tilde{\xi}^{v,1} \circ v''(l) + \tau \tilde{\xi}^{u,1} \circ u'(c)}^{-1}$, and $V_0 := F_L M_1 S. / ((1 - \tau) W \xi^y \circ \mathbf{y}^{-\tau} \circ l^{-\tau})$.

Equation (109) then implies:

$$\bar{\psi} = \hat{M}_0 \bar{\omega} + \hat{M}_1 \bar{\lambda}_c + \hat{M}_2 \bar{\lambda}_l - \mu S. \quad (117)$$

with: $\hat{M}_0 := D_{\xi^{u,0} \circ u'(c)}$, $\hat{M}_1 := -D_{\xi^{u,E} \circ u''(c)}(I - R\Pi)$, $\hat{M}_2 := (1 - \tau) W D_{\xi^y \circ (y \circ l)^{1-\tau} \circ \tilde{\xi}^{u,1} \circ u''(c)}$.

We obtain using (117) and (116):

$$\bar{\psi} = M_3 \bar{\omega} + M_4 \bar{\lambda}_c + \mu V_1, \quad (118)$$

where $M_2 := I - \hat{M}_2 M_1$, $M_3 := M_2^{-1}(\hat{M}_0 + \hat{M}_2 M_0)$, $M_4 := M_2^{-1} \hat{M}_1$, $V_1 := M_2^{-1}(\hat{M}_2 V_0 - S)$.

Furthermore, equations (110), (111), and (118) imply:

$$\bar{\lambda}_c = M_5 \bar{\omega} + \mu V_2, \quad (119)$$

where $\tilde{R}_5 := -((I - P) + P(I - \beta R \bar{\Pi}) M_4)^{-1} P(I - \beta R \bar{\Pi})$, $M_5 := \tilde{R}_5 M_3$, and $V_2 := \tilde{R}_5 V_1$.

Substituting (118) and (119) into (113), we deduce:

$$\mu = -L_1 \bar{\omega}, \quad (120)$$

where $C_1 := \tilde{a}^\top (V_1 + M_4 V_2) + (\xi^{u,E} \circ u'(c))^\top \Pi V_2$ and $L_1 := (\tilde{a}^\top (M_3 + M_4 M_5) + (\xi^{u,E} \circ u'(c))^\top \Pi M_5) / C_1$.

We deduce from (118) and (119):

$$\bar{\lambda}_c = (M_5 - V_2 L_1) \bar{\omega}, \quad (121)$$

$$\bar{\psi} = M_6 \bar{\omega}, \quad (122)$$

and from (116):

$$\bar{\lambda}_l = \hat{M}_6 \bar{\omega}. \quad (123)$$

We have defined $\hat{M}_6 := M_0 + M_1 M_6 - V_0 L_1$ and $M_6 := M_3 + M_4(M_5 - V_2 L_1) - V_1 L_1$.

Constructing the constraints. The constraint of equation (115) becomes after substituting the expressions (122) of $\bar{\psi}$ and (123) of $\bar{\lambda}_l$:

$$L_2 \bar{\omega} = 0, \quad (124)$$

where:

$$\begin{aligned} \mathbf{L}_2 &:= \left(\ln(\mathbf{y} \circ \mathbf{l}) \circ \boldsymbol{\xi}^y \circ (\mathbf{y} \circ \mathbf{l})^{1-\tau} \right)^\top \mathbf{M}_6 \\ &\quad + \left((1 + (1 - \tau) \ln(\mathbf{y} \circ \mathbf{l})) \circ \boldsymbol{\xi}^y \circ (\mathbf{y} \circ \mathbf{l})^{1-\tau} \circ \tilde{\boldsymbol{\xi}}^{u,1} \circ u'(\mathbf{c}) \right)^\top \hat{\mathbf{M}}_6. \end{aligned}$$

The constraint (112) becomes after substituting the expressions (122) of $\bar{\boldsymbol{\psi}}$ and (123) of $\bar{\boldsymbol{\lambda}}_l$:

$$\mathbf{L}_3 \bar{\boldsymbol{\omega}} = 0, \quad (125)$$

where:

$$\mathbf{L}_3 := \left(\boldsymbol{\xi}^y \circ (\mathbf{y} \circ \mathbf{l})^{1-\tau} \right)^\top \mathbf{M}_6 + (1 - \tau) \left(\boldsymbol{\xi}^y \circ (\mathbf{y} \circ \mathbf{l})^{1-\tau} \circ \tilde{\boldsymbol{\xi}}^{u,1} \circ u'(\mathbf{c}) \right)^\top \hat{\mathbf{M}}_6.$$

D.3.5 Constrained weights

Now, assume that there is a limited number of K weights, ω^U . Let $\mathbf{M}_7$ be an $N_{tot} \times Y$ and $\mathbf{I}_{Y \times Y}$ be a $Y \times Y$ identity matrix. Then we have

$$\bar{\boldsymbol{\omega}} = \mathbf{D}_S \mathbf{M}_7 \omega^U,$$

where $\mathbf{D}_S$ is the diagonal matrix of the the variable S_{yN} , and $\mathbf{M}_7$ is a matrix identifying the productivity level of each bin.

Finally, the weights for each productivity level are given as a solution to the following minimization problem:

$$\begin{aligned} \min_{\omega} \quad & \mathbf{S}^P \left\| \omega^U - \mathbf{1}_Y \right\|^2, \\ \text{s.t.} \quad & \mathbf{L}_2 \mathbf{D}_S \mathbf{M}_7 \omega^{U\top} = 0, \\ & \mathbf{L}_3 \mathbf{D}_S \mathbf{M}_7 \omega^{U\top} = 0, \\ & \mathbf{1}^\top \mathbf{D}_S \mathbf{M}_7 \omega^{U\top} = 1. \end{aligned} \quad (126)$$

In case we want to estimate the weights for each history, we just need to set $\mathbf{M}_7 = \mathbf{I}$ and solve the previous problem for N^{tot} weights.

E Marginal change in fiscal systems

In Figure 8, we test the robustness of our estimates through a simple exercise. We increase the government spending to output ratio of the United States by 10% and of France by 1%. We then conduct different experiments financing this increase using each of the instruments (i.e., τ_k, τ_c, κ , and τ), ensuring that the state budget constraint is still satisfied. For each instrument, we plot the difference in the weights using our previously discussed estimation strategy. Table 6 summarizes the changes in the instruments required to satisfy the state budget after an increase in G/Y . Note that this increase in G/Y is financed either by an increase in the tax on capital (τ_k), tax on consumption (τ_c), average tax on labor ($1 - \kappa$), or progressivity (τ). It is noteworthy that financing the increase in G by either τ_c or κ has the same general equilibrium effect, implying that the difference in weights is identical for both experiments. This result is linked to the

	US		France	
	Steady state	Increase in G/Y	Steady state	Increase in G/Y
τ_k	0.36	0.387	0.35	0.361
τ_c	0.05	0.076	0.18	0.19
κ	0.85	0.83	0.728	0.72
τ	0.16	0.22	0.23	0.25

Table 6: Changes in the fiscal instruments after an increase in G/Y for the United States and France.

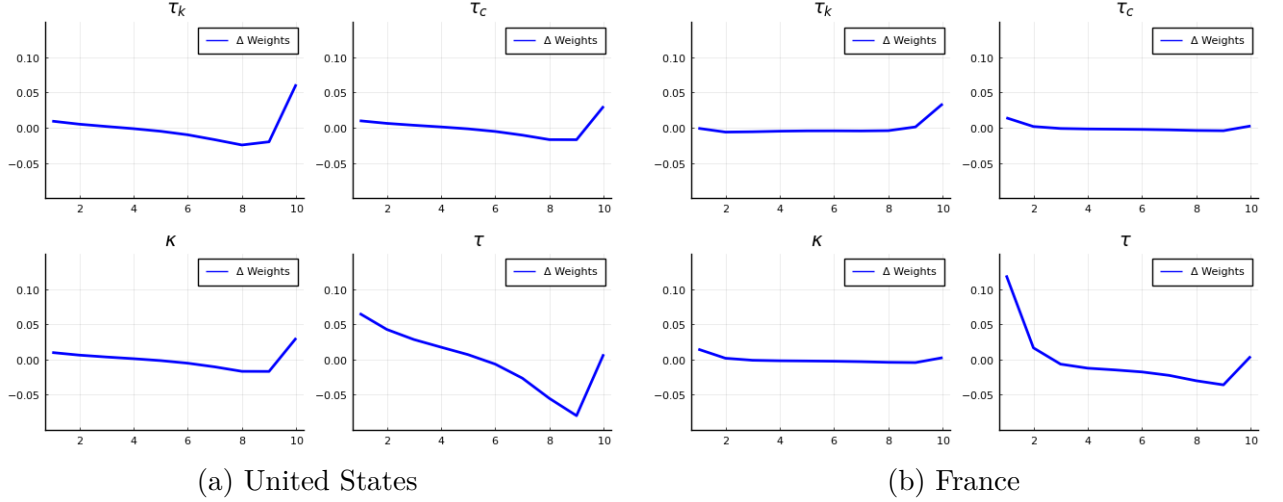


Figure 8: Change in weights by increasing G/Y for the United States and France.

redundancy result discussed in Section 5.3.

Finally, it can be observed that the instrument causing the greatest change in weights is the progressivity of the labor tax, supporting the idea that increasing progressivity benefits agents with lower productivity shocks at the expense of those with higher shocks.

F Changes in the fiscal system

Figure 9 shows how the weights, utility, labor, and capital income change when we alter the fiscal system of each country by that of the other under consideration. Panel (a) demonstrates the case where the United States adopts the tax system of France, while panel (b) illustrates the reverse scenario where France adopts the tax system of the United States. As expected, we observe effects occurring in opposite directions. The experiment depicted in Figure 9 is conducted as follows: once the capital-to-output ratio is set to its steady-state value, we iterate the value of κ such that G/Y remains constant. By conducting the experiment in this manner, not only are the model parameters unchanged, but also the main macroeconomic ratios.

Table 7 shows the new fiscal system for each country under consideration.

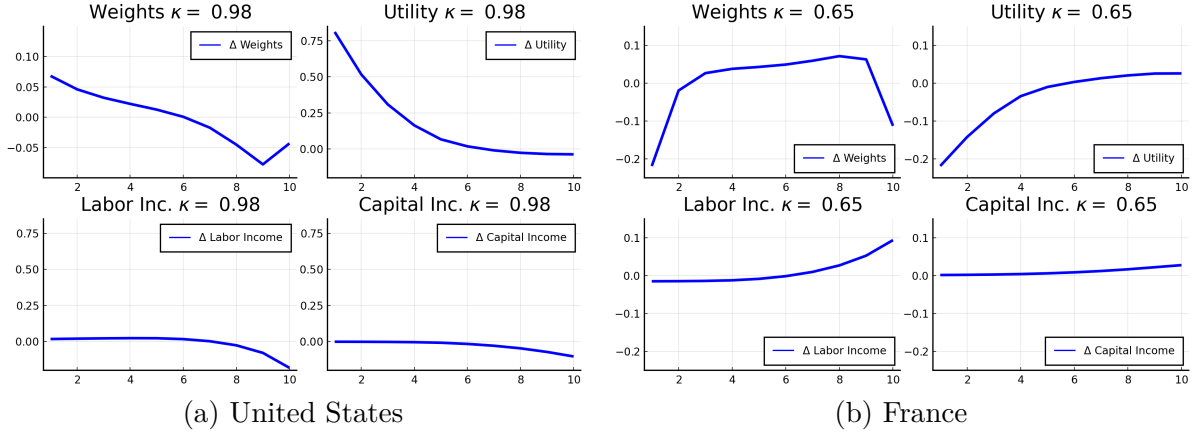


Figure 9: Difference in weights between the US and French fiscal systems.

	United States	France
τ_k	0.35	0.36
τ_c	0.18	0.05
κ	0.98	0.65
τ	0.23	0.16
B/Y	0.21	0.91

Table 7: Fiscal system for the United States and France under the experiment.

G Weights by History

We calculate the weights by history in Figure 10, where we represent the weights for the US in panel (a) and for France in panel (b) to ease comparison. We do the same exercise by considering the average wage per year of each agent and wealth associated with each history in the national currency of the respective country. This is also represented in Figure 10.²⁶

For the US, one can observe that the weights have a U-shape in the dimension of productivity/income (for low wealth levels), becoming more increasing as the wealth grows, and are slightly increasing in the dimension of wealth (for low productivity/income level), becoming more decreasing as the productivity/income level gets higher. This means the planner puts more weight on low productivity agents as long as they are not wealthy. The relatively low progressive tax for labor favors the high-wage and thus high-productivity agents, which implies bigger weights for these agents. However, for agents who are credit constrained, the presence of progressivity favors them

²⁶For the US we obtain the average hourly earnings of all employees from the *FRED* for the year 2007. This value is US\$ 20.17 per hour. For France we use the value for 2006 obtained in *Eurostat*, which is also the average hourly earnings of all employees. The value for France for the year 2006 is € 13.23 per hour. We then obtain the normalized wage using $\tilde{w}_t = \frac{\bar{w} \sum_{y^N \in \mathcal{Y}^N} S_{t,y^N} \tilde{l}_{t,y^N}}{\sum_{y^N \in \mathcal{Y}^N} \xi_{y^N}^y S_{t,y^N} (\tilde{l}_{t,y^N} y_{y^N})^{1-\tau_t}}$, where $\bar{w}$ represents the average wage. By doing this procedure we can, hence, calculate the total wage per year as being $\tilde{w}_t (\tilde{l}_{t,y^N} y_{y^N})^{1-\tau_t}$, where $\tilde{l}_{t,y^N} := 8760 l_{t,y^N}$ with 8760 being the total hours in a year and l_{t,y^N} the fraction of time the agent in this economy spends working. To normalize the wealth we use the fact that $\frac{A_t}{Y_t} = \frac{K_t}{Y_t} + \frac{B_t}{Y_t}$. Notice, we can define $A_t = GDP_t^i (\frac{K_t}{Y_t} + \frac{B_t}{Y_t})$, where the values of $\frac{K_t}{Y_t}$ and $\frac{B_t}{Y_t}$ were calculated previously. For both countries we use the GDP data measured at national currency, current prices. The data for the GDP was obtained in the OECD for the year 2007.

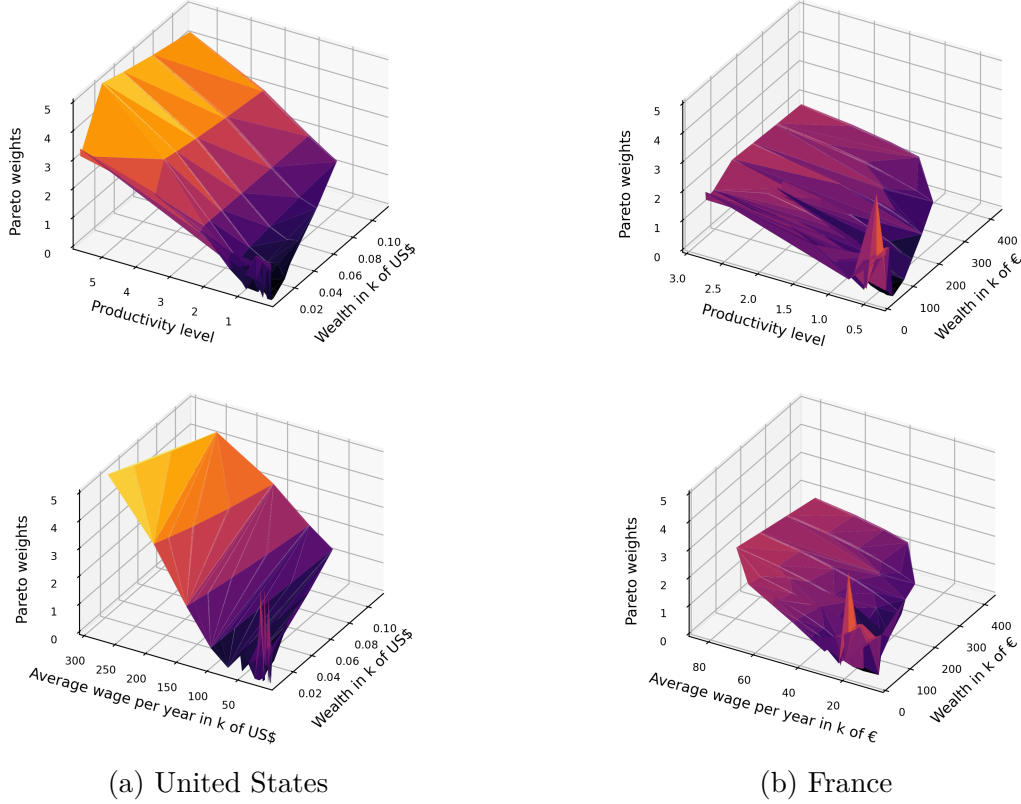


Figure 10: Pareto weights as a function of productivity (income per capita) and wealth for the US and France.

and does not cause distortions, which implies the planner gives a higher weight to those agents.

For France, the shape of the weights is different from the US. Although it also has a U-shape format, it favors low productivity agents more significantly. The weights decrease sharply with productivity and involve less heterogeneity in weights than the US distribution. This shape for the weights in the income dimension results from a higher progressivity in the labor tax. As with the US, the heterogeneity along the wealth dimension is much more limited, although it varies less than in the US case. Notice that for high levels of productivity/income, the weights along the wealth dimension become less heterogeneous. The same can be said when analyzing the evolution of the weights for higher levels of wealth, where the weights become more homogeneous as productivity/income increases. This result supports the idea that the fiscal system in France exhibits a higher inequality aversion component than that in the US.

Finally in Figure 11, we plot the weights for the US in panel (a) and France in panel (b) as a function of histories. The blue dashed line represents the history weights, whose computation is detailed in Appendix D.3.5. In this case, we have as many weights as the possible number of histories.

In Figure 12, we present the non-parametric Pareto weights as a solid blue line and the average weights by history as a dashed red line. The objective here is to take the history-specific weights depicted in Figure 11 and, for histories that share the same productivity level in the current period, calculate an average weight. These averages are derived by grouping histories that have the same productivity at the beginning of the truncation period and then computing

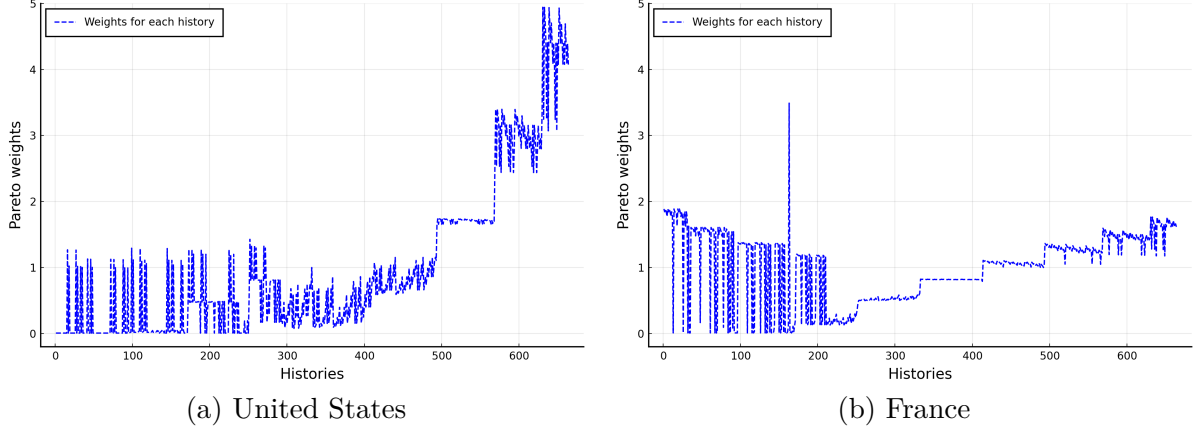


Figure 11: Period weights as a function of history for the US and France.

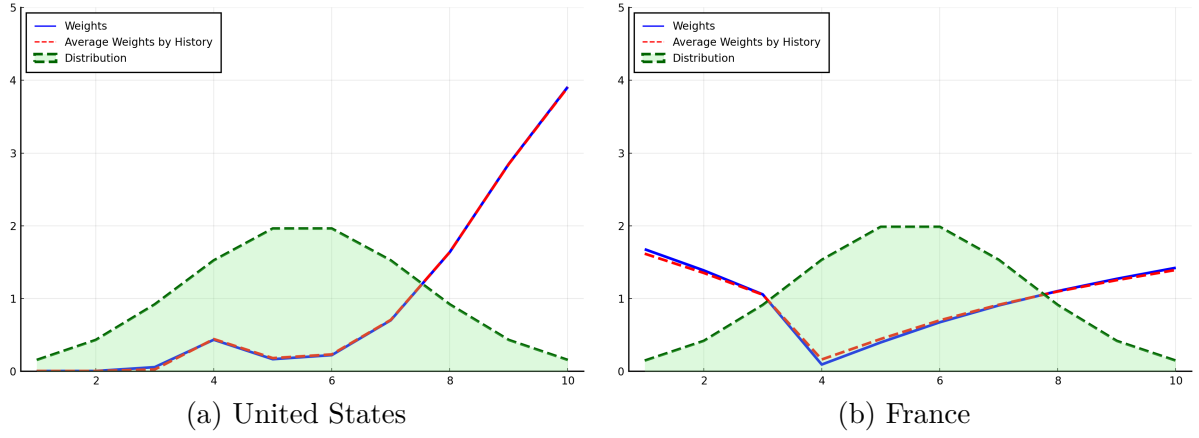


Figure 12: Comparison between Period weights and Average Weights by history for the US and France.

an average weight, taking into account the population size associated with each history as shown in Figure 11. As a result, we obtain 10 possible weights.

The histories are arranged in ascending order of productivity level at the start of the truncation, ensuring that the red dashed line closely approximates the non-parametric results, which is evident here. While differences do exist between these two methods of calculating weights, they are broadly similar overall.

H An alternative identification of the SWF

As explained in the Section 4.4, there are $n \geq 0$ constraints, written as $\sum_{y \in \mathcal{Y}} L_{k,y} \omega_y = 0$, where $L_{k,y} \in \mathbb{R}$ ($k = 1, \dots, n$ and $y \in \mathcal{Y}$).

The variance-minimizing weights are characterized by the vector $\hat{\omega}_y$, solving the following

program:

$$\hat{\omega}_y = \operatorname{argmin}_{\omega_y} \sum_{y \in \mathcal{Y}} \pi_y (\omega_y - 1)^2, \quad (127)$$

$$\text{s.t. } 0 = \sum_{y \in \mathcal{Y}} L_{k,y} \omega_y \text{ for all } k = 1, \dots, n, \quad (128)$$

$$1 = \sum_{y \in \mathcal{Y}} \pi_y \omega_y. \quad (129)$$